June 2016 group 1

Specimens:

A : germinating maize seeds. **E: earthworm**

B: liver fluke. **F: tilapia**

C: groundnut plant with nodules. **G: housefly maggot**

D: millipede. **H: garden snail**

1. (a)For each of the eight specimens (A to H) provided state:

i. The major group (phylum) to which it belongs,
ii. The sub-group (class) to which it belongs,
iii. The scientific or common name.

Your answer to this question should be written in your answer book in columns as shown below. (8 mks)

Specimen	Phylum	Class	Common or scientific name
A	Angiospermophyta	Monocotyledoneae	Maize or *Zea mays*
B	Platyhelminthes	Trematoda	Liver fluke or *Fasciola hepatica*
C	Angiospermophyta	Dicotyledoneae	Groundnut or *Arachnid hypogea*
D	Annelida	Oligochaeta	Earthworm or *Lumbricus terrestris*
E	Arthropoda	Diplopoda	Millipede or *Julius terrestris*
F	Chordata	Osteichthyes	Tilapia or *Tilapia zilli*
G	Arthropoda	Insecta	Housefly larva or *Musca domestica*
H	Mollusca	Gastropoda	Garden snail or *Helix aspersa*

(1/3 x 24=8 mks)

(b) Examine specimen C (groundnut), then discuss any feature that can improve soil fertility. (4mks)

- The presence of root nodules which contain nitrogen fixing bacteria that convert atmospheric nitrogen to soil nitrate
- When the leaves, roots and stems rot, they increase humus content in the soil
- The roots penetrate into the soil, thereby improving water infiltration and aeration.

(2x2 = 4 mks)

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NB: – The above features of groundnut also go for beans.

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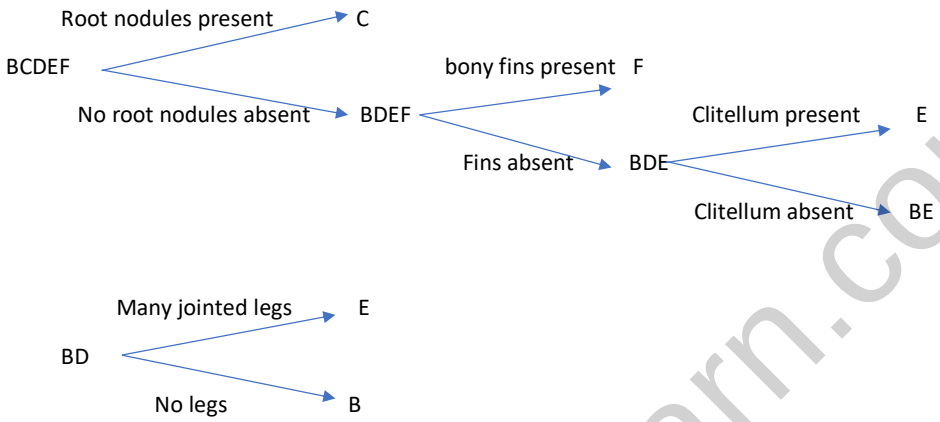
11(c) Make a simple dichotomous key to separate specimens B, C, D, E and F (excluding colour and size). (4 mks)



(d) State two roles of specimens E and G in the land ecosystem. (4 mks)

<u>Role of E (earthworm)</u>

- It creates many channels in the soil, thereby aerating the soil for good root growth.
- The channels also improve soil drainage for plant growth
- Their nitrogenase wastes mainly urea increases soil nitrogen level.
- Mixes up top soil and deep soil which reduces soil acidity and eases farming
- They drag leaves into soil burrows, thereby increasing soil humus.
- Their dead body increase soil humus content.
- They send out carbon dioxide used by plant for photosynthesis.
- They serve as food to some animals. (2 x2=4 mks)

<u>Role of G (housefly)</u>

- They are vectors of diseases like cholera.
- They are as scavengers, clearing off debris.
- They release carbon dioxide used by plants for photosynthesis.
- They cause food spoilage. (2x2=4 mks)

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2. (a) Make large labelled drawings of the lateral and ventral views of specimen D (earthworm). Annotate the parts used for

i. **Locomotion.**
ii. **Reproduction.** **(12 mks)**

Answer Page: **Be sure that students know the correct labels.**

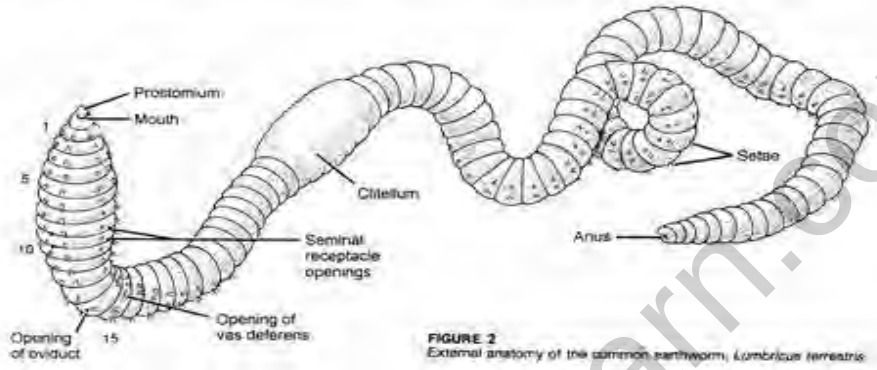
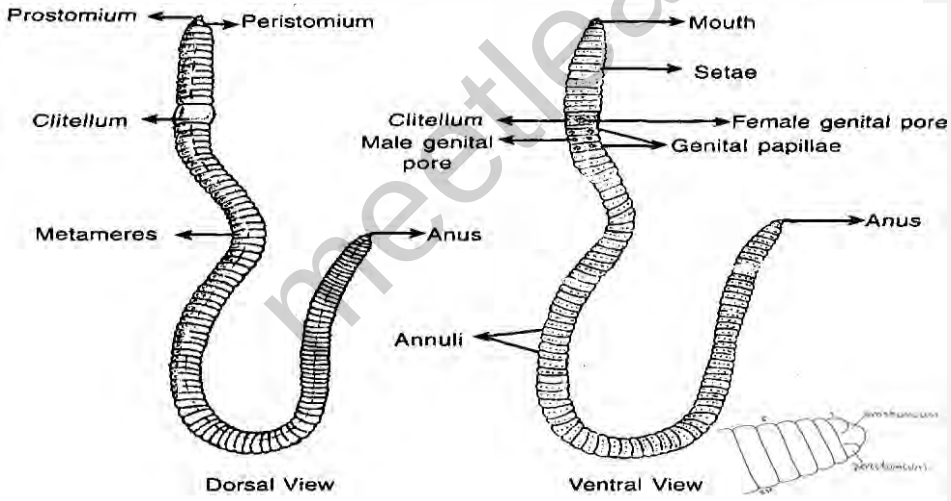


FIGURE 2
External anatomy of the common earthworm, Lumbricus terrestris



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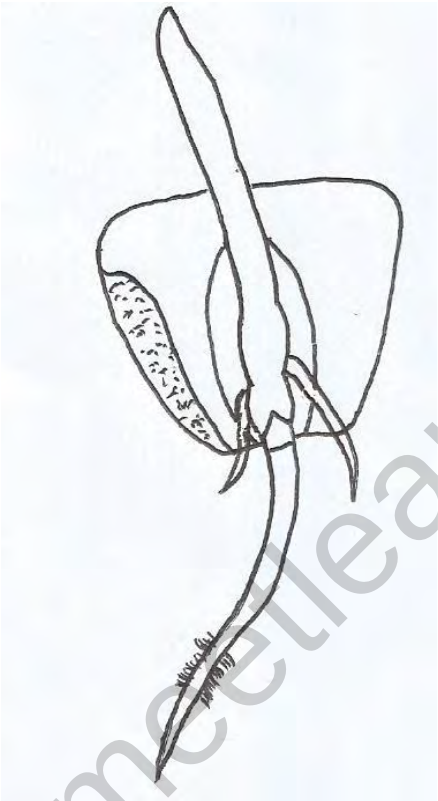
(b) (i) Using a razor blade carefully cut off specimen A longitudinally into two equal halves. Draw the two halves lying side by side and label their parts. (6 mks)

(ii) Annotate two parts that play a role in germination. (2 mks)



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3. (a) Dissect the thorax and abdomen of the bird to display its digestive system. Make a drawing of your dissection and label fully. (15 mks)

(b) Annotate

i. **Two structures concerned with mechanical digestion**
ii. **Three structures concerned with chemical digestion (10 mks)**

Leave your dissection properly displayed for assessment after the examination.

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The digestive system of a fowl

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4. You are provided with a branch of a mango leaf plant. Carefully remove five fresh leaves from the branch and immediately tie them apart on a string (thread). Weigh them together and record the weight. Tie the string containing the leaves near a window. Weigh the leaves after every 10 minutes for one hour. Record your result in the following table.

	X0	**X1**	**X2**	**X3**	**X4**	**X5**	**X6**
Time (minutes)	**0**	**10**	**20**	**30**	**40**	**50**	**60**
Initial weight (g)							
Final weight (g)							
Amount of water loss (g)							

(a) Describe your procedure and explain your result fully. (15mks)

Aim: To investigate the rate of water loss by evaporation from leaves (**the figures are just an example)**

	X0	**X1**	**X2**	**X3**	**X4**	**X5**	**X6**
Time (minutes)	**0**	**10**	**20**	**30**	**40**	**50**	**60**
Initial weight (g)	**48**	**48**	**40**	**34**	**29**	**25**	**22**
Final weight(g)	**48**	**40**	**34**	**29**	**25**	**23**	**22**
Amount of water loss (g)	**0**	**8**	**14**	**19**	**23**	**26**	**26**

(6x1/2=3 mks)

Procedure	Observation	Inference

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• 5 leaves were detached from a mango plant. • They were immediately tied apart on a string. • The leaves were weighed and the weight noted. • The string containing the leaves was tied near the window. Weighing of the leaves was done every 10 minutes for one hour.	• The leaves were becoming weaker. • The leaves were losing water with time • After some time, the weight was no more changing.	• Water was being list through the stomata and cuticle of the leaves. • The cells of the leaves were being plasmolysed, causing the leaves to wilt. • The air current near the window increased the rate of transpiration by removing the diffusion shells. • Sunlight and the high temperature in the room also encouraged transpiration. • Transpiration was complete when the weight was stagnant. Stomata were closed. • Wilting was very pronounced. • The cuticle on the upper leaf surface minimized water loss.

(12x1 mks)

(b) (i) calculate the percentage change in mass of the leaves from the beginning to the end of the experiment. (5 mks)

Percentage change of weight of leaves= $\frac{\text{weight at } t_0 - \text{weight at } t_{60}}{\text{Weight at } t_0} \times \frac{100}{1}$

$= \frac{48 - 22}{48} \times \frac{100}{1} = 58.17\%$

(ii) Draw a graph of the amount of water loss per minutes versus time. (5 mks)

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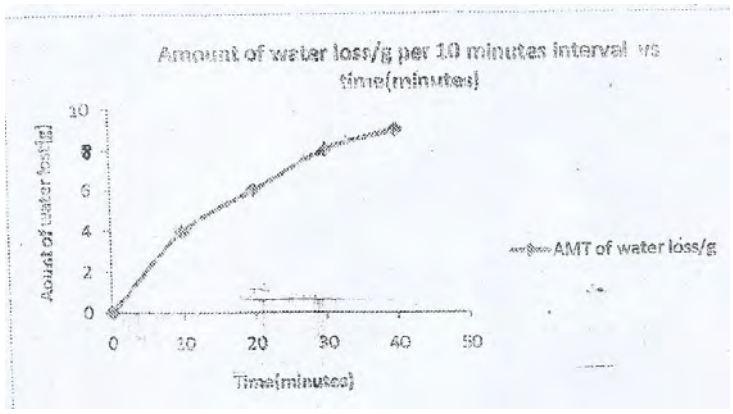

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June 2016 group 4

Specimens:

A : grasshopper. **E: mushroom**

B: garden snail. **F: spider**

C: toad. **G: fern**

D: cockroach. **H: hibiscus**

1. (a)For each of the eight specimens (A to H) provided state:

i. The major group (phylum) to which it belongs,
ii. The sub-group (class) to which it belongs,
iii. The scientific or common name.

Your answer to this question should be written in your answer book in columns as shown below. (8 mks)

Specimen	Phylum	Class	Scientific or common name
A	Arthropoda	Insecta	Grasshopper or Valanga nigricomis
B	Mollusca	Gastropoda	Snail or Helix aspersa
C	Chordata	Amphibia	Toad or Bufo bufo
D	Arthropoda	Insecta	Cockroach or Periplaneta americana
E	Basidiomycota	Basidiomycetes	Mushroom or Agaricus campestris
F	Arthropoda	Arachnida	Spider or Miranda aurantia
G	Filicinophyta	Filicinae	Fern or Dryopteris filix-mas
H	Angiospermophyta	Dicotyledoneae	Hibiscus or Hibiscus rosa-sinensis

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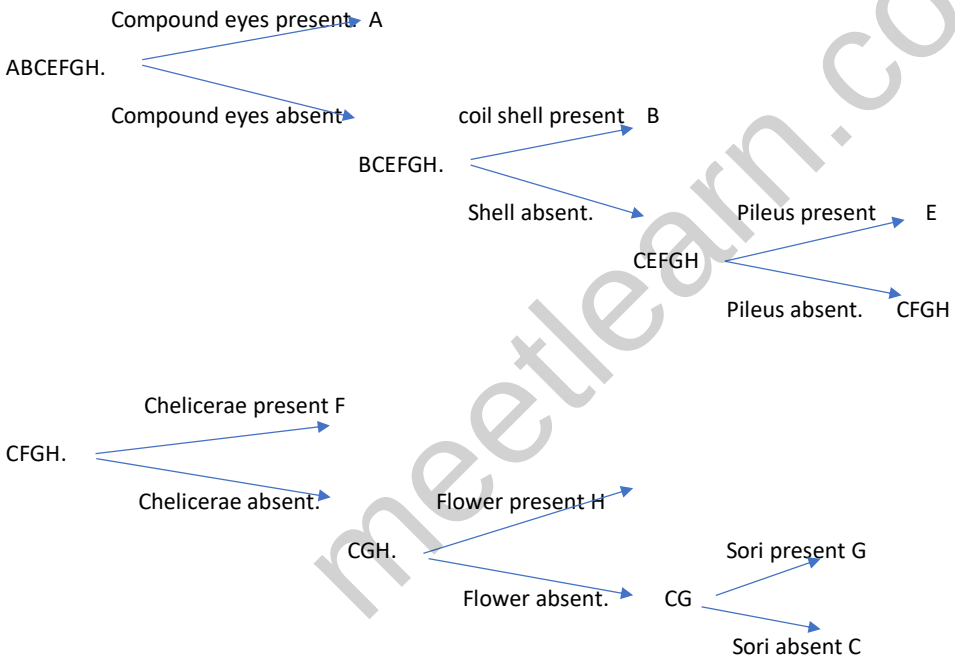
(b) (i) construct a dichotomous key to separate the specimens A, B, C, E, F, G and H. (6 mks)



(ii) Why is colour and size not considered when constructing a dichotomous key? (1 mk)

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- This is because these characteristics vary with time, season and the feeding habit of the organism.

(c) (i) Give the habitat of specimen E (mushroom) (1 mks)

- Moist humus soil.
- Dead organic remains.
- Rotting tree trunk

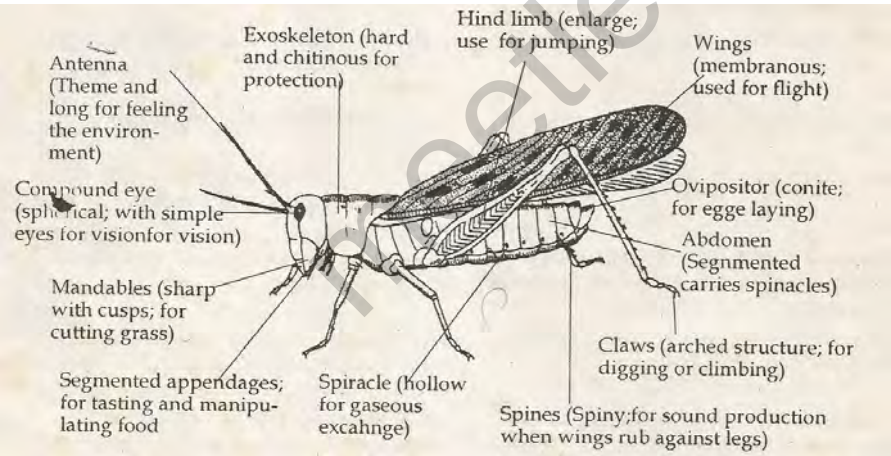
(ii) how is specimen E adapted to its environment? (4 mks)

- Has permeable hyphae (Rhizoid) for absorption of soluble nutrients
- Secretes extracellular enzymes to hydrolyse dead organic matter.
- Has basidia form the production of haploid basidiospores.
- Has a stipe for support of pileus.
- Has gills which support the sterigmata and basidia
- Basidiospores for asexual reproduction.

2. (a) observe specimen A (grasshopper) carefully and make a large labelled diagram. (7 mks)

(b) annotate on the diagram the parts concerned with

(i) respiration. (ii) locomotion. (iii) sensitivity. (6 mks)



External structure of a grasshopper

(c) Tabulate 4 differences between specimens D and F (4 mks)

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Specimen D (cockroach)	Specimen F (spider)
Two pairs of wings	No wings
Has compound eyes	Has simple eyes
Lacks chelicerae	One pair of chelicerae
Lacks Pedipalps	Has one pair of Pedipalps
Has 3 pairs of jointed legs	Has 4 pairs of jointed legs
Body divided into head, thorax and abdomen	Body divided into cephalothorax and abdomen
Has trachea for gaseous exchange	Has both book lungs and trachea for gaseous exchange
Lacks silk gland	Has silk glands
Lacks spinnerets	Has spinnerets

(d) How can a cockroach be considered a vector of diseases? (3 mks)

- It lives in toilets and other filthy places containing large amounts of decaying organic matter
- Picks up germs on its appendages like antennae, jointed legs and hairs on the body.
- It visits and settles on human food.
- May have carried with it germs for diseases like leprosy

3. (a) cut out one operculum to expose the opercular cavity. Make a drawing of the whole organism, showing the opercular cavity. Annotate one structure concerned with gaseous exchange (12 mks)

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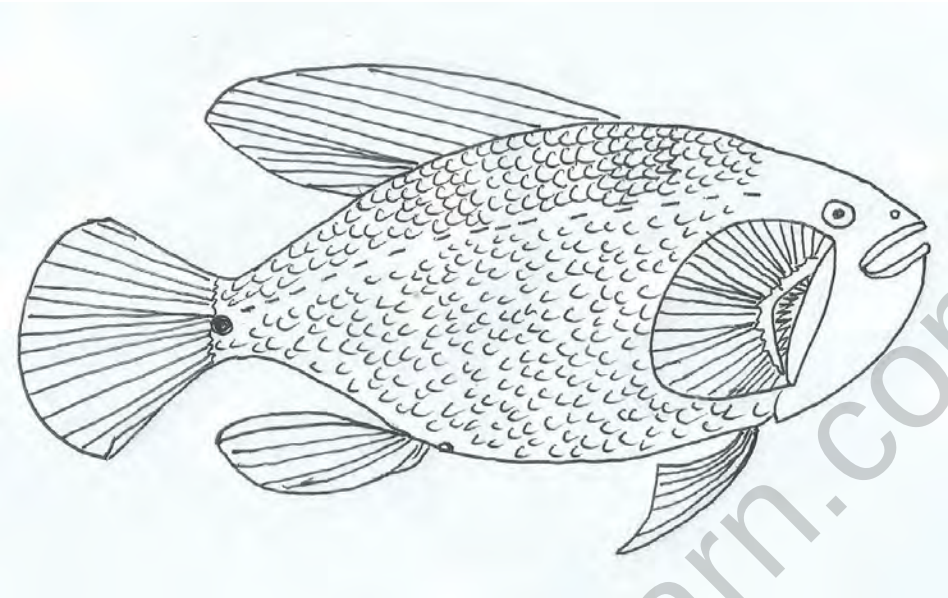

(b) Dissect the abdominal cavity and display the digestive system. Annotate two structures concerned with digestion. (13 mks)

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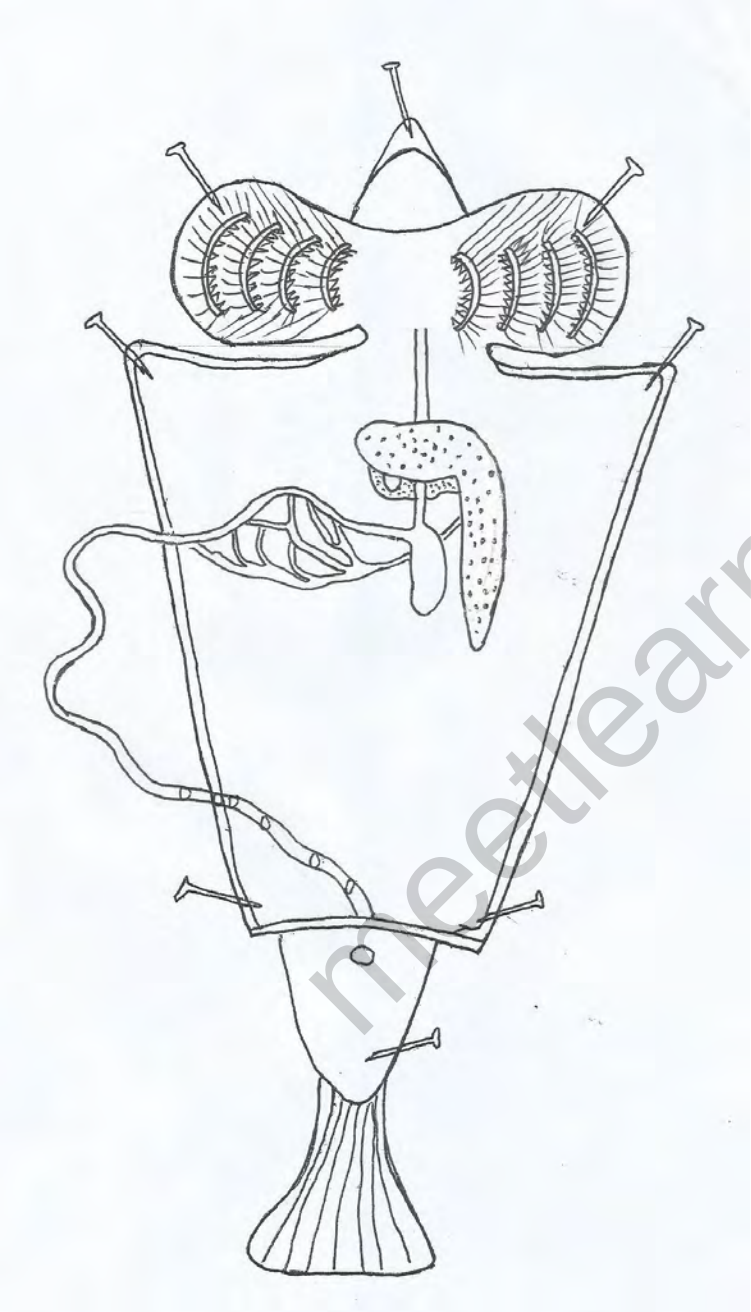

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4. Given solution J (made up of 1% soya beans, 1% sucrose and 1% starch).

Use solution J to test for the presence of

(i) carbohydrates. (17 mks)

(ii) proteins. (4 mks)

(iii) Fats. (4 mks)

<u>Answer</u>

(i) test for carbohydrates

*** test for starch**

Procedure	Observation	Inferences
• 5ml of solution J was put into a clean test tube using a syringe. • 5 ml of iodine solution were added into the test tube, while shaking after each drop.	• A blue black colouration was seen	• Solution J contains a high concentration of starch

***Test for reducing sugars**

Procedure	Observation	Inferences
• 2ml of solution J were put into a clean test tube using a syringe. • 2ml of Benedict's solution were added into the test tube. • The mixture was shakened and boiled in a warm water bath for 5 minutes, while shaking at regular interval	• The blue colour of the Benedict's solution / mixture did not change.	• No reducing sugar is present in solution J.

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(ii) Test for proteins.

Procedure	Observation	Inferences
• 2ml of solution J were put into a clean test tube using a syringe. • 1ml of dilute HCl was added. • The mixture was boiled in a warm water bath for 3 minutes. • The mixture in the test tube was allowed to cooled and Li_2CO_3 or $NaHCO_3$ powder was added until fizzing stopped, or 1ml of dilute HCl • 2ml of Benedict's solution were added into the mixture. • The mixture was boiled in a hot water bath for 5 minutes, while shaking at intervals	• The colour of the mixture changed from blue to green, to greenish yellow, to orange and then to brick red.	• Non- reducing sugars are present in solution J. • The HCl hydrolysed the glyosidic bond holding the two monosaccharides together in the disaccharide. • The resulting monosaccharides later gave the positive reducing sugar test.
Procedure	**Observation**	**Inferences**
• 2ml of solution J were put into a clean test tube. • 1ml of dilute NaOH was added. • 1% $CuSO_4$ was added drop by drop into the mixture, while shaking after each drop.	• On adding 2 drops of 1% $CuSO_4$,purple or mauve or violet colour was seen.	• Much protein is present in solution J

Commented [1]:

(iii) Test for fats

Procedure	Observation	Inferences
• 2ml of solution J were put into a clean test tube using a syringe.	• The floating fat droplets at the top of the mixture picked	• Solution J contains fat.

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• 2ml of Sudan lll stain/solution was added into the test tube. • 2ml of water was added into the mixture. • The mixture was shaken vigorously and then allowed to stand for 5 minutes.	up the red stains of Sudan lll solution	

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June 2017 group 1

Specimens:

A: woodlouse. **E: clam**

B: bean plant with roots. **F: garden snail**

C: star fish. **G: Guinea pig**

D: housefly. **H: chicken**

1. . (a)For each of the eight specimens (A to H) provided state:

i. **The major group (phylum) to which it belongs,**
ii. **The sub-group (class) to which it belongs,**
iii. **The scientific or common name.**

Your answer to this question should be written in your answer book in columns as shown below. (8 mks)

Specimen	Phylum	Class	Common or scientific name
A	Arthropoda	Malacostraca	Woodlouse or Oniscus asselus
B	Angiospermophyta	Dicotyledoneae	Bean plant or Phasealus vulgaris
C	Echinodermata	Stelleroidea	Star fish or Asteria rubens
D	Arthropoda	Insecta	Housefly or Apis mellifera
E	Mollusca	Pelecypoda/ Bivalvia	Clam or Mytilus edulis
F	Mollusca	Gastropoda	Garden snail or Helix aspersa
G	Chordata	Mammalia	Guinea pig or Cavia cobaye
H	Chordata	Aves/ Avia	Chicken/ domestic fowl

(b) Give four external features each of specimens G and H that make them to adapt to their respective habitats (8 mks)

Adaptations of specimen G (Guinea pig)

- Fur/ hair on the body to maintain a constant body temperature.

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- Pinnae to collect sound waves into the ear drum.
- Presence of whiskers to feel the environment in dark.
- Muscular hind limbs for hopping
- Claws to grip the ground during locomotion
- Eyes for vision
- Pair of nostrils for smelling and breathing.

Adaptations of specimen H (chicken)

- Fore limbs modified to wings for flight.
- Keel feathers for flight.
- Pointed sharp horny beak/beck for picking grains.
- Horny scale on legs for protection
- Sharp claws for scratching the ground for food.
- Feathers for protection and insulation.
- Nostrils for breathing.
- Eyes with nictitating membrane for vision.

(c) How is specimen H economically important? (2 mks)

- They are reared for sale (generate income)
- They are a source of food (protein)
- Their eggs are used for food and for sale.
- They are a source of employment e.g. poultry farming
- They destroy crops
- Fowl drop (their faeces) is used as manure
- Feathers are used for decoration.

(d) Give two structural differences between specimens C and E. (2 mks)

Specimen C (star fish)	**Specimen E (clam)**
Pentaradial symmetry	Bilateral symmetry
Pedicellariae to protect the body	Bivalve shells to protect the body
Aboral and oral sides	Laterally compressed

2. (a) Draw the dorsal view of specimen A (woodlouse). (6 mks)

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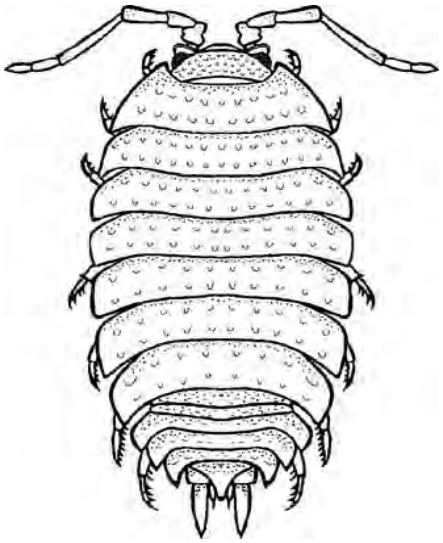

Dorsal view of wood louse

(b) (i) Why are specimens A (woodlouse) and D (housefly) grouped in the same major group (phylum). (3 mks)

- Both have segmented bodies.
- Both have jointed legs
- Their bodies are covered by chitinous exoskeleton
- Both have bilateral symmetry
- Both have jointed antennae

(ii) Why are specimens A and D grouped in different sub-groups (classes). (4 mks)

Specimen A (woodlouse)	**Specimen D (housefly)**
No wings	Wings for flight
Many pairs (7) of jointed legs	3 pairs of jointed legs
Chewing mouth part (mandible)	Sucking mouth part (proboscis)
Thorax not conspicuous (cephalothorax)	Thorax is conspicuous
Compound eyes only	Compound and simple eyes

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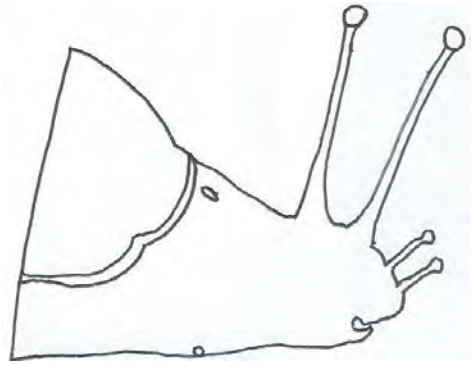
Haltere or balancer absent	Haltere or balancer present

(c) Draw the head region of specimen F (garden snail) . (4 mks). Annotate any two parts to show their functions.(4 mks)



Head region of garden snail

3. (a) Dissect the bird provided and display the digestive, circulatory, respiratory, excretory and reproductive systems. (13 mks)

(b) Annotate two parts each concerned with the following:

(i) for respiration. (ii) for mechanical digestion. (iii) for chemical digestion. (12 mks)

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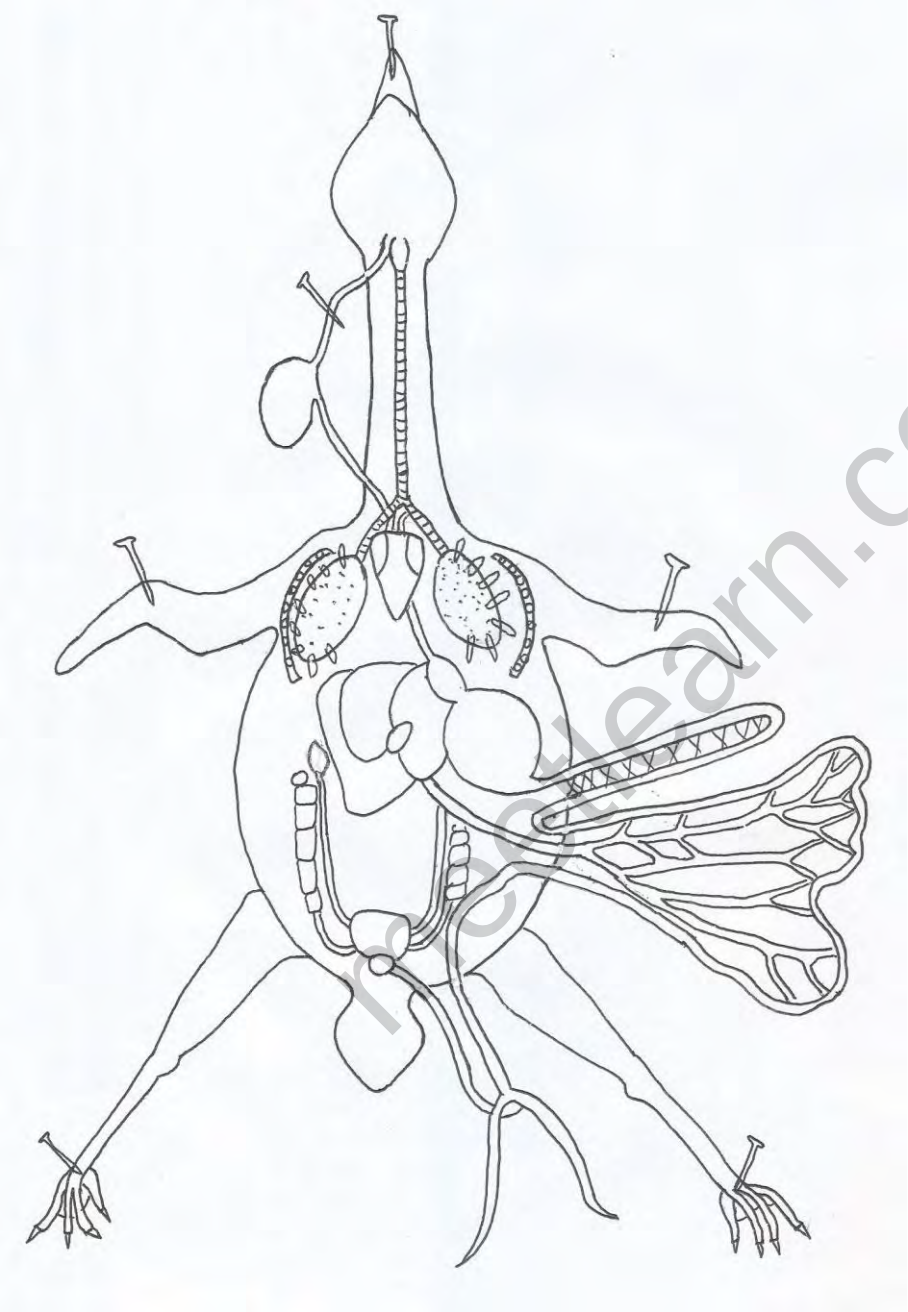

Respiratory, circulatory, digestive, excretory and reproductive systems of a fowl

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4. (a) Macerate the bean seeds provided and form a solution. Use the solution to test for the presence of the following: Starch, protein, reducing sugars and non-reducing sugars. (20 mks)

Answer

- Ten grains of germinating bean seeds were macerated using a mortar and a pestle.
- 12ml of distilled water was added to the crushed grains to make a solution pool
- The solution was filtered using a funnel and filter leper.

Test aim	Procedure	Results	Inferences
Starch	• 2ml of solution was put into a clean test tube using a syringe • 3 drops of iodine solution were added	• A dark yellow colouration observed	• No starch present
Protein	• 2ml of solution was put into a clean test tube using a syringe • 1ml of NaOH was added • 3 drops of $CuSO_4$ was added, while shaking after each drop.	• A violet or purple colouration seen	Proteins present
Reducing sugars	• 2ml of solution was put into a clean test tube using a syringe. • 2ml of Benedict's solution was added. • The mixture was heated in a hot water bath for 5 minutes, while shaking at intervals.	• Greenish yellow or orange colouration observed	• Reducing sugars present
Non-reducing sugars	• 2ml of the solution was put into a clean test tube. 5 drops of HCl was added and the mixture shakened	• Greenish yellow to brick red colouration observed.	• Non-reducing sugars present

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	• The mixture was heated in a hot water bath for 3 minutes. • The mixture is cooled under running tap water. • Solid Na2CO3 was added to neutralize the acid. 2ml of Benedict's solution was added and shaken • The mixture was heated in a warm water bath for 5 minutes, while at intervals.		

(b) Critically comment on the results obtained. (5 mks)

- Seeds are storage organs for food.
- In beans/ dicots, food is stored in the cotyledons
- Protein and small amount of starch are stored in the seeds.
- On germination, starch is converted to sugars.
- The sugars are oxidized to produce energy for germination
- Proteins are used for growth.

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June 2017 group 2

Specimens:

A: Toad. **E: Pine**

B: Cockroach. **F: Mushroom**

C: Obelia. **G: Moss**

D: Hibiscus. **H: lizard**

1. (a) For each of the specimens (A to H) provided, state:

i. The major group (phylum) to which it belongs,
ii. The Sub-group (class) to which it belongs,
iii. The scientific or common name

Your answer should be written in your answer book in columns as shown below:
(8 mks)

Specimen	Phylum	Class	Scientific or common name
A	Chordata	Amphibia	Toad or Bufo bufo
B	Arthropoda	Insecta	Cockroach or Periplaneta americana
C	Cnideria	Hydrozoa	Obelia or Obelia geniculata
D	Angiospermophyta	Dicotyledoneae	Hibiscus or Hibiscus rosa-sinensis
E	Coniferophyta	Coniferinae/Coniferae	Pine or Pinus sylvestris
F	Basidiomycota	Basidiomycetes	Mushroom or Agaricus campestris
G	Bryophyta	Musci	Moss or Polytrichum commune
H	Chordata	Reptilia	Lizard or Agama agama

(b) Give one characteristic each common to the phyla A, B, C, D and E. (5 mks)

Specimen	Characteristic feature of phylum

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A (toad)	• Presence of paired limbs
B (cockroach)	• Jointed appendages • Body covered with cuticle • Segmented body.
C (Obelia)	• Radial symmetry.
D (hibiscus)	• Flowers for reproduction. • Net Venation. • Fibrous root system
E (pine)	• Cones for reproduction. • Needles-like leaves

(c) (i) Give the ecological niche of specimen B (cockroach). (2 mks)

- Lives in dark corners like in cupboards and pit toilets.
- It lays eggs in cupboards or crevices
- Feeds on variety of organic wastes.
- Serves are food to many animals.

(ii) Give the effect of cockroach to its environment. (5 mks)

- Destroys cloths in cupboards.
- Serve as food to animals like chicken.
- Agent of decay.
- Can transmit diseases like leprosy.
- Release carbon dioxide used by autotrophs for photosynthesis.

2**. (a) (i) Observe specimen H (lizard) carefully and make a large labelled drawing of the lateral view. (8 mks)**

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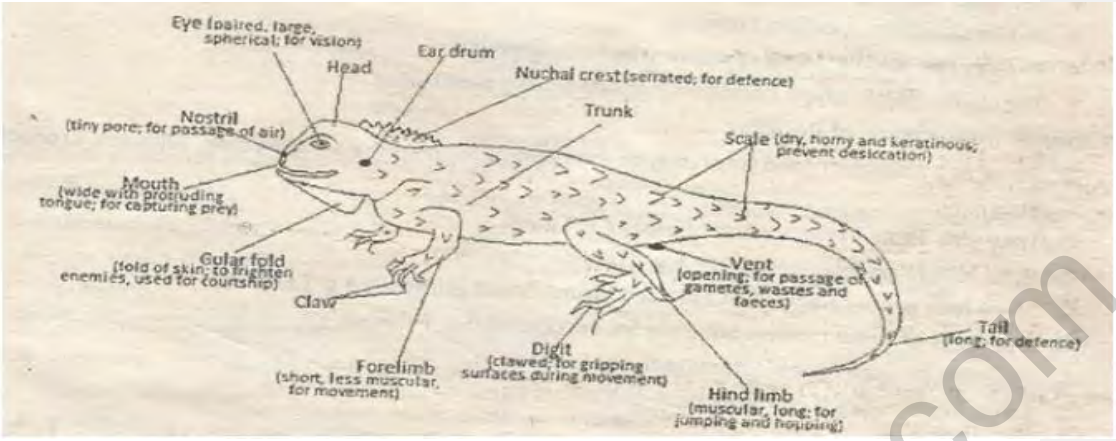

Lateral view of a lizard

(ii) How is specimen H (lizard) adapted to its environment? (4 mks)

- Body is covered with horny scale to protect against mechanical injury and water loss.
- Has eyes with eyelids for vision.
- Has muscular fore and hind limbs for fast movement.
- Has ear for sound detection.
- Has hard and sharp claws for gripping the surface when moving.
- Has protrusible tongue to capture preys.
- Has gular fold and nuchal crest for courtship.

(b) Give three visible structural differences between specimens A (toad) and H (lizard). (3 mks)

Specimen A (toad)	**Specimen H (lizard)**
Tail absent	Tail present
Wide gaping mouth	Mouth is not so wide
No scale on body	Body covered with horny scales
Muscular long hind limbs	Hind limbs not so muscular or long

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No gular fold	Gular fold present
No nuchal crest	Nuchal crest present
Webbed feet	Feet are not webbed
Poison glands on the back	No poison glands
Ear drum	Ear tube
Eye with no eyelids	Eyes with eyelids

(c) Make a large labelled and annotated diagram of the reproductive structure of specimen C (Obelia) (6 mks)

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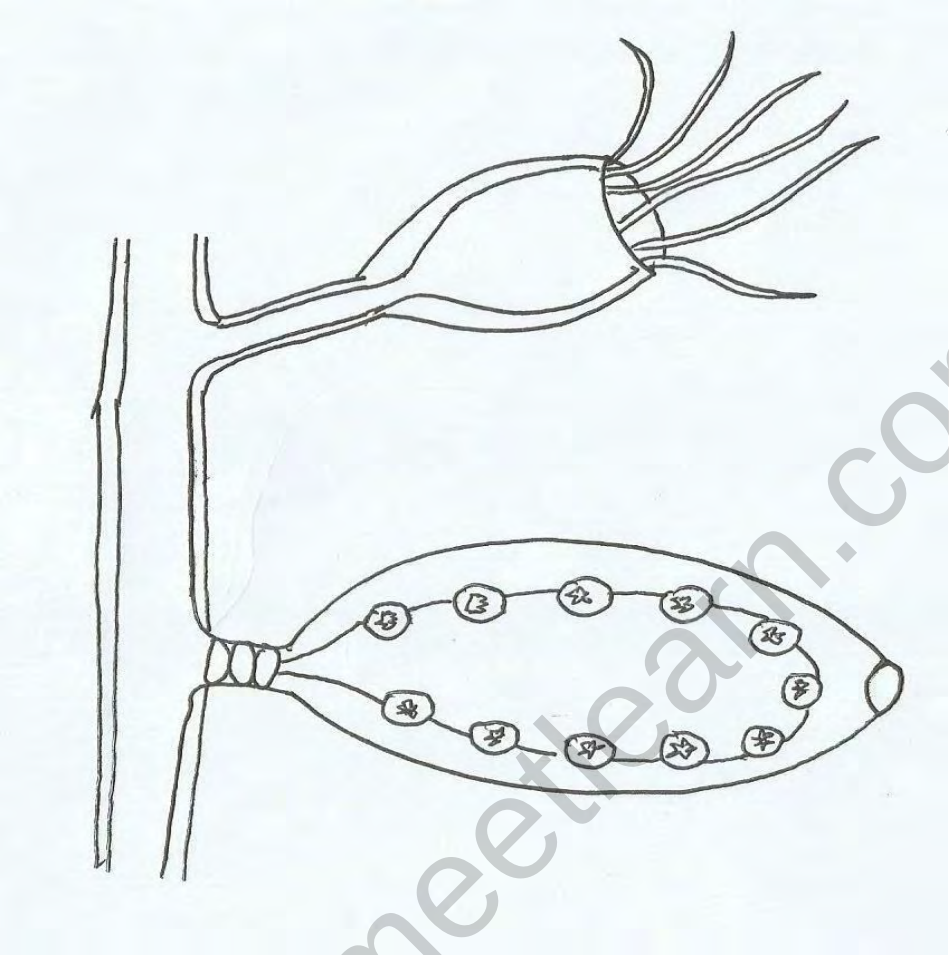

Reproductive structure of specimen C (Obelia)

3. (a) Dissect the fish provided and display the digestive and reproductive systems. (13 mks)

(b) Annotate three structures concerned with

(i) reproduction. (ii) excretion (12 mks)

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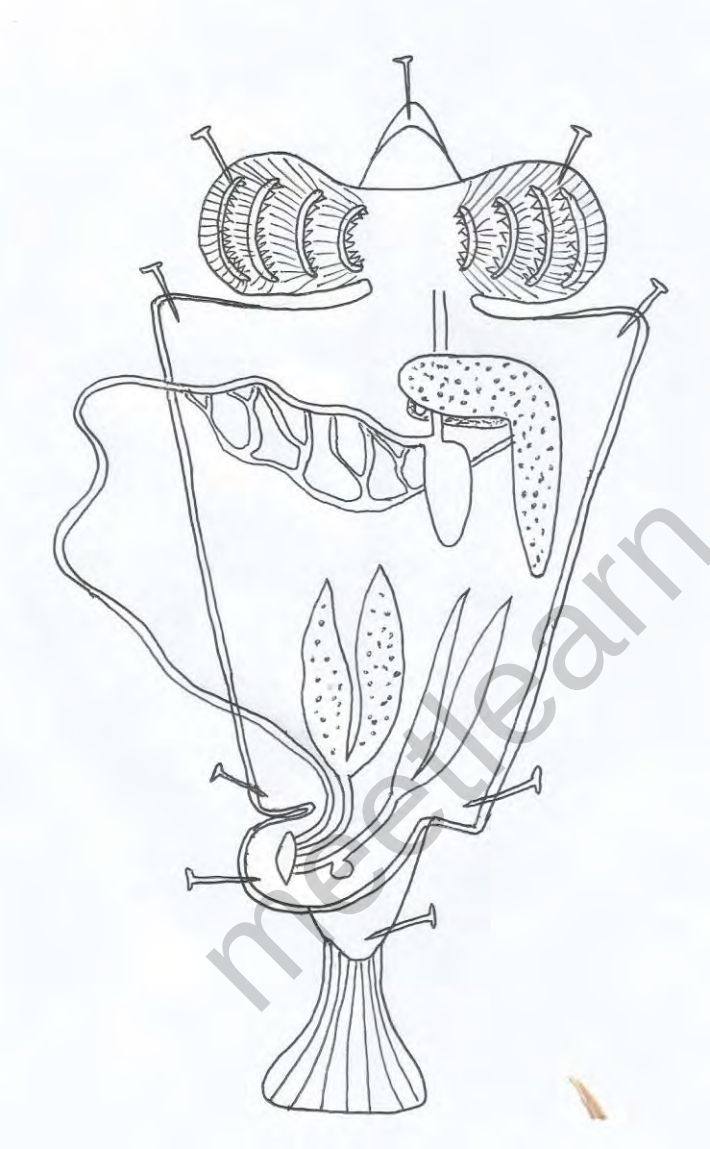

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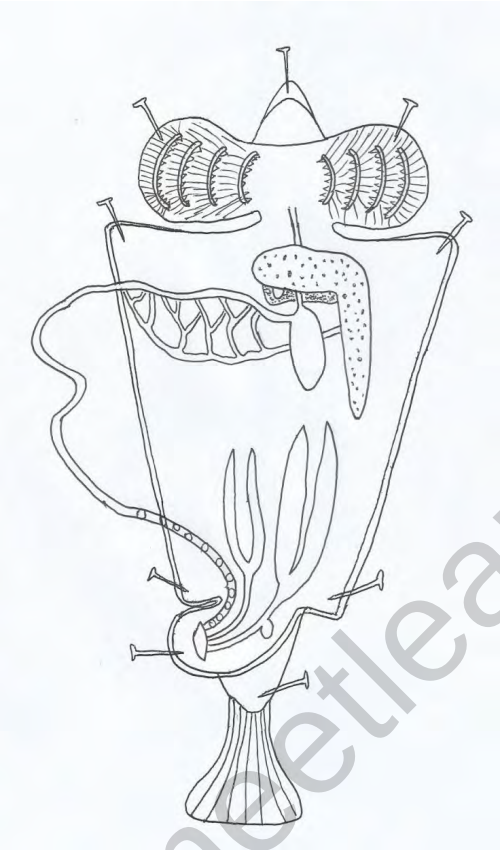

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4. Solution S = fresh palm wine.

<u>Answer</u>

(a) (i) Preparation of serial dilutions (e.g. for a volume of 10ml)

four solutions of concentrations 5%, 25%, 50% and 75% were prepared as follows:

5% = 0.5ml of solution S + 9.5ml of distilled water

25% = 2.5ml of solution S + 7.5ml of distilled water

50% = 5ml of solution S + 5ml of distilled water

75% = 7.5ml of solution + 2.5 of distilled water

Precaution: All the volumes should be 10ml

(ii) Procedure.

- 1ml of each solution (5%, 25%, 50 and 75%) was put into separate clean labelled test tubes using a syringe.
- 5 drops of methylene blue was added.
- The test tubes were allowed for 60 minutes.
- The time taken for each solution to decolorize was noted.

Observations / results.

Conc. of solution S (% dilution)	75%	50%	25%	5%
Time for decolorization	15 minutes	25 minutes	50 minutes	60 minutes

Explanation of results.

- Active palm wine had live yeast cells.
- The yeasts ferment the wine sugar.
- Electrons are released.
- The electrons are picked up by the dye methylene blue.
- The dye is reduced by reductase enzymes.
- The dye is decolorized as it is reduced.

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- The highest concentration 75% has more yeast cells, and therefore much reductase for reduction.
- The least concentration 5% has the least yeast cells, therefore less reductase for reduction.

(b) Control experiment to show that enzymes are involved.

Procedure	Observation	Inferences
• 2ml of solution S was put into a clean test tube. • The test tube was heated for 5 minutes in a hot water bath. • The solution was cooled under running tap water. • 5 drops of methylene blue was added into the test tube. • The mixture was shaken and allowed to stand for 60 minutes	• No decolorization took place indefinitely.	• The enzyme reductase that would have caused the decolorization of methylene blue or caused the reduction was denatured by heat.

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June 2017 group 3

Specimens:

A: crayfish. **E: clam**

B: elephant grass. **F: earthworm**

C: sea urchin. **G: chicken**

D: grasshopper. **H: lizard**

1. (a) For each of the specimens (A to H) provided, state:

iv. The major group (phylum) to which it belongs,
v. The Sub-group (class) to which it belongs,
vi. The scientific or common name

Your answer should be written in your answer book in columns as shown below:
(8 mks)

Specimen	Phylum	Class	Scientific or common name
A	Arthropoda	Malacostraca	Crayfish or Cambarus affinis
B	Angiospermophyta	Monocotyledoneae	Elephant grass or Festuca pratensis
C	Echinodermata	Echinoidea	Sea urchin or Echinus esculentus
D	Arthropoda	Insecta	Grasshopper or Valanga nigricomis
E	Mollusca	Bivalvia	Clam or Mytilus edulis
F	Annelida	Oligochaeta	Earthworm or Lumbricus terrestris
G	Chordata	Aves or Avia	Chicken or Gallus domestica
H	Chordata	Reptilia	Agama lizard or Agama agama

(b) Give the external features of specimens G and H that make them adapt to their respective modes of life or habitats. (8 mks)

*** External structural adaptations of specimen G (Chicken)**

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- Has keel or filoplume feathers for flight.
- Pointed horny beak for picking grains.
- Has horny scales on legs for protection.
- Has sharp claws for scratching the soil for food.
- Has eyes with eyelids for vision.
- Has nostrils for breathing and smelling.
- Body is covered with feathers for protection and temperature regulation.
- Fore limbs are modified to form wings for flight.

*** External structural adaptations of specimen H (Lizard)**

- Body is covered with horny scales for protection against mechanical injury and water loss.
- Has eyes with eyelids for vision.
- Has muscular fore and hind limbs for fast movement.
- Has ear for sound detection.
- Has hard and sharp claws for gripping the surface when moving.
- Has protrusible tongue to capture preys.
- Has gular fold and nuchal crest for courtship.

(c) Give 2 structural differences between specimens C and E. (2 mks)

Specimen C (sea urchin)	**Specimen E (clam)**
Has pentaradial symmetry	Bilateral symmetry
Has long spines to protect the body	Has bivalve shell to protect the body
Has Aboral and oral sides	Compressed laterally

2. (a) Observe specimen A carefully. Make a large labelled drawing of the dorsal view of specimen A (8 mks)

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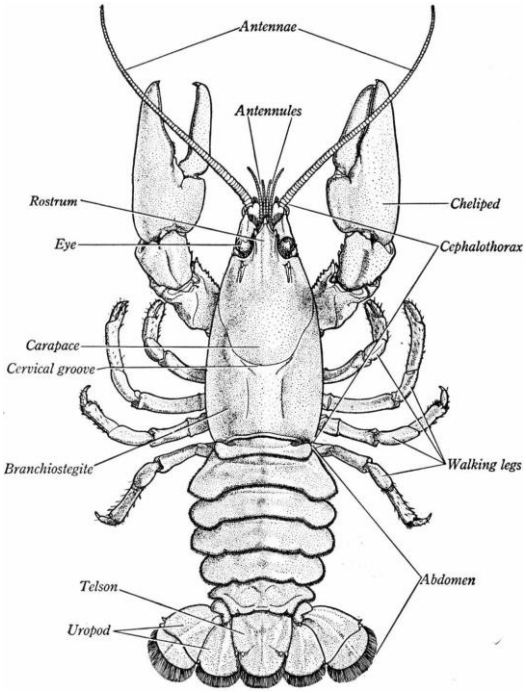

Structure of the Dorsal view of crayfish

(b) (i) Why are specimens A (crayfish) and D (grasshopper) grouped in the same major group/phylum? (3 mks)

- Both have segmented bodies.
- Both have jointed legs/appendages.
- Their bodies are covered by chitinous exoskeleton.
- Both have a bilateral symmetry. 1q

(ii) Why are specimens A and D grouped in different sub-groups (class). (4 mks)

Specimen A (crayfish)	**Specimen D (grasshopper)**
No wings	Wings present for flight

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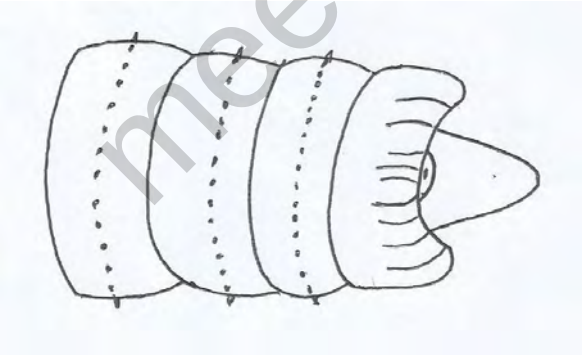
Many pairs of jointed legs	3 pairs of jointed legs
Pincer present	Pincer absent
Thorax if not conspicuous, but has cephalothorax	Thorax is conspicuous
Stalked compound eyes	Sessile compound eyes
Antennules present	No Antennules

(c) Draw the head region of specimen F (earthworm) and annotate two parts concerned with feeding. (5 mks)



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3. (a) Dissect the bird provided and display the digestive, circulatory, respiratory, excretory and reproductive systems. (13 mks)

(b) On your diagram, annotate three structures each concerned with

(i) blood circulation. (ii) excretion. (12 mks)

Leave your dissection properly displayed for assessment after the examination.

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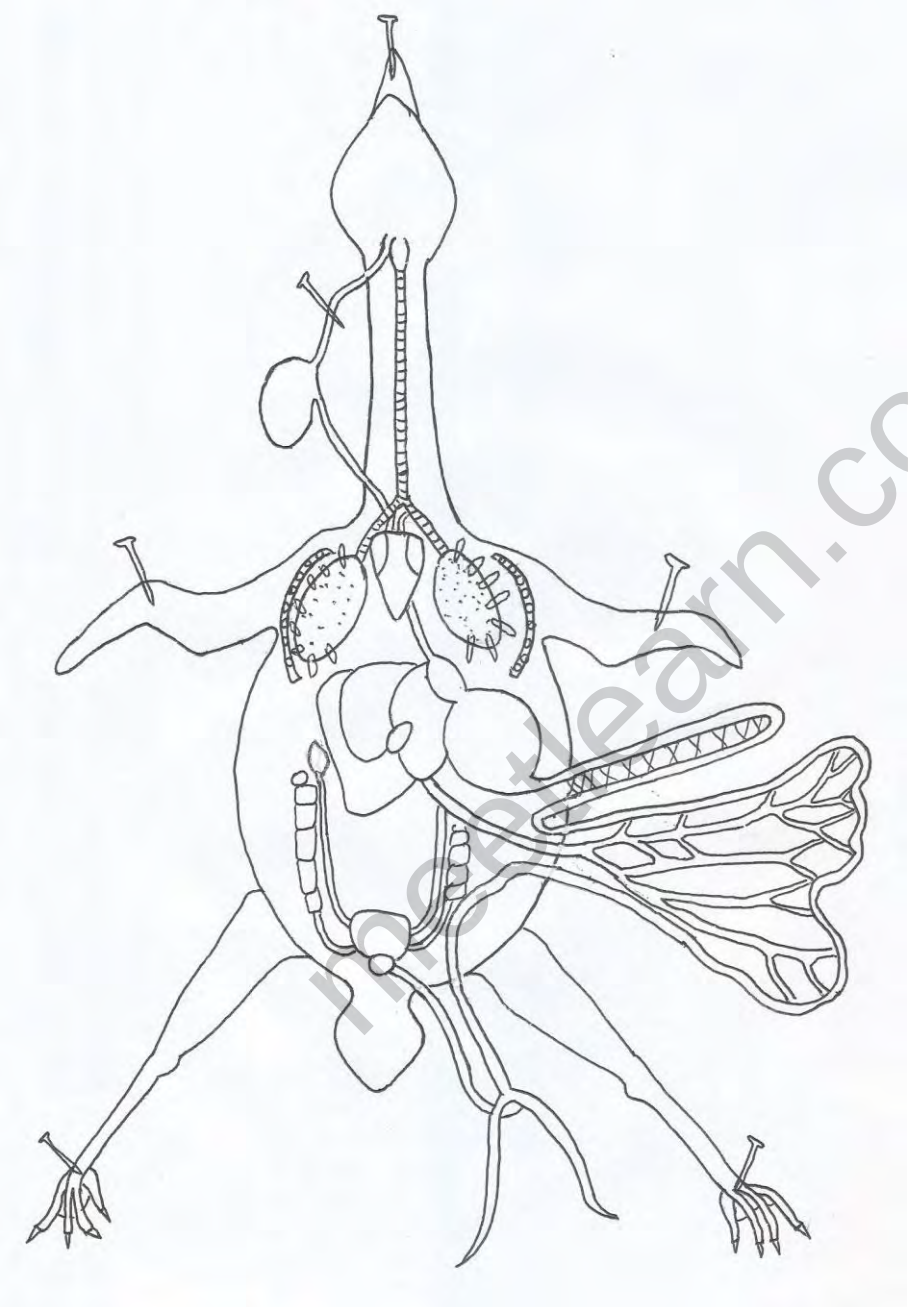

Structure of Respiratory, circulatory, digestive, excretory and reproductive systems of a fowl.

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4. (a) Macerate the germinating maize seeds provided and form solution S. Use the solution S to test for the presence of the following: Starch, protein, reducing sugars and non-reducing sugars. (20 mks)

Answer

- Ten grains of germinating maize seeds were macerated using a mortar and a pestle.
- 12ml of distilled water was added to the crushed grains to make a solution pool (solution S)
- The solution was filtered using a funnel and filter leper.

Test aim	Procedure	Results	Inferences
Starch	• 2ml of solution S was put into a clean test tube using a syringe • 3 drops of iodine solution were added	• A blue black colouration was seen	• Solution S contains starch
Protein	• 2ml of solution S was put into a clean test tube using a syringe • 1ml of NaOH was added • 3 drops of $CuSO_4$ was added, while shaking after each drop.	• A violet or purple colouration seen	Proteins is present in solution S
Reducing sugars	• 2ml of solution S was put into a clean test tube using a syringe. • 2ml of Benedict's solution was added. • The mixture was heated in a hot water bath for 5 minutes, while shaking at intervals.	• Greenish yellow or orange colouration observed	• Reducing sugars is present in solution S.

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Non-reducing sugars	• 2ml of the solution S was put into a clean test tube. 5 drops of HCl was added and the mixture shakened • The mixture was heated in a hot water bath for 3 minutes. • The mixture is cooled under running tap water. • Solid Na2CO3 was added to neutralize the acid. 2ml of Benedict's solution was added and shaken • The mixture was heated in a warm water bath for 5 minutes, while at intervals.	• Greenish yellow to brick red colouration observed.	• Non-reducing sugars is present in solution S

(b) Critically comment on the results obtained. (5 mks)

- Seeds are storage organs for food.
- In maize, food, principally starch is stored in the endosperm.
- Small amount of proteins are stored in the aleurone layer
- On germination, starch is converted to sugars.
- The sugars are oxidized to produce energy for germination
- Proteins are metabolized for growth and cellular synthesis.

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June 2013 group 3.

Specimens:

A: mango. **F: earthworm**

B: toad. **G: tilapia**

C: fruit fly. **H: water flea**

D: selaginella. **I: dog fish**

E: snail. **J: maize plant**

1. (a) For each of the specimens (A to J) provided, state:

vii. The major group (phylum) to which it belongs,
viii. The Sub-group (class) to which it belongs,
ix. The scientific or common name

Your answer should be written in your answer book in columns as shown below:
(15 mks)

Specimen	(i)	(ii)	(iii)
A	Angiospermophyta (Anthophyta)	Dicotyledoneae	Mango twig or *Mangifera indica*
B	Chordata	Amphibia	Toad or *Bufo regularis*
C	Arthropoda	Insecta	Fruit fly or *Drosophila melanogaster*
D	Lycopodophyta or Lycophyta	Lycopodinae	Selaginella or *Selaginella densa*
E	Mollusca	Gastropoda	Snail or *Helix aspersa*
F	Annelida	Oligochaeta	Earthworm or *Lumbricus terrestris*
G	Chordata	Osteichthyes	Tilapia or *Tilapia zilli*
H	Arthropoda	Branchiopoda	Water flea or *Daphnia pulex*
I	Chordata	Chondrichthyes	Dog fish or *Squalus acanthias*
J	Angiospermophyta	Monocotyledoneae or Liliopsida	Maize plant or *Zea mays*

(½ x 30=15 mks)

(b) Draw a well labeled diagram specimen D (selaginella)

Annotate one feature concerned with reproduction. (7 mks)

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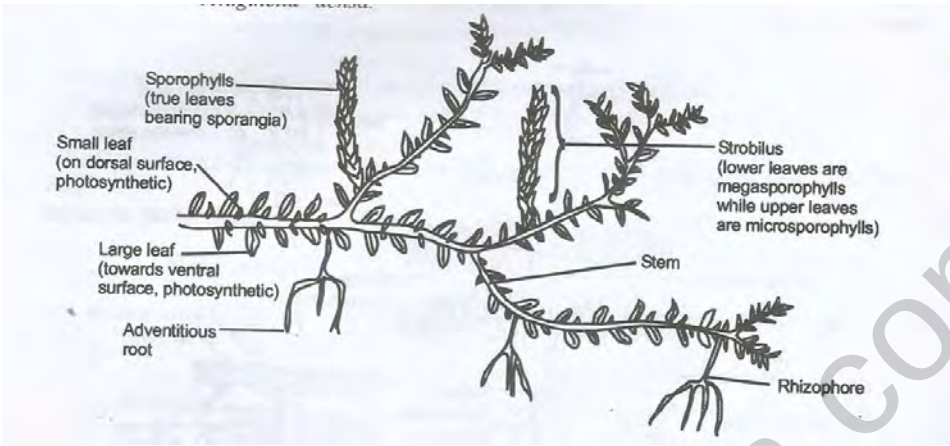

(c) How does specimen F improve soil fertility ? (8 mks)

- Mixes up top soil and deep soil which reduces soil acidity and eases farming.
- Burrows aerate plant roots and other soil organisms.
- Borrows help to improve soil drainage. The water is taken up by plants for photosynthesis.
- Worm casts add humus and increase soil fertility.
- They carry leaves into the soil through borrows that decay and add to soil fertility.
- When they die, their bodies decompose and add to humus content of soil.
- Their nitrogenous wastes mainly urea increase nitrogen level in the soil.

2. Examine specimens E and F closely.

a. For each specimen state four external structural adaptations to its mode of life. (8 mks)

<u>Specimen E (snail)</u>

- Has tentacles with small eyes for vision.
- Hard calcareous shell protects the body against mechanical injury and water loss.
- Has a large muscular foot for locomotion.

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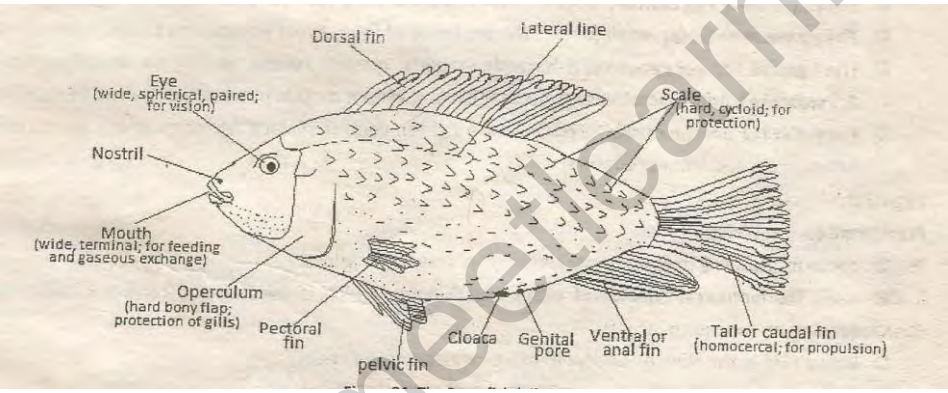
- Show torsion of visceral mass at some stage of development to allow for easy movement

Specimen F (earthworm)

- Streamlined body allows easy passage through the borrows.
- Has chaetae used for locomotion and hold two worms together during copulation.
- Has clitellum which helps in copulation and also secretes the cocoon in which eggs develop
- Presence of a terminally located mouth for feeding and anus for sending out of undigested materials.
- Moist glandular body for gaseous exchange and smooth movement in the soil.

b. Draw a well labeled diagram of specimen G (tilapia) and annotate clearly 3 features concerned with movement. (12 mks)



Structure of tilapia

3. (a) Draw and annotate the head region of the bird provided to show how this animal is adapted to life in its environment. (5 mks)

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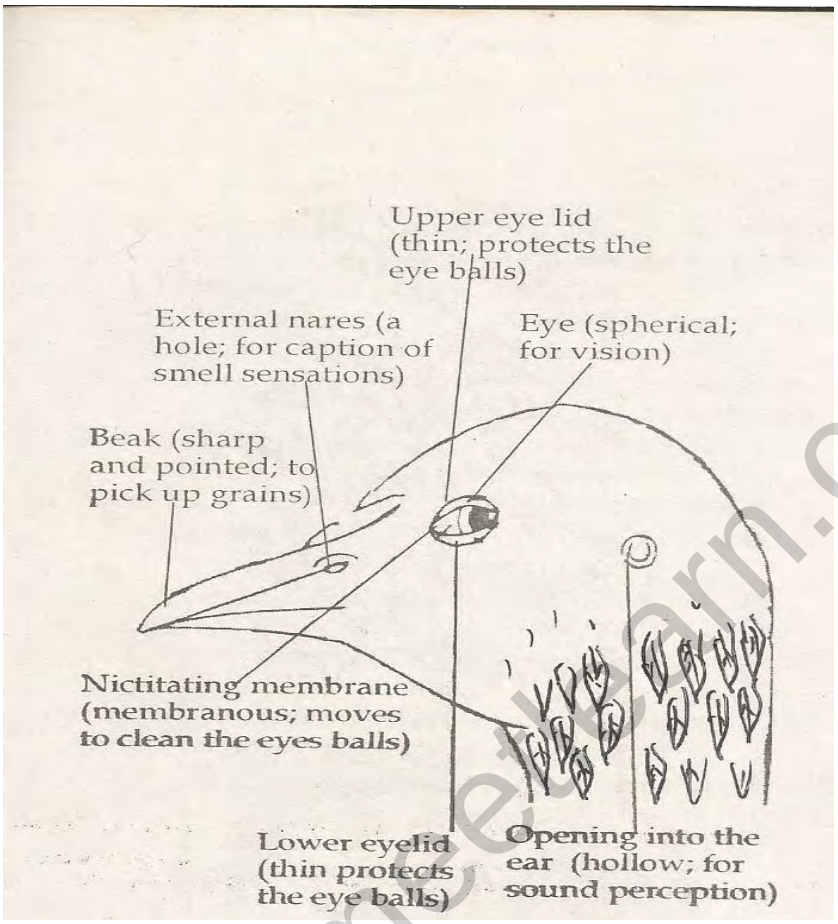

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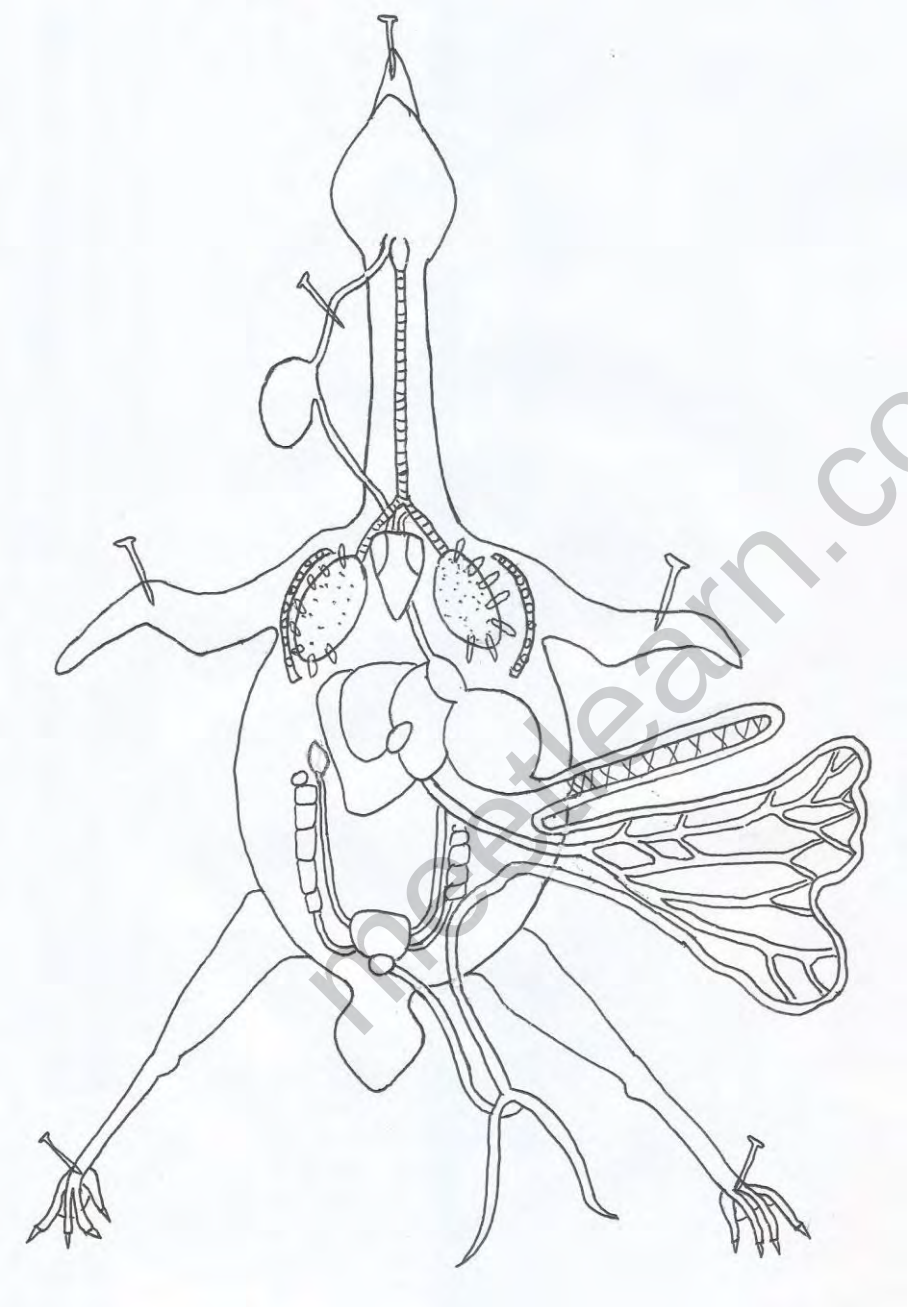
Structure of the head region of a fowl

(b) (i) Dissect into the perivisceral cavity of the bird provided to display organs concerned with respiration, circulation, digestion, excretion and reproduction. (15 mks)

(ii) Annotate five structures concerned with Rep. (5 mks)



Structure of respiratory, circulatory, digestive, excretory and reproductive systems of a fowl

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4. Use four leaves from the branch labeled F to investigate the rate of water loss from the leaves. Detach the leaves and weigh them carefully, and then weigh them for a further three times, allowing 10 minutes between each weighing for 40 minutes. Suspend the leaves on the line provided between the weighings.

(a) Tabulate and plot a graph of your result. (9 mks)

(b) Using the final set of data, calculate the overall percentage change in mass. (2 mks)l

(c) Comment critically on the validity of your method and on the information it gives regarding the rate of water loss from the leaves.

(a)

	X0	X1	X2	X3	X4
Time (minute)	0	10	20	30	40
Weight loss (g)	20	16	14	12	11
Amount of water loss (g)	0	4	6	8	9

Graph

(b) calculation of the percentage change in mass of leaves from the beginning to the end of the experiment.

Initial weight (Xo) = 20g

Final weight (X 4) = 11g

$$\% \text{ change in mass} = \frac{Xo-X4}{X0} \times \frac{100}{1} = \frac{20-11}{20} \times \frac{100}{1} = 45\%$$

(c)

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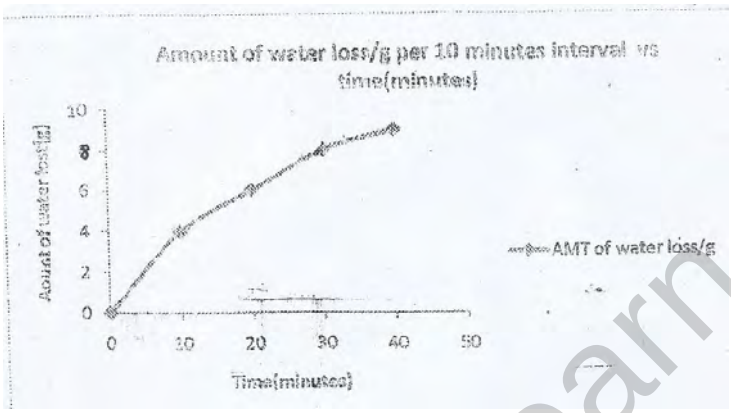

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• Four leaves were detached from the branch F provided . • They were immediately tied apart on a thread and weighed, and the mass noted. • The leaves were tied on a line beside the window. • The leaves were weighed after every 10 minutes for 40 minutes, and the masses noted. • The temperature of the environment was noted for every 10 minutes.	• There was a great drop of mass. • The greatest water loss occurred in the first 10 minutes. • As time was passing, the rate of water loss was decreasing.	• The leaves were fresh and contained water. • Stomata were opened and the leaves lost water by transpiration. • Transpiration rate was fast in the first 10 minutes. • Light, high temperature and wind increased the rate of transpiration. • At the 40th minutes, the rate of transpiration decreased because the leaves were stressed and the stomata closed.

Short comings:

- The balance was not very sensitive.
- There humidity affected the result.
- The leaves were not kept under water before the start of experiment.
- There were poor measurements.

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North West regional mock 2013 group 1

Specimens:

A: domestic fowl/chicken **F: crayfish**

B: fern. **G: tilapia**

C: Spirogyra **H: hibiscus**

D: star fish **I: Chordata**

E: earthworm **J: octopus**

1. (a) For each of the specimens (A to J) provided, state:

i. The major group (phylum) to which it belongs,
ii. The Sub-group (class) to which it belongs,
iii. The scientific or common name

Your answer should be written in your answer book in columns as shown below:
(15 mks)

Specimen	**Phylum**	**Class**	**Scientific or common name**
A	Chordata	Aves / Avia	Domestic fowl or Gallus domestica
B	Filicinophyta	Filicinae	Fern Dryopteris felix-mas
C	Chlorophyta	Chlorophyceae	Spirogyra or Spirogyra jogensis
D	Echinodermata	Stelleroidea	Star fish or Asteria rubens
E	Annelida	Oligochaeta	Earthworm or Lumbricus terrestris
F	Arthropoda	Malacostraca	Crayfish or Cambarus affinis
J	Chordata	Chondrichthyes	Dogfish Scyliorhinus caniculus
H	Angiospermophyta	Dicotyledoneae	Hibiscus or Hibiscus rosa-sinensis
I	Chordata	Osteichthyes	Tilapia or Tilapia zilli
J	Mollusca	Cephalopoda	Octopus or Octopus vulgaris

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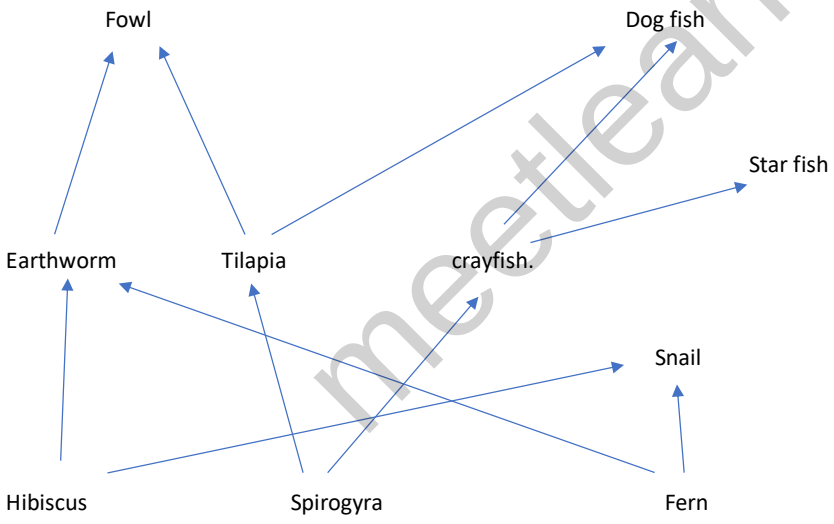
(b) Give one diagnostic features each of the members of the class of specimens A, B, F and H. (4 mks)

Specimen	Diagnostic feature
A (domestic fowl / chicken)	• presence of feathers • presence of beak/ beck • fore limbs modified to form wings • presence of gizzard for mechanical digestion
B (fern)	• presence of sori under the frond
F (crayfish)	• presence of cephalothorax • presence of carapace • has stalked compound eyes
H (hibiscus)	• Presence of flowers for sexual reproduction. • Tap root system • Net venation on leaves

(c) Construct a food web to show how specimens A to I are related. (4 mks)



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(d) Draw specimens B (fern) and E (earthworm) and annotate structures concerned with reproduction. (7 mks)

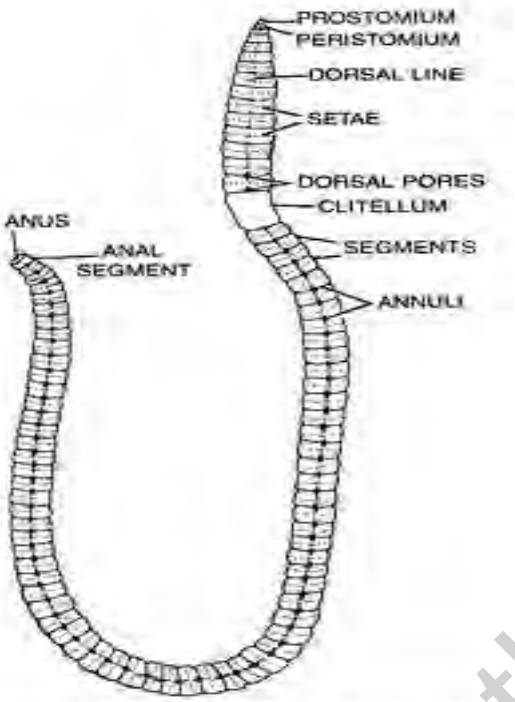

Fig. 7A.1. Earthworm in dorsal view.

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2. (a) (i) Open the left wing of the bird provided and draw a large labelled diagram of the anterior region of the bird together with the opened wing. Annotate the four types of feathers. (11 mks)

Anterior region of a fowl with an opened wing

(b) Bring out five structural differences between specimens G and I (5 mks)

Specimen G (dogfish)	**Specimen I (tilapia)**
Ventrally located mouth	Terminally located mouth
Heterocercal tail	Homocercal tail

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Placoid scales	Cycloid scales
No operculum	Presence of operculum
Flashy fins	Fins supported by bony rays
Five gill slits	Four gill slits
External gill slits present	No external gill slit

(c) Give the role of the different fins in movement. (2 mks)

- The dorsal and ventral fins reduce yawing (prevent rolling)
- The pelvic and pectoral fins function for steering, balance, change of direction, control pitching and serve as brakes.
- The caudal or tail fin provides a powerful movement (thrust) , control direction of movement and keeps the upright.

(d) Give the habitats of specimens C, E, G and J. (2 mks)

Specimen	**Habitat**
C (Spirogyra)	On slow flowing streams / banks of streams / on ponds
E (earthworm)	In moist humus soil / decaying organic matter
G (dogfish)	In sea / ocean (marine waters)
J (toad)	Banks of streams / swampy areas / under old damp blocks

3. (a) Draw the leg of the bird and annotate the parts. (6 mks)

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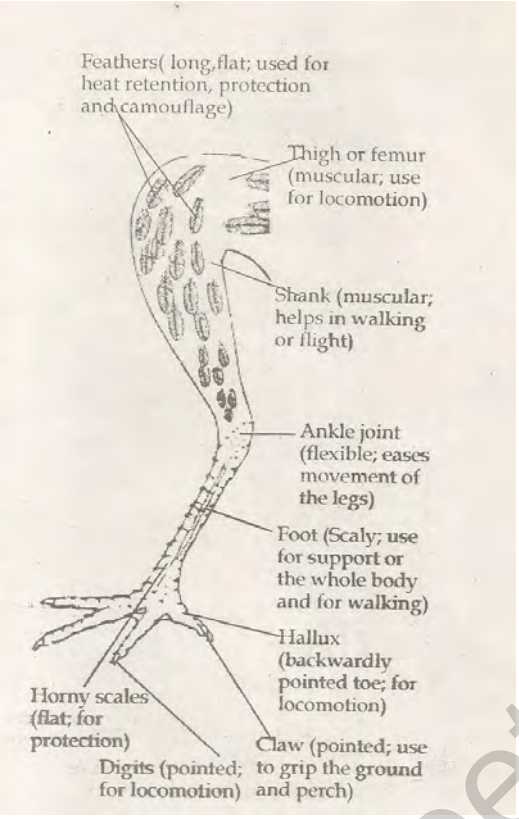

Structure of the leg of a fowl

Dissect the bird provided and display the digestive system. Make a large labelled drawing of your dissection. Annotate any four parts concerned with digestion. (16 mks)

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(b) Bring out 3 differences between the digestive system of bird and that of guinea pig. (3mks)

Digestive system of bird	Digestive system of Guinea pig
Gizzard present	Gizzard absent
Proventriculus present	Proventriculus absent
Diverticulum present	Diverticulum absent
Beak present	Beak absent
Crop present	Crop absent
Caecum absent	Caecum present
Teeth absent	Teeth present
Appendix absent	Appendix present
Rectum absent	Rectum present

4. You are provided with tradescantia leaves and solutions P and Q and also distilled water. Investigate the effect of the three different solutions on the epidermal cells of the tradescantia. Report your results are fully as possible, give the procedure, observations and the inferences. Draw the cells as seen under the electron microscope. (25 mks)

Answer.

Aim: To investigate the effect of distilled water and epidermal cells of tradescantia

Procedure	Observation	Inferences
• A small strip was peeled off from the lower epidermis of the tradescantia leave. • The strip wad placed on a clean virgin slide • 5 drops of solution distilled water was put on the strip and covered with a cover slid. • It wad allowed to stand for 15 minutes. • The prepared slide was then put on the microscope stage and viewed under low power (X 40)	• The cells bulged or larger (turgid) • The guard cells were more curved • The stomata were widely open	• Water entered the cells by osmosis. • The cells became turgid. • Uneven expansion or the outer and inner walls of guard cells caused the guard cells to curve and the stomata to open widely. • The distilled water is hypotonic while the cell sap is hypertonic.

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*** Cells in solution P (saturated salt solution)**

Procedure	Observation	Inferences
• A small strip was peeled off from the lower epidermis of the tradescantia leave. • The strip wad placed on a clean virgin slide • 5 drops of solution P was put on the strip and covered with a cover slid. • It wad allowed to stand for 15 minutes. • The prepared slide was then put on the microscope stage and viewed under low power (X 40)	• The cells became smaller (flaccid) • The cells looked weak. • The guard cells became less curved. • The stomata were seen closed.	• Water leaves the cells by osmosis. • The cells became flaccid. • The uneven expansion of the outer and inner walls of the guard cells led to closing of stomatal pores. • Solution P is a hypertonic solution e.g. saturated salt solution, while the cell sap is hypotonic.

*** Cells in solution Q (0.1M NaOH)**

Procedure	**Observation**	**Inferences**
• A small strip was peeled off from the lower epidermis of the tradescantia leave. • The strip wad placed on a clean virgin slide • 5 drops of solution Q was put on the strip and covered with a cover slid. • It wad allowed to stand for 15 minutes. • The prepared slide was then put on the microscope stage and viewed under low power (X 40)	• The cells became smaller (flaccid) • The cells looked disrupted. • The chloroplasts in the guard cells were destroyed and chlorophyll leaked out, staining all the preparation. • The guard cells and other neighbouring cells were seen wavy in nature.	• The solution Q is a very corrosive solution. E.g. concentrated NaOH. • The cell walls were disrupted and destroyed. • Chloroplasts membranes and cell surface membranes were destroyed. • Osmosis did not take place because there was no selectively permeable membrane, since the cell membranes were destroyed.

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JUNE 2014 GROUP 3.

Specimens

A: Wood louse. **F: cypress**

B: Hydra. **G: domestic fowl**

C: Hibiscus. **H: selaginella**

D: Earthworm. **I: dog fish**

E: cockroach. **J: Guinea pig**

1. (a) For each of the specimens (A to J) provided, state:

(i) the major group (phylum) to which it belongs;

(ii) the sub-group (class) to which it belongs;

(iii) the scientific or common name.

You answer to this question should be written in your answer book in columns as shown below. (15 mks)

Specimen	Phylum	Class	Scientific or common name
A	Arthropoda	Malacostraca	Wood louse or *Oniscus asselus*
B	Cnideria	Hydrozoa	Hydra or *Hydra fusca*
C	Angiospermophyta	Dicotyledoneae	Hibiscus or *Hibiscus rosa-sinensis*
D	Annelida	Oligochaeta	Earthworm or *Lumbricus terrestris*
E	Arthropoda	Insecta	Cockroach or *Periplaneta americana*
F	Coniferophyta	Coniferinae	Cypress or *Cupressus arizonica*
G	Chordata	Aves	Chicken or *Gallus domestica*
H	Lycopodophyta	Lycopodinae or Lycopsida	Selaginella or Spike moss or *Selaginella densa*
I	Chordata	Chondrichthyes	Dog fish or *Squalus acanthias*

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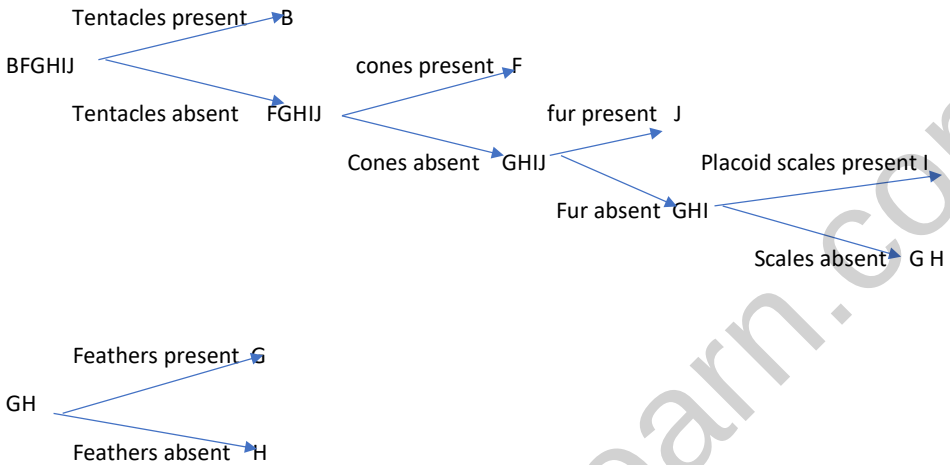
J	Chordata	Mammalia	Guinea pig or *Cavia porcellus*

(b) Construct a dichotomous key to separate specimens B, F, G, H, I and J, using only visible diagnostic features. (5 mks)



(c) (i) make a well labelled drawing of the half flower of specimen C. (6 mks)

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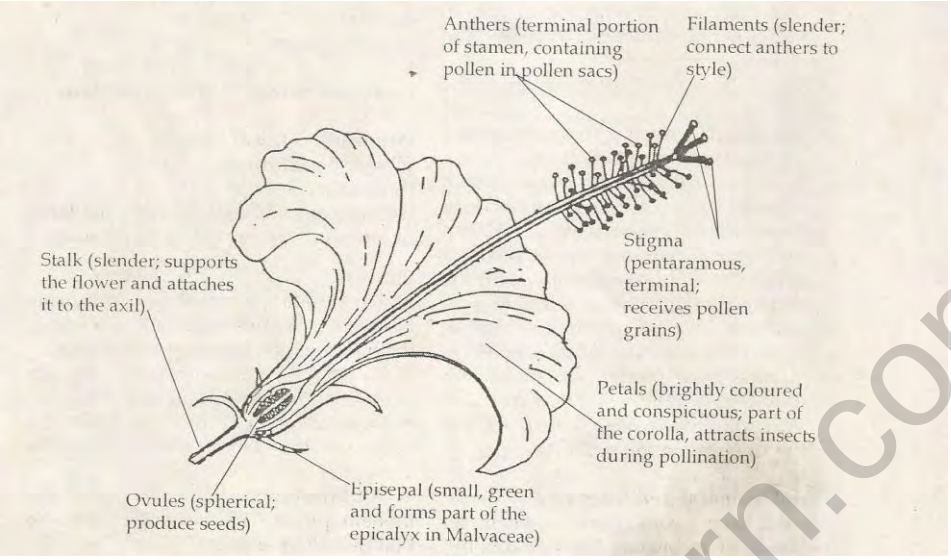

(ii) How is it adapted to its mode of pollination? (4 mks)

- It has brightly coloured petals to attract insects and birds
- Petals produces a good scent to attract pollinators like insects and birds.
- It produces nectar to attract pollinators like insects and birds
- Stigma is broad on a firm style.
- Has five broad sticky stigma to trap pollen grains that adhere to it
- Pollen grains are sticky to adhere on stigma and on pollinators
- Stigma is at the tip to easily trap pollen grains
- Nectarines are found at the base of the petals to ensure that pollen is collected before insect reaches there
- Petals have nectar guide lines that direct proboscis for nectar collection.

2. (a) (i) Make a large labelled drawing of the vertical view of specimen A (wood louse) (9 mks)

(ii) Annotate your drawing to show structures concerned with sensory perception. (4 mks)

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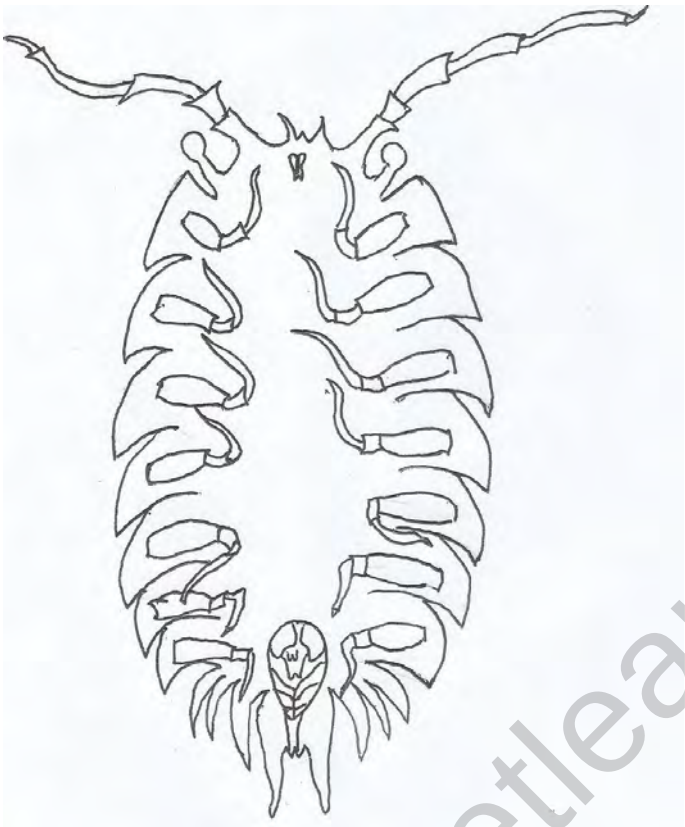

Structure of the ventral view of wood louse

(b) (i) State three structural similarities between specimens A (wood louse) and E (cockroach). (3 mks)

- Both have a pair of compound eyes
- Both have a pair of jointed antennae
- Both have chewing mouth parts (mandibles)
- Their bodies are covered by a chitinous exoskeleton
- Both have segmented bodies
- Both have spiracles for breathing
- Both are dorsoventrally flattened

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(ii) Tabulate four differences between specimens A and E. (4 mks)

Specimen A (wood louse)	Specimen B (cockroach)
Body divided into two, that is cephalothorax and abdomen	Body divided into three parts that is head, thorax and abdomen
Antennules present	Antennule absent
No wings	Two pairs of wings present
Has short antennae	Has long antennae
Has uropod and telson	Lacks uropod and telson
Lacks cerci and style	Cerci and style present
Many pairs of jointed legs	Three pairs of jointed legs

3. (a) Pin the fish provided on its back on the dissecting board or dish. Cut through the muscle wall into the perivisceral cavity to display the digestive and Urinogenital systems. Make a fully labelled drawing of your dissection. (19 mks)

(b) Make annotations on your drawing to indicate three structures concerned with reproduction and three concerned with excretion. (6 mks)

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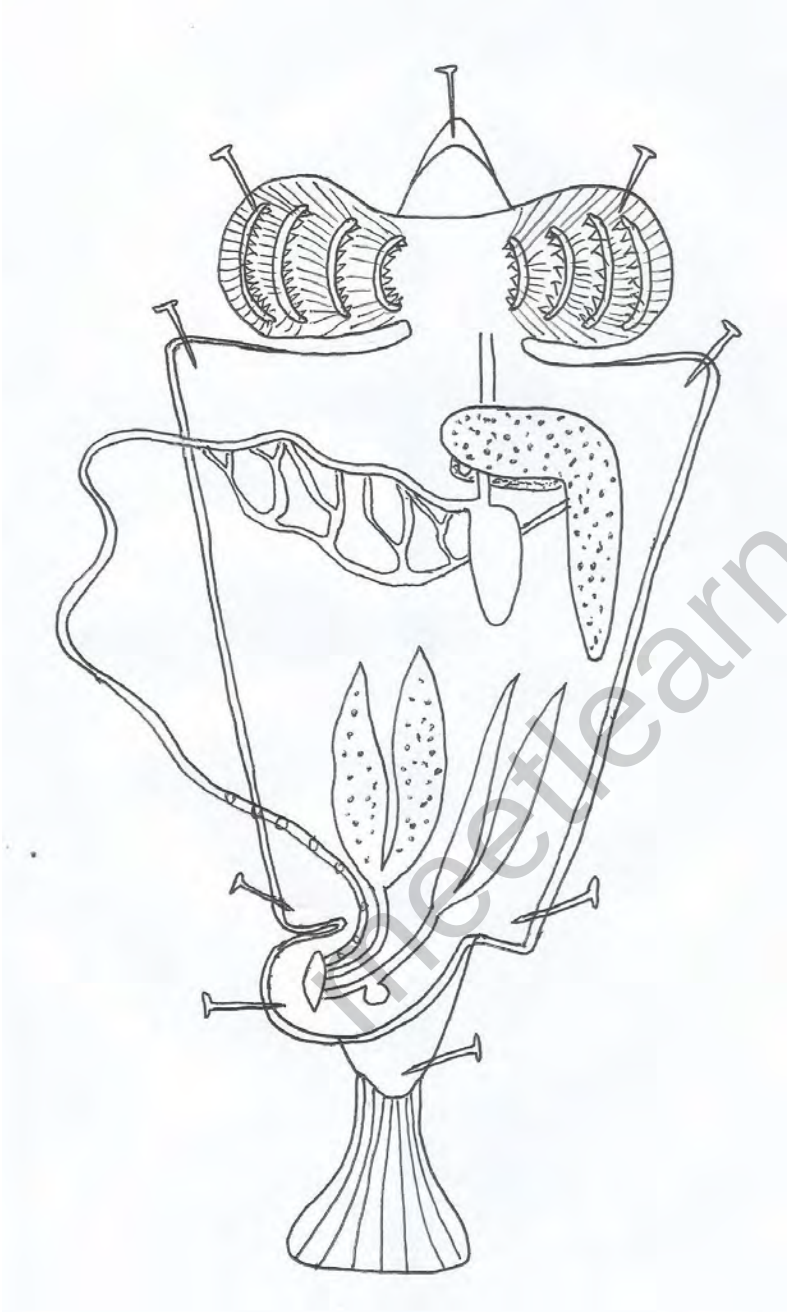

Structure of the digestive and female Urinogenital systems of a fish

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Structure of the digestive and male Urinogenital systems of a fowl

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4. (a) Catalase is an enzyme which liberates oxygen as shown by effervescence when hydrogen peroxide is added.

(i) Remove the gills from the fish you have dissected. Macerate in a mortar, and add some water to obtain a concentrated solution of the gill tissue. Prepare the following dilutions of your solution: 100%, 75%, 50% and 25%. Explain your method in each case. (7 mks)

ANSWER.

- The gills of the fish was removed and macerated using a mortar and a pistil
- 10ml of water was added to the ground tissue.
- The concentrated solution of the gill tissue was filtered using a funnel and a filter paper.
- All the solution was 100% concentration
- Other percentage dilution were prepared as follow
- 75% = 1.5ml of gill solution + 0.5ml of water
- 50% = 1ml of gill solution + 1ml of water
- 25% = 0.1ml of gill solution + 1.5ml of water.
- All the percentage dilutions were 2ml each

(ii) Investigate the effect of these solutions on hydrogen peroxide. Give a concise account of your procedure and explain your observations as fully as you can. (13 mks)

ANSWER

- The different % dilutions were put into separate test tubes
- The total volume of solution in each test tube was 2ml
- 2ml of hydrogen peroxide was added into each of the test tubes.
- The test tubes were allowed to a and for 2 minutes

Observation.

% concentration	100%	75%	50%	25%
Rate of fuzzing	++++	+++	++	+

Explanation of result

- Active tissues produce hydrogen peroxide as a by-product of tissue respiration
- Hydrogen peroxide is very toxic, and must be converted to non toxic products.
- Catalase is an enzyme produced by all living cells that helps to convert hydrogen peroxide to water and oxygen
- $H_2O_2 \xrightarrow{\text{Catalase}} 2H_2O + O_2$
- Fizzing indicates the release of oxygen gas

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- The more concentration the enzyme is in solution, the faster the rate of reaction, and the more the rate of fizzing. This explains why 100% solution produced more fizzing because it has more enzyme Catalase.
- The 25% solution gave the least fizzing because it has the least amount or concentration of enzyme Catalase.

(b) Suggest an experiment that could be carried out to confirm that your solution contains the enzyme. (5 mks)

ANSWER

NB: This question requires that the candidate design a control experiment. It requires heating of the solution or changing the pH.

- 2ml of the 100% concentration of gill tissue was transferred into a clean test tube using a syringe.
- The solution was heated for 5 minute in a hot water bathe
- 2ml of H_20_2 was added into the test tube.
- The test tube and content was allowed to stand in a test tube rag for 5 minutes.
- No fizzing was seen

  Explanation of result and Conclusion.
- Enzymes are protein, and are denatured at very high temperature
- The active sites of the enzyme were destroyed, hence no enzyme-substrate complex was formed.
- Hydrogen peroxide was not converted to water and oxygen, hence no fizzing was produced.

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BAMENDA MOCK 2014 GROUP 2

Specimens

A: Spirogyra. F: medicinal leech

B: Fern. G: crayfish

C: lizard. H: cockroach

D: butterfly. I: chicken

E: Lycopodium. J: morning glory

1. (a) for each of the specimens (A to J) provided, state;

(i) the major group (phylum) to which it belongs,
(ii) the sub-group (class) to which it belongs
(iii) the scientific or common name

You answer should be written in your answer book in columns as shown below. (15 mks)

Answer.

Specimen	(i)	(ii)	(iii)
A	Chlorophyta	Chlorophyceae	Spirogyra or Spirogyra jogensis
B	Filicinophyta	Filicinae	Fern or Dryopteris felix-mas
C	Chordata	Reptilia	Agama lizard or Agama agama
D	Arthropoda	Insecta	Butterfly or Pieris brassicae
E	Lycopodophyta	Lycopodinae	Lycopodium or Lycopodium clavatum
F	Annelida	Hirudinae	Medicinal leech Nereis diversicolor
G	Arthropoda	Malacostraca	Crayfish or Cambarus affinis
H	Arthropoda	Insecta	Cockroach or Periplaneta americana
I	Chordata	Aves or Avia	Chicken or Gallus domestica

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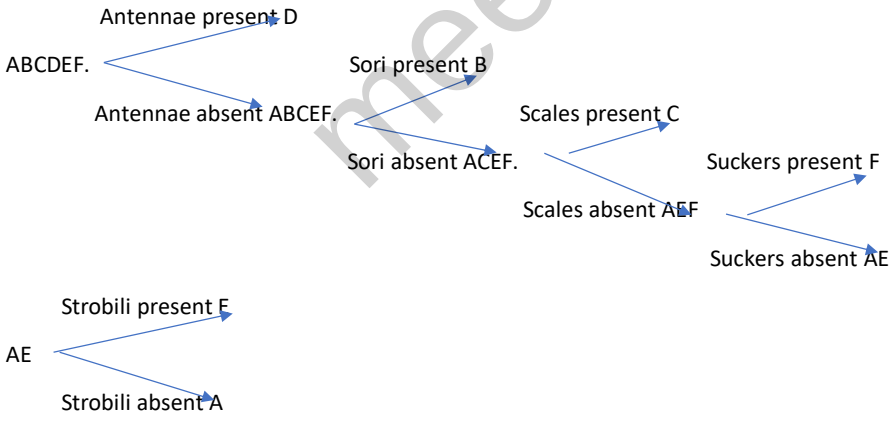
J	Angiospermophyta	Dicotyledoneae	Morning glory or Ipomea purpurea

(b) (i) For each of the specimens B, C, D and E, state one diagnostic feature pertaining to its sub-group (class). (4 mks)

<u>Answer</u>

Specimen	Diagnostic feature
B	▪ Presence of sori ▪ Leaves form fronds
C	▪ Body covered with waterproof scale ▪ Forked tactile sticky tongue ▪ Five clawed digits on limbs
D	▪ A pair of compound eyes ▪ 3 pairs of jointed legs ▪ Two pairs of wings ▪ A pair of antennae ▪ Body divided into head, thorax and abdomen
E	▪ Sporangia in strobili ▪ Microphyllous and homosporous

(ii) Using only external visible features, excluding size and colour, construct a dichotomous key to separate specimens A-F. (5 mks)



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(iii) How are specimens F and I adapted to their mode of lives. (6 mks)

Answer

Adaptations of specimen F (medicinal leech)

- Muscular pharynx pumps body fluids of the prey into its system
- Anterior small sucker for attaching to prey
- Mouth with 3 horny chitinous jaws for cutting flesh of host
- Large posterior sucker for looping movement

Adaptations of specimen I (domestic fowl)

- Short strong and horny beak (bill) for feeding
- Feathers for warmness and protection
- Forelimbs modified to form wings for flight
- Large flight pectoral muscles that help in flight
- Strong hollow light bipedal limbs for movement (walking and running)

2. (a) Make a large labelled drawing of specimen H (cockroach) . On your drawing, annotate one feature concerned with

(i) Nutrition
(ii) Movement. (10 mks)

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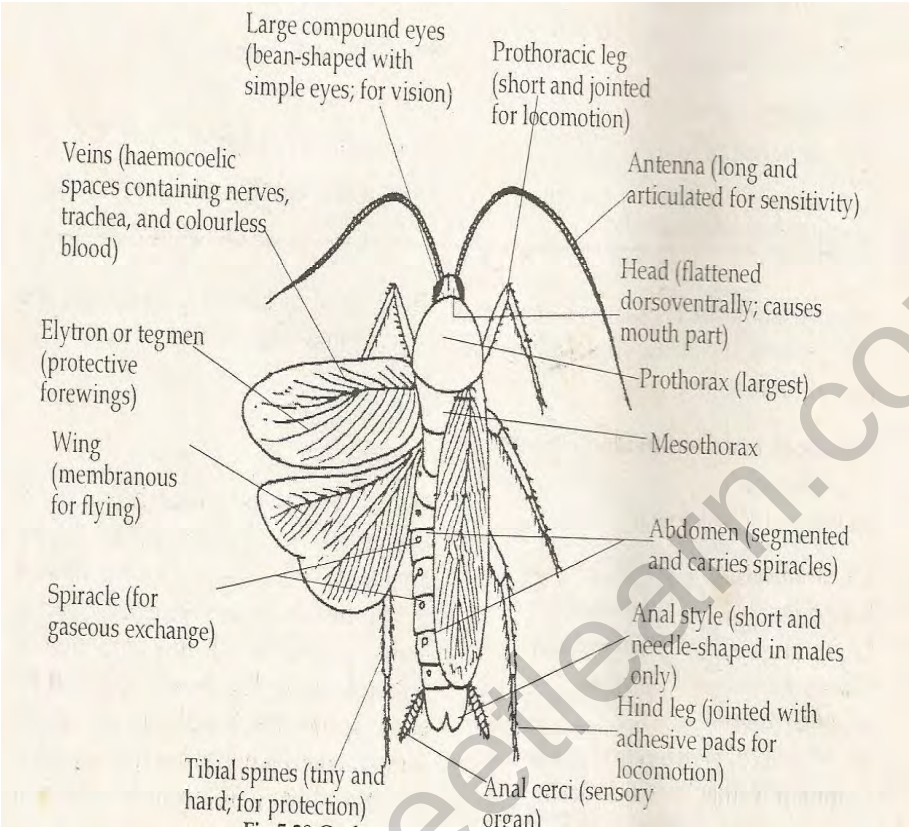
Answer



Structure of a cockroach

(b) List two similarities and three differences between specimens E and J. (5 mks)

Answer

Similarities between specimens E (Lycopodium) and J (morning glory)

- Both have a conspicuous sporophyte
- Both have true roots, leaves and stems
- Both contain the vascular bundles

Differences between specimens E and J

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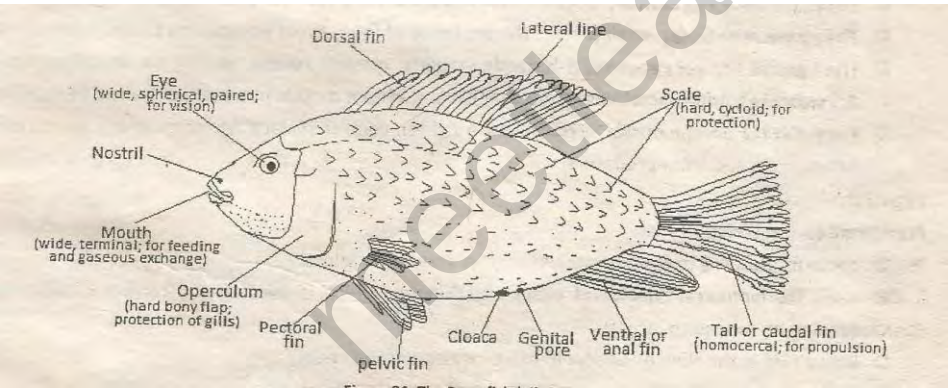
Lycopodium	Morning glory
Strobili for reproduction	Flower for reproduction
Rhizophore present	No Rhizophore
Two types of leaves	One type of leaf

(c) Identify from specimens A to J which of them are producer, primary consumer and decomposer. (5 mks)

Answer

Producer	Primary consumer	Decomposer
▪ Spirogyra ▪ Fern ▪ Lycopodium ▪ Morning glory	▪ Butterfly	▪ Cockroach

3. (a) Make a labeled diagram of the lateral view of the bony fish provided. Annotate four features that adapt the organism to survive in its habitat. (9 mks)



Structure of the lateral view of a tilapia

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(b) (i) Dissect the fish to expose clearly the respiratory and digestive systems. Make a fully labelled drawing of your dissection. (12 mks)

(ii) Annotate two structures concerned with respiration (4 mks)

Leave your dissection for grading at the end of the examination.

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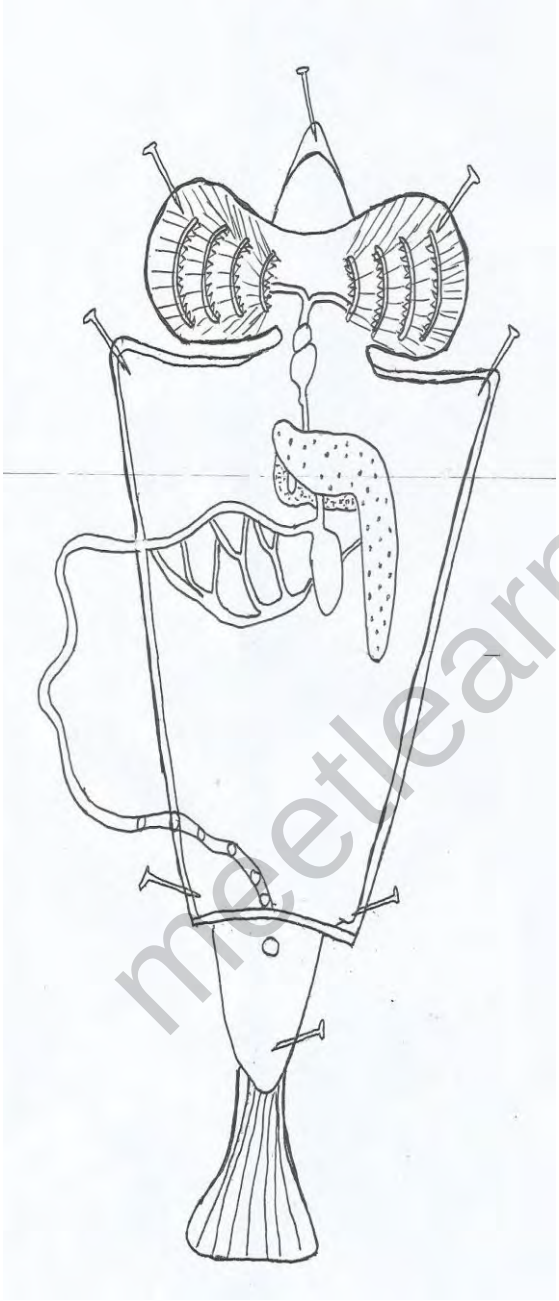

Respiratory and digestive systems of a fish

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4. Many tissues contain the enzyme Catalase, that catalyzes the breakdown of hydrogen peroxide to oxygen and water. The release of oxygen produces a slight fizzing

(a) Remove a sample of

(i) blood (ii) liver and (iii) muscle from your dissected specimen. Using only the materials provided, test each sample for the presence of Catalase. Record your procedure, results and conclusions. (13 mks)

Answer

Procedure

- Samples of blood, liver and muscle were removed from the dissected organisms, and were crushed separately using a mortar and a pistil
- 4ml of water was added it each tissue to form a paste
- The crushed samples were filtered to obtain enzyme pools
- Using a syringe, 2ml of each tissue solution was transferred into separate test tubes.
- 2ml of hydrogen peroxide was added into each of the test tubes
- The test tubes were allowed to stand for 3 minutes.

Results or observations

Sample	**Effervescence or length of foaming**
Blood	+++++ or 9cm
Liver	++++++++ or 14cm
Muscle	+++ or 5cm

Explanation of result and conclusions

- Catalase is present in all living tissues.
- The most metabolically active tissues is the liver, that is why it produces the highest foaming
- The less metabolically active tissues is the muscle, that is why is produces the least foaming
- Catalase converts the toxic hydrogen peroxide to harmless products water and oxygen.

- $2H_2O_2 \xrightarrow{\text{Catalase}} 2H_2O + O_2$
- Fizzing indicates the release of oxygen gas.

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(b) Test the content of beaker P for the presence of starch, reducing sugars and protein. Record your procedure, results and conclusions. (12 mks)

Answer

NB. Solution P is milk solution.

Food type	**Procedure**	**Results, discussion and conclusions**
Starch	• 2ml of solution P was put in a test tube using a syringe • 3 drops of iodine solution were added, while shaking after each drop	• No blue black colour was seen, indicating the absence of starch
Reducing sugars	• 2ml of solution P was put in a clean test tube using a syringe • 2ml of Benedict's solution was added and the test tube shakened •The test tube was heated for 5 minutes, while shaking at intervals	• The blue colour of the solution changed to green, yellow, orange and to brick red. This indicates the presence of reducing sugar
Protein	• 2ml of solution P was put in a clean test tube. • 1ml of dilute NaOH solution was added. • 3drops of 5% $CuSO_4$ solution was added, while shaking after each drop.	• A purple colour was seen, indicating the presence of protein.

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June 2015 group 2

Specimens:

A: bread mould. **E: rag worm / Nereis / sand worm**

B: moss. **F: Agama lizard**

C: fern. **G: tadpole**

D: honeybee. **H: spider**

1. (a) For each of the specimens (A to H) provided, state:

(i) the major group (phylum) to which it belongs;
(ii) the sub-group (class) to which it belongs;
(iii) the scientific or common name

Your answer should be written in your book in columns as shown below. (12 mks)

Specimen	**Phylum**	**Class**	**Scientific or common name**
A	Zygomycota	Zygomycetes	Bread mould or Rhizopus nigricans
B	Bryophyta	Musci	Moss or Polytrichum commune
C	Filicinophyta	Filicinae	Fern or Dryopteris felix-mas
D	Arthropoda	Insecta	Honeybee Apis mellifera
E	Annelida	Polychaeta	Nereis/ ragworm or Nereis diversicolor
F	Chordata	Reptilia	Agama lizard or Agama agama
G	Chordata	Amphibia	Tadpole or Bufo regularis
H	Arthropoda	Spider	Spider or Araneus diadematus

(b) (i) Give four similarities and five differences between specimens D and H. (9 mks)

Similarities between specimens D (honeybee) and H (spider)

- Both have jointed legs / appendages.
- Their bodies are covered by chitinous exoskeleton.
- Both have spiracles for breathing.

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- Both have abdomen.
- Both have hairs on the body.

Differences between specimens D (honeybee) and H (spider)

Specimen D (honeybee)	**Specimen H (spider)**
Three pairs of jointed legs	Four pairs of jointed legs
Body divided into head, thorax and abdomen	Body divided into prosoma and opisthosoma (abdomen)
Sucking mouth parts	Prehensile Pedipalps
Wings for flight	No wings present
Compound eyes	Simple eyes
Spiracles for breathing	Spiracles and book lungs for breathing
Antennae present	No antennae
No spinnerets	Spinnerets present

(ii) How is specimen D adapted to collect pollen grains? (9 mks)

- Fore legs with comb for cleaning of pollen from the head and wings of the bee.
- Mid legs with prong for digging pollen out from pollen basket.
- Hind legs with pollen basket to carry pollen.
- Hairy body for collecting pollen.
- Short tongue for collecting nectar.

2. (a) Draw the anterior region of specimen E (Nereis) and annotate the parts concerned with

(i) gaseous exchange and

(ii) irritability. (10 mks)

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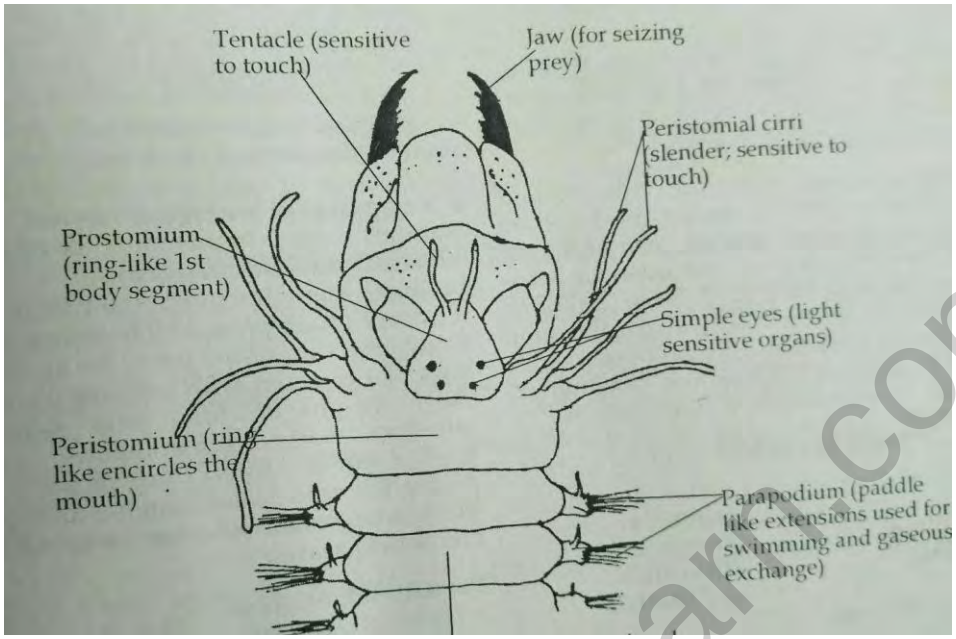

Head region of Nereis

(b) make a large labelled drawing of specimen B (moss plant) and annotate the parts concerned with nutrition and reproduction. (10 mks)

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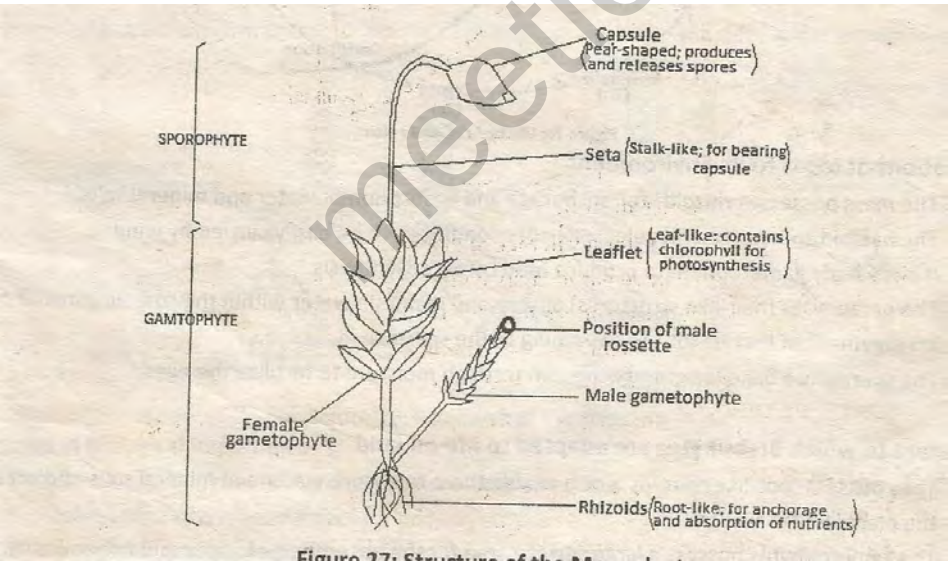

Figure 27: Structure of the M

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3. (a) Dissect the mammal provided and display the Urinogenital system. Make large labelled drawing of your dissection. (13 mks)

(b) Annotate three parts concerned with

(i) excretion. (ii) reproduction. (12 mks)

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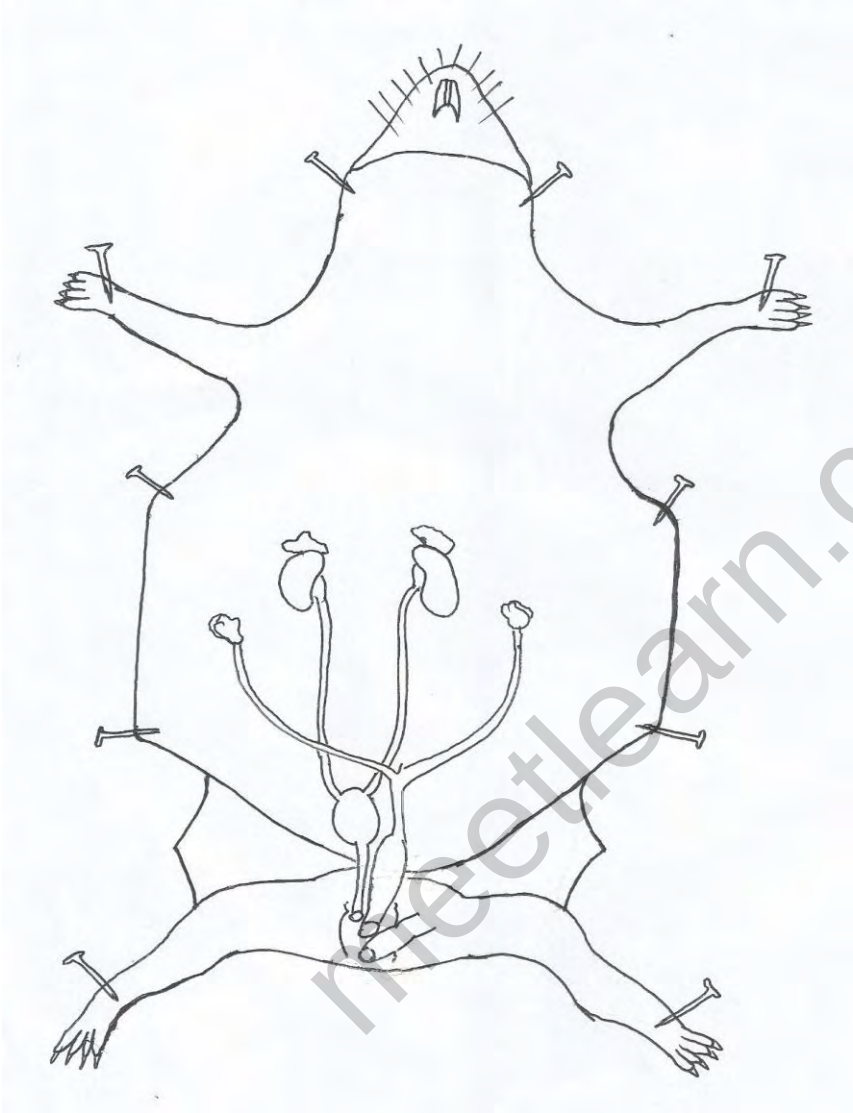

Female Urinogenital system of a guinea pig

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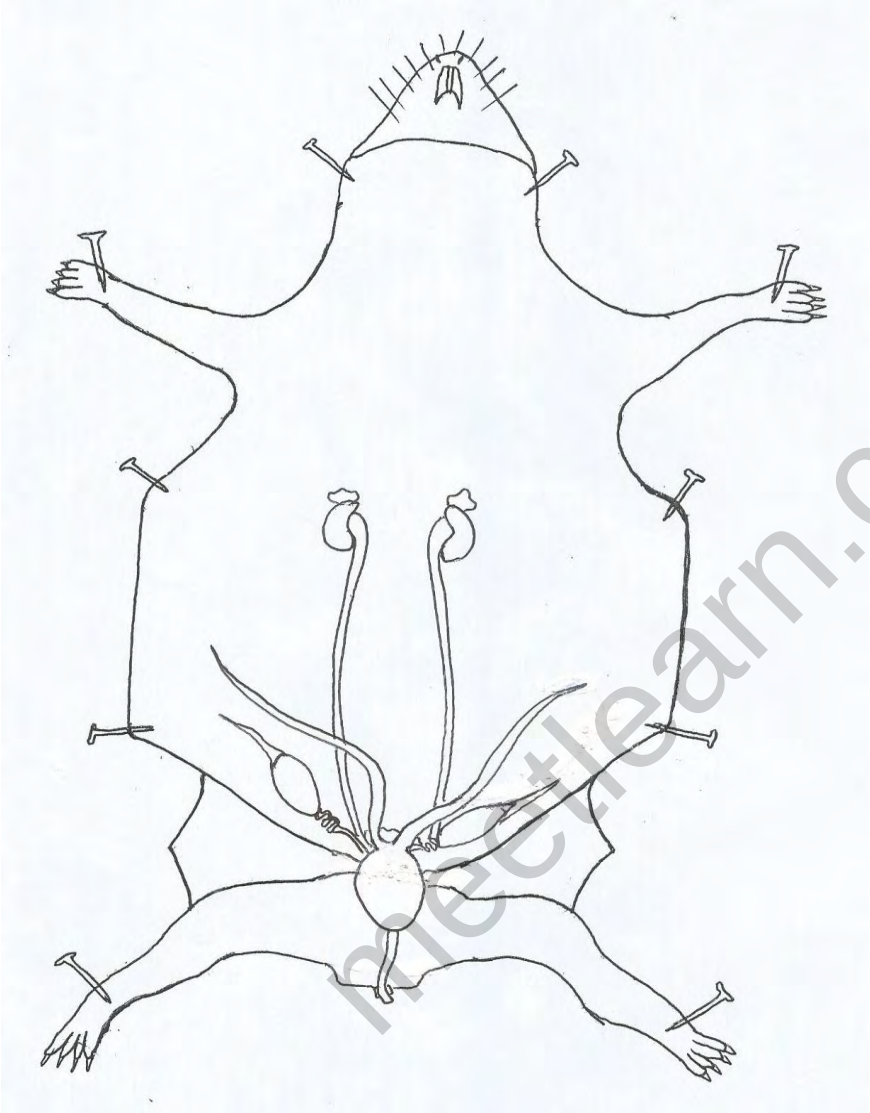

Male Urinogenital system of a guinea pig

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4. (a) Remove the liver, pancreas and rectum and macerate them separately to form three separate solutions. Use this solutions to show the action of the enzyme Catalase in the different organs with hydrogen peroxide. Report your results fully . represent your result on a bar chart. (20 mks)

Answer.

Title: Action of liver Catalase with hydrogen peroxide

Procedure	Observation	Inferences
• A small piece of liver was removed from the dissected animal • It was macerated using a mortar and a pestle. • 5ml of distilled water was added. • The solution was filtered using a filter paper and a funnel. 2ml of the enzyme pool was transferred into a clean test tube using a syringe. • 2ml of hydrogen peroxide was added to the test tube. • The test tube was allowed to stand for 3 minutes.	• Foaming or fizzing was seen. • the length of foam was 12cm	• liver contains the enzyme Catalase. • The Catalase breaks down toxic hydrogen peroxide to harmless products water and oxygen.

Title: Action of pancreas Catalase with hydrogen peroxide

Procedure	Observation	Inferences
• A small piece of pancreas was removed from the dissected animal • It was macerated using a mortar and a pestle. • 5ml of distilled water was added. • The solution was filtered using a filter paper and a funnel. 2ml of the enzyme pool was transferred into a clean test tube using a syringe.	• Foaming or fizzing was seen. • the length of foam was 8cm	• Pancreas contains the enzyme Catalase. • The Catalase breaks down toxic hydrogen peroxide to harmless products water and oxygen.

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• 2ml of hydrogen peroxide was added to the test tube. • The test tube was allowed to stand for 3 minutes.		

Title: Action of rectum Catalase with hydrogen peroxide

Procedure	**Observation**	**Inferences**
• A small piece of rectum was removed from the dissected animal • It was macerated using a mortar and a pestle. • 5ml of distilled water was added. • The solution was filtered using a filter paper and a funnel. 2ml of the enzyme pool was transferred into a clean test tube using a syringe. • 2ml of hydrogen peroxide was added to the test tube. • The test tube was allowed to stand for 3 minutes.	• Foaming or fizzing was seen. • the length of foam was 5cm	• Rectum contains the enzyme Catalase. • The Catalase breaks down toxic hydrogen peroxide to harmless products water and oxygen.

Tissue type	Liver	Pancreas	Rectum
Length of foam (cm)	12	8	5

Bar chart.

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Conclusion:

- The enzyme Catalase is found in all living tissues.
- Catalase is an enzyme that converts or breaks down toxic hydrogen peroxide produced as a metabolic waste to harmless products that is water and oxygen.
- The concentration of Catalase in a tissue depends on the metabolic rate of the tissue.
- Liver produced more foaming because it metabolically very active, followed by pancreas and then the least rectum.
- $2H_2O_2 \xrightarrow{\text{Catalase}} 2H_2O + O_2$

(b) Device another experiment to show that the liver truly contains the enzyme Catalase. (5 mks)

Title: Denaturation of the enzyme Catalase

Procedure	Observation	Inferences
• A small piece of rectum was removed from the dissected animal • It was macerated using a mortar and a pestle. • 5ml of distilled water was added. • The solution was filtered using a filter paper and a funnel. 2ml of the enzyme pool was transferred into a clean test tube using a syringe. • The test tube was heated for 5 minutes. • It was then cooled under running tap water. • 2ml of hydrogen peroxide was added to the test tube. • The test tube was allowed to stand for 3 minutes.	• No foaming or fizzing was seen.	• The heat denatured the enzyme Catalase. • Active sites were destroyed and substrate (hydrogen peroxide) could not fit in to form enzyme-substrate complex. • No products were formed.

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June 2015 group 3

Specimens:

A: Obelia. **E: selaginella.**

B: cypress. **F: ragworm**

C: grasshopper. **G: tilapia**

D: spider **H: mushroom**

1. (a) For each of the specimens (A to H) provided, state:

(i) the major group (phylum) to which it belongs;
(ii) the sub-group (class) to which it belongs;
(iii) the scientific or common name

Your answer should be written in your book in columns as shown below. (12 mks)

Specimen	Phylum	Class	Scientific or common name
A	Cnideria	Hydrozoa	Obelia or Obelia geniculata
B	Coniferophyta	Coniferae/Coniferinae	Cypress or Cupressus arizonica
C	Arthropoda	Insecta	Grasshopper or Dissosteira carolina
D	Arthropoda	Arachnida	Spider or Areneus diadematus
E	Lycopodophyta	Lycopodinae	Selaginella
F	Annelida	Polychaeta	Ragworm or Nereis diversicolor
G	Chordata	Osteichthyes	Tilapia or Tilapia zilli
H	Basidiomycota	Basidiomycetes	Mushroom or Agaricus campestris

(b) State the specific habitat of each of specimens A to H. (4 mks)

Specimen	Specific habitat
A (Obelia)	Sea bed
B (cypress)	Sand uplands / damp humus soil
C (grasshopper)	Grass lawns / farm / bushes

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D (spider)	Walls of buildings / gardens
E (selaginella)	Moist humus soil / road sides
F (Nereis or ragworm or sand worm)	Sandy shores
G (tilapia)	Ponds / rivers / streams
H (mushroom)	Farm land / rotting logs of

(c) Select two visible features in each of the specimens A to H and use them to describe how they are adapted to their respective environments. (14 mks)

Specimen	**Adaptive features**
A (Obelia)	• chitinous exoskeleton for support and protection. • finger-like tentacles for feeding
B (cypress)	• needle-like leaves to reduce transpiration. • cones for spore production
C (grasshopper)	• barbed legs for protection. • compound eyes for vision. • exoskeleton for protection. Wings for flight. • antennae for feeling and smelling.
D (spider)	• spinnerets for web construction. •jointed legs for movement.
E (selaginella)	• Rhizophore for support. • Microphyllous leaves for food production.
F (Nereis)	• parapodia for movement in sand. • long tentacles for feeding.
G (tilapia)	• cycloid scales for protection. • homocercal tail for swimming/steering • dorsal fin for balance and stability
H (mushroom)	• cap-like pileus for spore production. • rhizomorph for support and feeding

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2. (a) Make a large labelled drawing of specimen A (Obelia) (8 mks)

On your drawing, annotate

(i) one part for feeding.

(ii) one part for reproduction.

(iii) one part for protection. (6 mks)

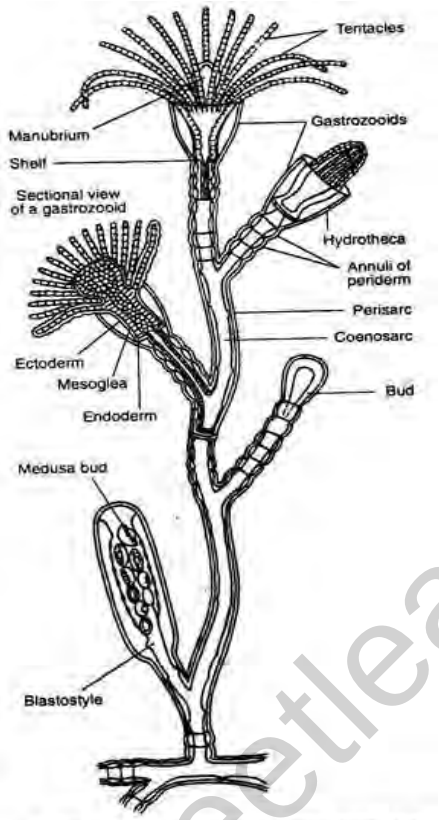

Fig. 12.14: Enlarged view of an *Obelia* colony. (after Parker and Haswell)

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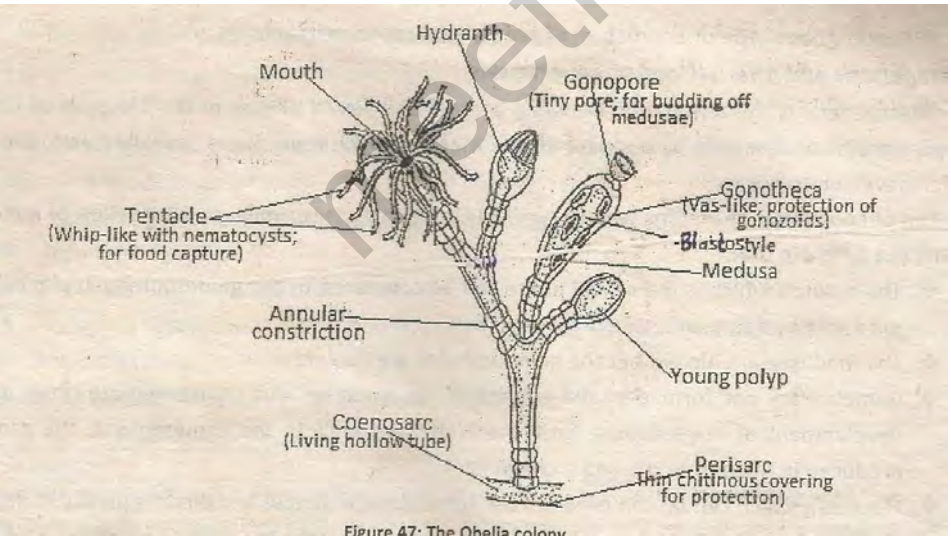

Figure 47: The Obelia colony

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(b) Specimens B and E belong to two different phyla. Using a table construct three prominent differences between these specimens. (6 mks)

Specimen E (selaginella)	**Specimen B (Cypress)**
Microphylous leaves	Scaly leaves
Herbaceous stem	Woody stem
Fibrous root system	Tap root system
Creeping stem	Erect stem
Does not produce seeds	Produce seeds

3. (a) pin the freshly killed fish on its back on a dissecting board. Carefully dissect the fish by cutting into the perivisceral cavity. Display the Urinogenital system, respiratory system and circulatory system. Make a large labelled drawing of your dissection. (13 mks)

(b) Annotate on your drawing

(i) One part each concerned with reproduction and excretion of nitrogenous waste products.

(ii) One visible blood vessel of the animal.

(iii) Two structures concerned with gaseous exchange. (12 mks)

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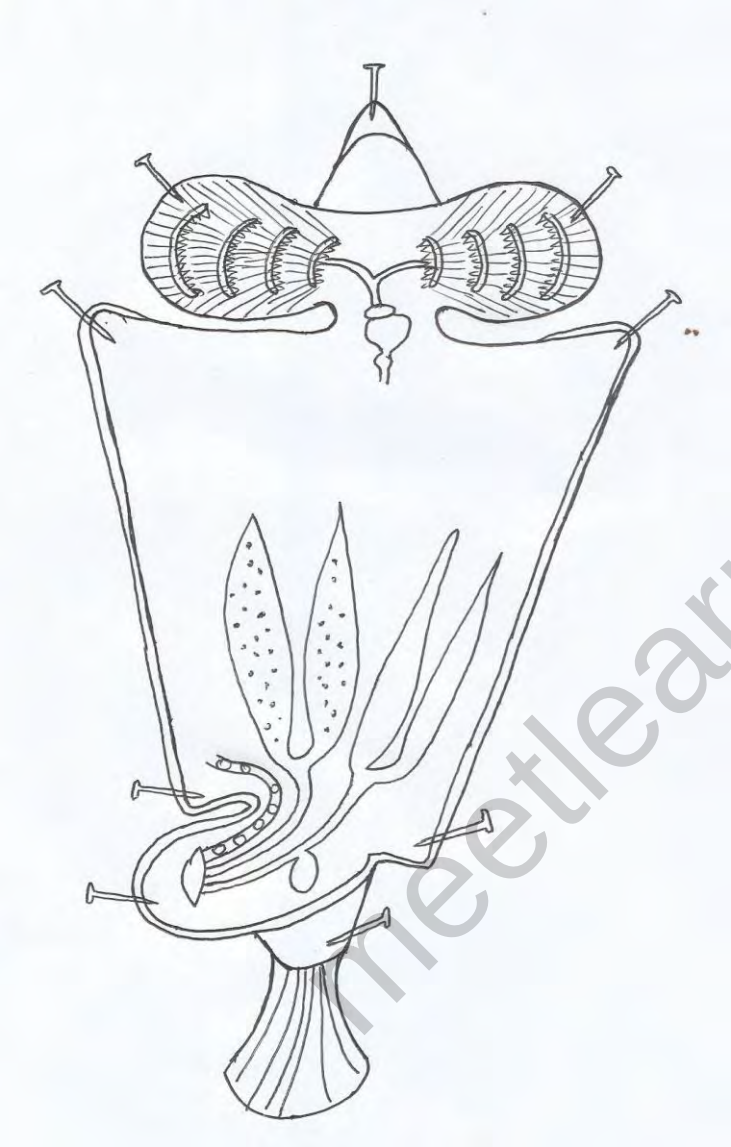

Structure of the respiratory, circulatory and female Urinogenital systems of a fish

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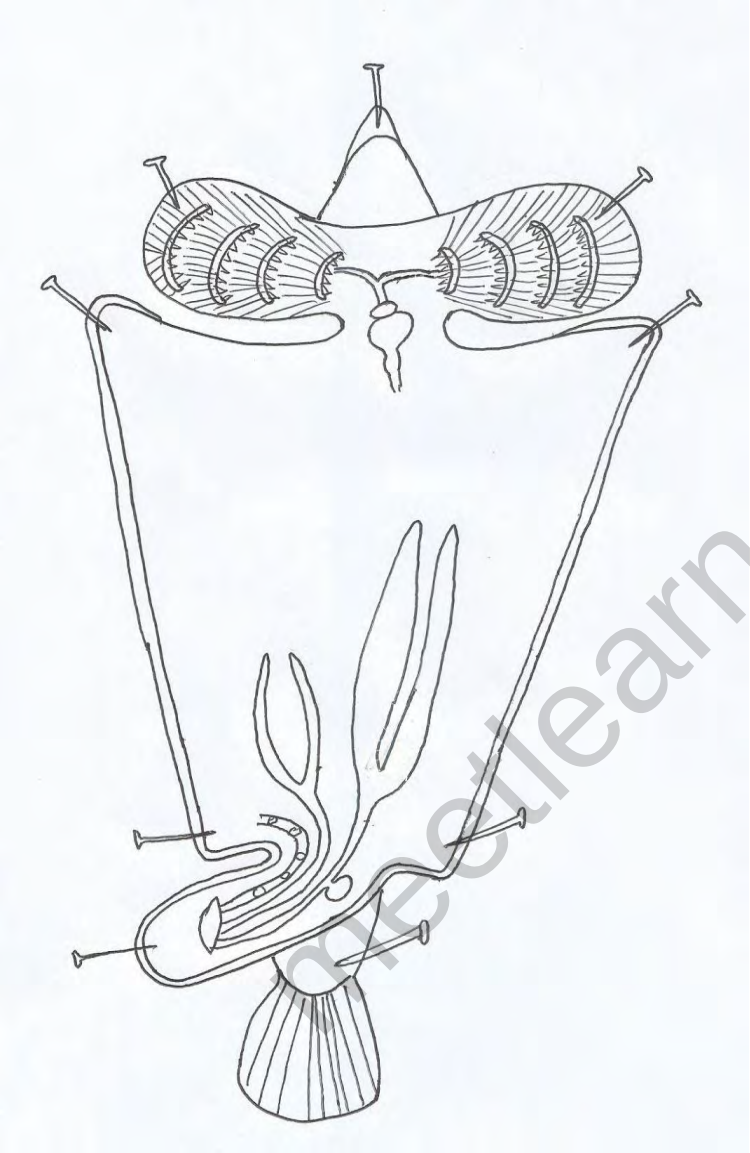

Structure of the replicatory, circulatory and male Urinogenital systems of a fish

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4. Make strips is 2mm and of maximum length as is possible with the tuber provided (when you choose a length, all the strips should be of that length). Note the textures of the strips carefully. Immerse the strips in the different solutions P, Q and R provided. After 30 minutes, remove the strips and blot them. Note their new lengths and textures.

Record your procedures and results and comment fully on your findings. (25 mks)

Answer

Procedure

- The potato tuber provided was washed and peeled using a knife.
-

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North west regional mock 2015 group 1

Specimens:

A: elephant grass. **F: liverwort**

B: fruit fly **G: liver fluke**

C: woodlouse. **H: spider**

D: lizard. **I: fern with sori**

E: crayfish **J: sea urchin**

1. (a) For each of the specimens (A to H) provided, state:

(i) the major group (phylum) to which it belongs;
(ii) the sub-group (class) to which it belongs;
(iii) the specific microhabitat

Your answer should be written in your book in columns as shown below. (15 mks)

Specimen	Phylum	Class	Microhabitat
A	Angiospermophyta	Monocotyledoneae	Farms/gardens/bushes
B	Arthropoda	Insecta	Kitchen/ dust bins/decaying ripe fruits
C	Arthropoda	Malacostraca	Inside barks of logs/ under stones/kitchen
D	Chordata	Reptilia	Rocks/ walls/ tree trunks/ on blocks
E	Arthropoda	Malacostraca	Rivers/streams/ marine waters
F	Bryophyta	Hepaticae	Moist humus soil/ along banks of slow flowing streams/ around taps/ moist verandas
G	Platyhelminthes	Trematoda	Liver of sheep
H	Arthropoda	Arachnida	Gardens/ under stones/corners of the house
I	Filicinophyta	Filicinae	Farm/walls of buildings/tree trunks
J	Echinodermata	Echinoidea	Sea bottom

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(b) (i) State any two features common to specimens B (fruit fly) and H (spider). (2 mks)

- Both have segmented bodies
- Both have jointed legs/appendages.
- Both have tracheal system for gaseous exchange.
- Both have chitinous exoskeleton covering the body.

(ii) Give any three structural adaptations of H to its environment. (3 mks)

- It has 4 pairs of jointed legs for movement.
- Has sticky claws which ease movement upside down on ceilings.
- Has 8 simple eyes for vision.
- Has spinnerets to spin webs to trap preys.
- Has chelicerae with poison glands to paralyze preys.
- Body covered with chitinous exoskeleton to protect and also reduce water loss.

(c) Make an annotated drawing of the ventral view of specimen D (lizard) to show how it is adapted to its environment. 10 mks)

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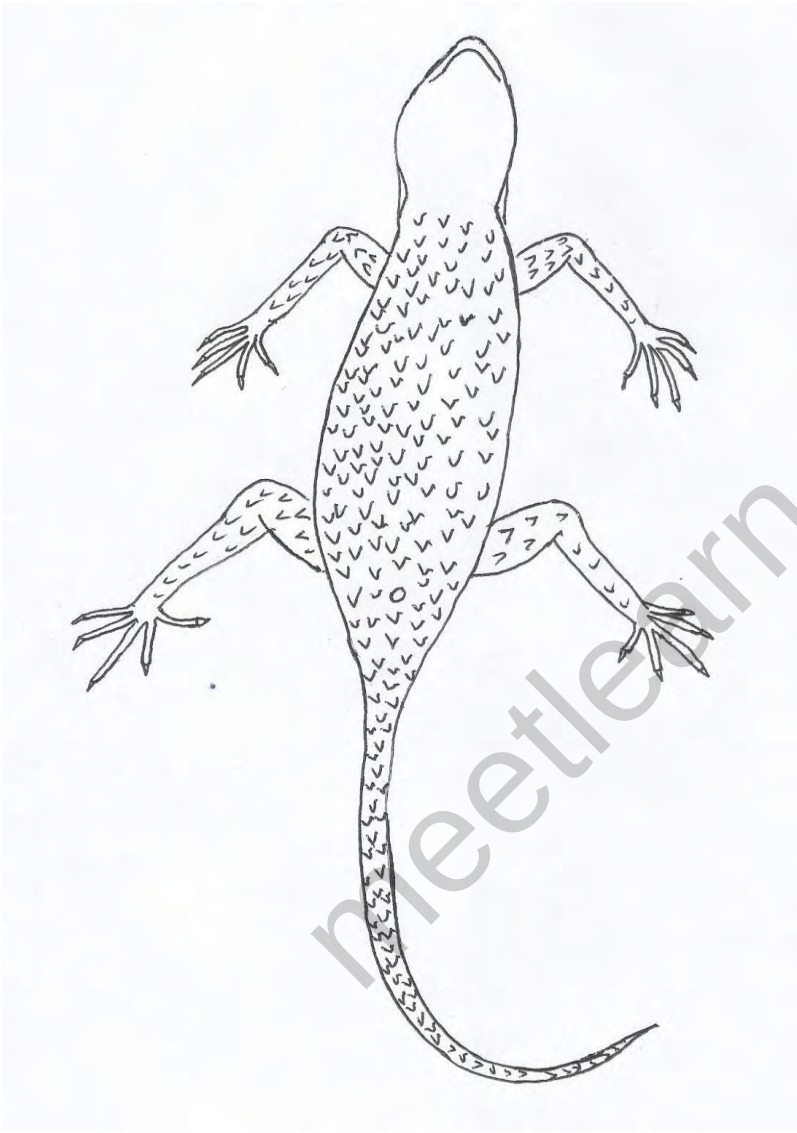

Structure of the ventral view of a lizard

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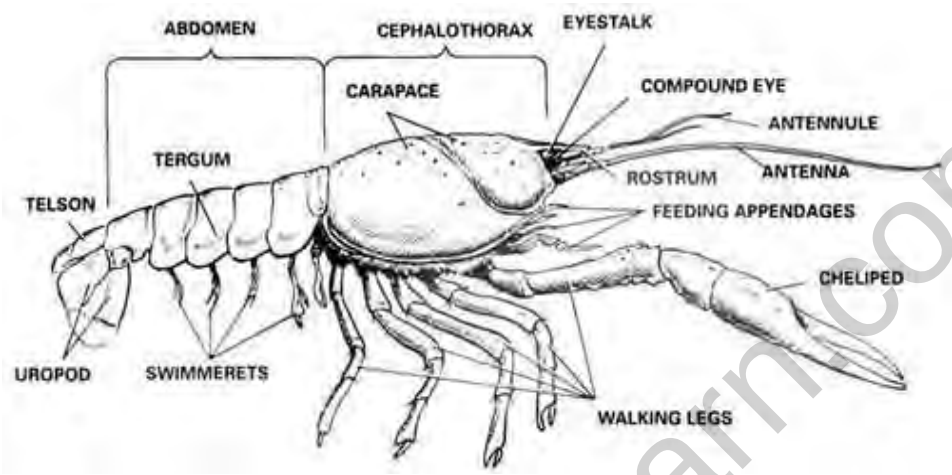
2.

(a) Draw and labelled the lateral view of specimen E (crayfish) (8 mks)

(b) Annotate any five appendages to show their functions. (5 mks)



Structure of the lateral view of a crayfish

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(c) State 2 similarities and 3 differences between specimens F (liverwort) and I (fern). (5 mks)

Similarities:

- Their cells have chloroplasts with chlorophyll for photosynthesis.
- They both produce spores.
- Both need water for fertilization.

Differences:

Specimen F (liverwort)	Specimen I (fern)
Gemma cup present	Gemma cup absent
No rhizome	Contains rhizome
Lacks true leaves	Has true leaves
Gametophyte phase is dominant	Sporophyte phase is dominant
Lacks true roots	Has true roots
So sori	Contains sori

(d) Why are specimens F and I classified under different phyla? (2 mks)

- F is has a dorsoventrally flattened thallus body, while I does not.
- F has dichotomous branching, while I does not.
- I contains sori, while F does not.
- Gemmae (spores) are dispersed by water in F, while in I, spores are dispersed by wind.
- Rhizome present in I but absent in F.

3. Pin the fish provided on its back on a dissecting board. Cut through the muscle wall and into the perivisceral cavity to display the circulatory system, respiratory system and the alimentary canal. Make a drawing of your dissection and label fully.

Make annotations on your drawing to include the role of two structures each concerned with gaseous exchange, blood circulation, nutrition and reproduction.

Leave your dissection properly displayed for assessment at the end of the examination. (25 mks)

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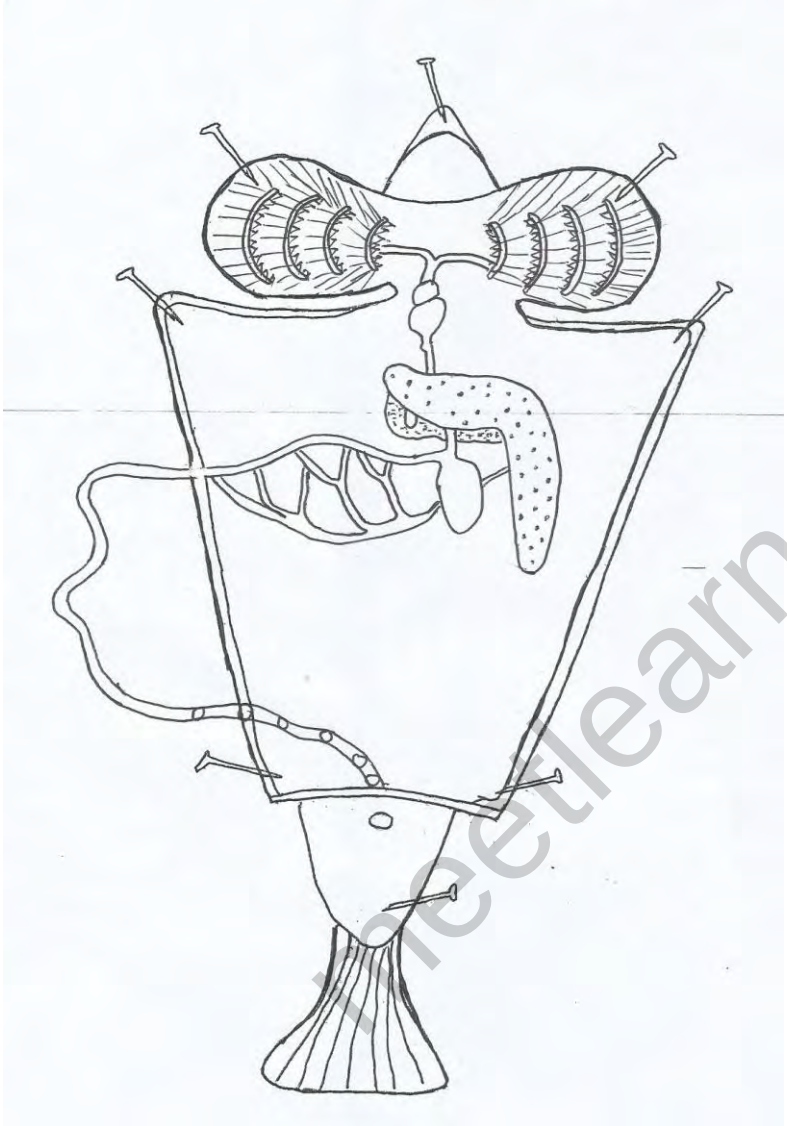

Structure of the circulatory, respiratory and digestive systems of a fish

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4. Solution S = 1M NaCl

Solution T = 1M NaOH

Solution X = Distilled water

You are provided with fresh cocoyam stalks and solutions S, T and X. Investigate the effects of immersing strips from cocoyam stalk into S, T and X.

(a) Describe your method, record your result and plot them on the graph paper provided. (15 mks)

(b) Explain the changes that have taken place in the strips immersed in the different solutions. What processes are responsible for the changes in shape, texture and length of these strips. (10 mks)

Answer

(a)

Procedure/method.

• Three strips of 50mm X 5mm X 5mm were prepared from the cocoyam stalk provided.

• The strips were blotted lightly using tissue paper.

• There textures were noted with the fingers.

• The strips were completely immersed in three separate beakers containing equal volumes different solutions labelled S, T and X.

• After 30 minutes, they were removed using a forceps and blotted with tissue paper.

• They were re-measured and their new textures felt.

Observations.

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• Before immersion, all the strips were curved outward and were hard.

• On removal,

Results in solution S (NaCl)	**Results in solution T (NaOH)**	**Results in solution X (water)**

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• The strip was flaccid or softer. • The strip was convex (or curved inward and straightened) • The length was shorter and measured 47mm. • Strip was flexible	• The strips appeared cooked, slimy or slippery. • The strip was slightly curved. • The strip was shorter and measured 49mm	• The strip was hard or turgid • The strip was concave or curves outward • The length of strip increased and measured 53mm.

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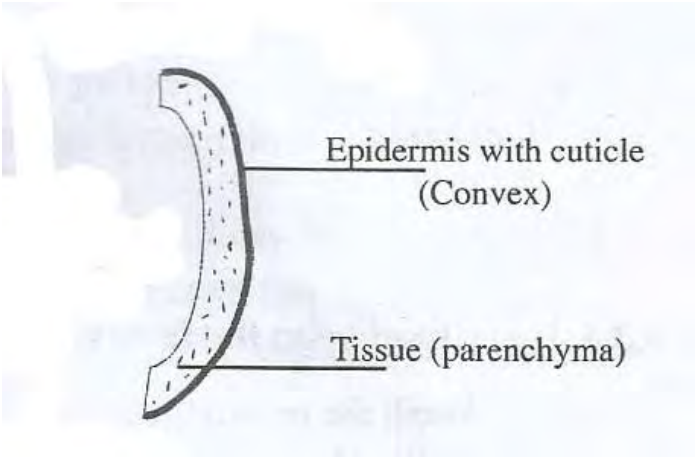

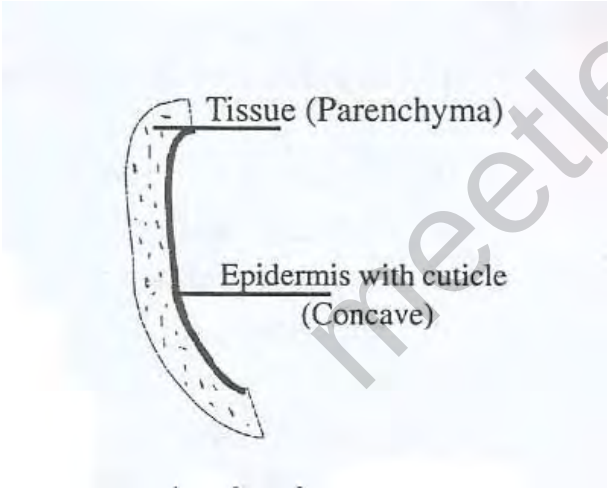

Graphs

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(b)

• The sides of strips without epidermis is freely permeable.

• The epidermis is hard and more resistant to pressure. It limits water movement, and this causes curvature.

Explanation of result of solution S.

• Solution S is having a lower water potential compared to the cell sap. It's a hypertonic solution.

• Water moves out of the cells by osmosis (exosmosis)

• The cells became plasmolyzed and flaccid.

• Resistance of epidermis caused the strip to become convex.

• Loss of water by the cells caused the decrease in length of the strip.

Explanation of result of solution T (NaOH)

• The solution is corrosive and destroyed the cells.

• Partial permeability of the cell membrane was destroyed.

• There was uncontrolled movement of solution. Cytoplasm and cell wall were destroyed.

• The cells lost their turgidity and became flaccid.

Explanation of result of solution X (water)

• Solution X is hypotonic. It has a higher water potential than the cell sap.

• Water moves into the cells by osmosis (endosmosis).

• The cells expanded and became turgid.

• Expansion of individual cells caused an overall increase in length.

• The resistance of the epidermis to expand caused the strip to become concave.

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June 2018 group 1

Specimens:

A:Spirogyra. **E: crayfish**

B:millipede. **F: Guinea pig**

C:mushroom. **G: Honey bee**

D: maize plant. **H: bean plant**

1. (a) For each of the specimens (A to H) provided, state:

(i) the major group (phylum) to which it belongs;
(ii) the sub-group (class) to which it belongs;
(iii) the scientific or common name

Your answer should be written in your book in columns as shown below. (8 mks)

A	Chlorophyta	Chlorophyceae	Spirogyra or Spirogyra jogensis
B	Arthropoda	Diplopoda	Millipede or Julus terrestris

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C	Basidiomycota	Basidiomycetes	Mushroom or Agaricus campestris
D	Angiospermophyta	Monocotyledoneae	Maize plant or Zea mays
E	Arthropoda	Malacostraca	Crayfish or Cambarus affinis
F	Chordata	Mammalia	Guinea pig or Cavia cobaye
G	Arthropoda	Insecta	Honey bee or Apis mellifera
H	Angiospermophyta	Dicotyledoneae	Bean plant or Phasealus vulgaris

(b) State the habitats of specimens A, B, C, D, E and H. (6mks)

Answer

Specimen	Habitat
A (Spirogyra)	Fresh water pond, ditches, slow flowing stream
B (millipede)	Forest floor, under decaying leaves, under stones
C (mushroom)	Moist humus soil, decaying log of wood, decaying organic matter
D (maize)	Moist humus soil, garden, cultivated farm land
E (crayfish)	Sea, ocean, fresh waters like streams and rivers
H (bean plant)	Moist humus soil, cultivated farm land, garden

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(c) state three ways each in which specimens G and H are useful to their environments. (6 mks)

Answer.

Usefulness of specimen G (Honey bee)

- They pollinate flowers to ensure sexual reproduction in plants
- They produced honey used as food for animals
- They produced wax used by man to produce candles
- Their dead body add soil nutrients

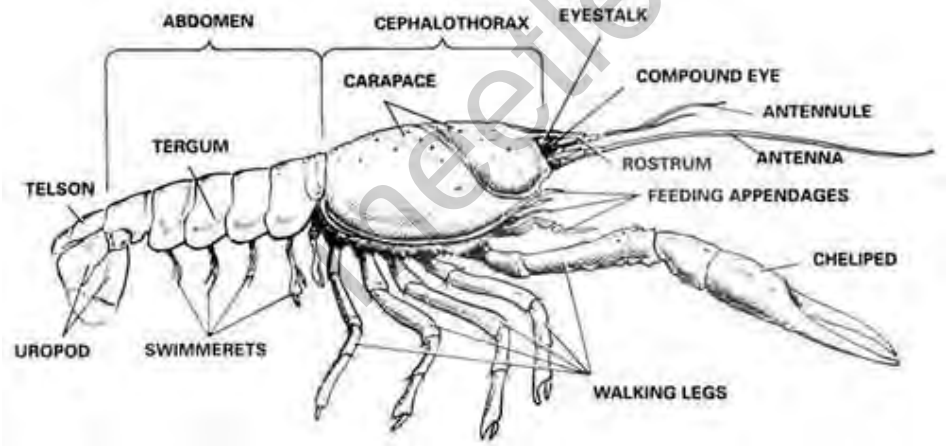
- The release carbon dioxide used by plants for photosynthesis
- They serve as food to other animals

Usefulness of specimen H (bean plant)

- They produced nectar used as food by honey bee and butterfly
- Their leaves, roots and stems decay and add soil humus
- Their root nodules house nitrogen fixing bacteria which fix nitrogen and increase soil nitrate
- They release oxygen used for respiration by animals
- Bean seeds are eaten by man as a source of protein
- Their leaves serve as hiding and protection sites for some animals.

2. (a) (i) Make a fully labelled drawing of the lateral view of specimen E (crayfish)

(ii) Annotate one feature each concerned with food capture and protection. (12 mks)



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(b) How are specimens B, C, G and H related ecologically? (6mks)

Answer.

Feeding relationship

- Honey bee feeds on the nectar and pollen produced by bean plant
- Millipede feeds on the dead leaves of bean plant
- The dead remains of honey bee, millipede and bean plant produce organic matter for the growth of mushroom
- The burrowing action of millipede helps to aerate soil for the growth of bean plant
- When Millipede, honey bee and mushroom die, they form humus which promotes the growth of bean plant

Reproductive relationship

- Honey bee pollinates the flowers of bean plant

Gaseous exchange relationship

- Millipede, honey bee and mushroom send out carbon dioxide used by bean plant for photosynthesis
- Bean plant sends out oxygen used by millipede, honey bee and mushroom for aerobic respiration.

Habitat/shelter

- Bean plant serves as shelter for millipede and honey bee

(c) State two external features of specimen F that are characteristics of its class. (2 mks)

Answer

The external features of specimen F (guinea pig) characteristic of class Mammalia include:

- The body is covered with fur
- The presence of mammary glands
- Presence of external ear flaps called **pinnae.**
- Presence of heterodont dentition
- Presence of large cranial volume
- Presence of pentadactyl limbs.

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3. (a) Dissect the bird to display all the organs in the abdominal cavity. Make a well labelled drawing of your dissection. (13 mks)

(b) Annotate two structures each involved in digestion, reproduction and excretion. (12 mks)

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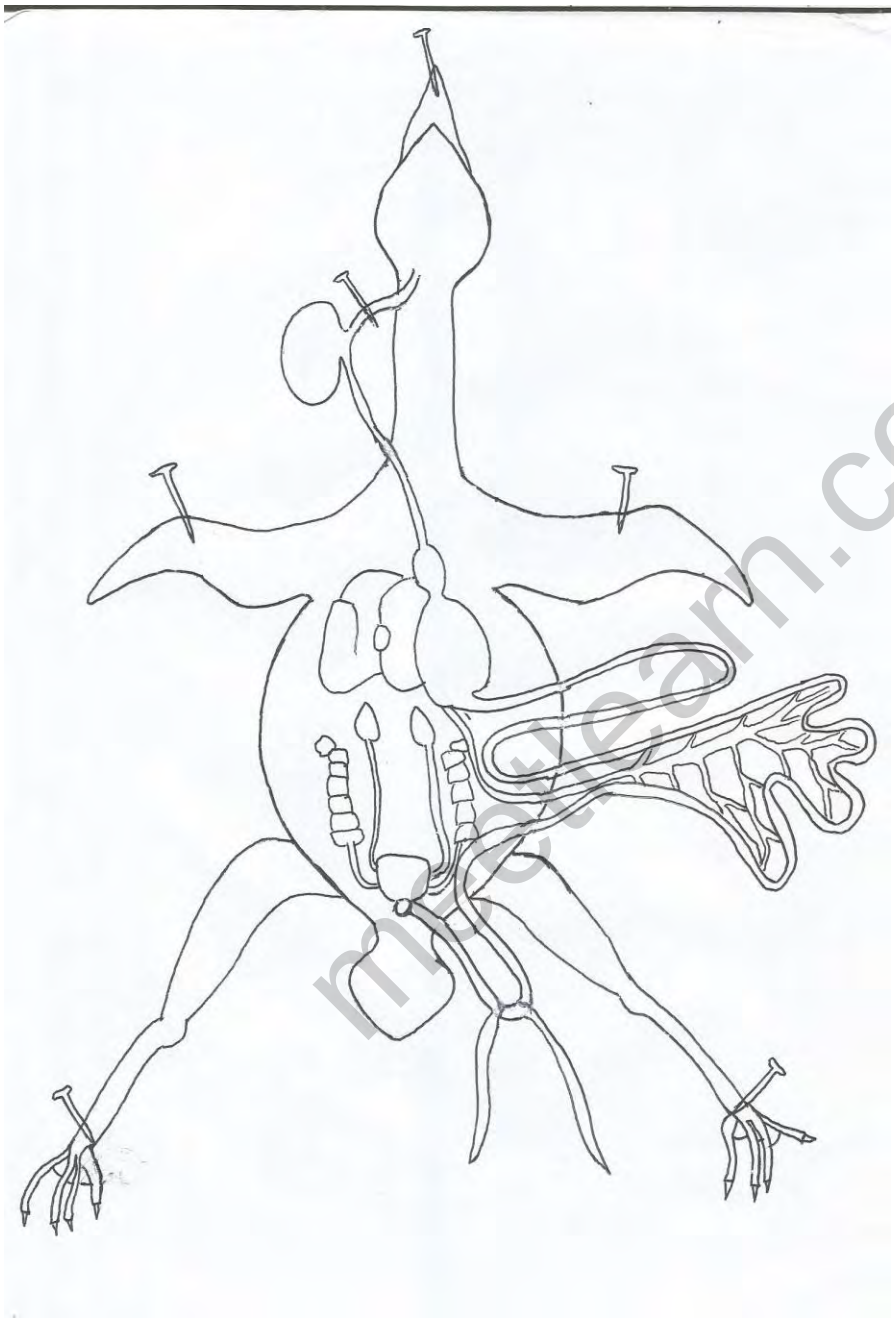

Structure of the organs of the abdominal cavity

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4. The enzyme Catalase present in both plant and animal tissues catalyzes the breakdown of hydrogen peroxide, with the evolution of oxygen gas shown by effervescence (fizzing)

Using the potato tuber provided first obtain an enzyme pool.

Describe your method, observations and explain your results as fully as possible. (25 mks)

Answer

Making an enzyme pool.

- The potato tubers were washed, peeled and chopped into pieces using a knife
- The chopped pieces were ground in a mortar using a pestle
- 50ml of water was added into the ground paste and mixed
- The enzyme was pool was filtered out using a filter paper and funnel

Experiment 1 : Reaction of enzyme Catalase with hydrogen peroxide

Procedure	Observation	Explanation and conclusions
• 2ml of Catalase was put into a test clean test tube using a syringe • 2ml of hydrogen peroxide was added • The mixture was left to stand for 5 minutes	• The mixture started fizzing and produced foams • The foam gave a length of 8cm	• The enzyme Catalase works best with the substrate hydrogen peroxide. • The active sites were not affected • The enzyme converted the hydrogen peroxide to water and oxygen. The released of oxygen gave the foaming

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Experiment 2: Effect of HCl on the reaction of the enzyme Catalase with hydrogen peroxide

Procedure	Observation	Explanation and conclusions
• 2ml of Catalase was put into a test clean test tube using a syringe	• The mixture didn't produced any foams	• The HCl destroyed the active sites of the enzyme,

• Five drops of conc. HCl was added into the enzyme, and the test tube shaken. • 2ml of hydrogen peroxide was added and shaken • The mixture was left to stand for 5 minutes		hence the enzyme was denatured • The substrate could not bind to the active sites to form an enzyme-substrate complex. • No reaction took place and no foams were produced

Experiment 3. Effect of NaOH on the reaction of the enzyme Catalase with hydrogen peroxide

Procedure	Observation	Explanation and conclusions
• 2ml of Catalase was put into a test clean test tube using a syringe • 5 drops of NaOH was added into the enzyme, and the test tube shaken. • 2ml of hydrogen peroxide was added and shaken • The mixture was left to stand for 5 minutes	• The mixture started fizzing slowly and produced small quantity of foams • The foam gave a length of 1.5cm	• The NaOH interfered with the enzyme and hydrogen peroxide was not completely broken down, hence the reaction produced little foams. • NaOH is an inhibitor of Catalase activity • NaOH slightly affected the enzyme active sites • NaOH didn't completely interfere with the working of the enzyme. •The enzyme Catalase possibly has a wide range of pH on alkaline media.

Experiment 4: Effect of heat on the reaction of the enzyme Catalase with hydrogen peroxide

Procedure	Observation	Explanations and conclusions
• 2ml of Catalase was put into a clean test tube using a syringe • The test tube was heated for 3 minutes in a hot water bath • The test tube was cooled under running tap water	• The mixture did not produce any foams	• Heat destroyed the active sites of the enzyme and the substrate could not fit into the active sites to form an enzyme-substrate complex. • The enzyme was denatured and no reaction took place, hence no foams were produced.

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• 2ml of hydrogen peroxide was added and the mixture shaken. The mixture was allowed to stand for 5 minutes.		

Precautions.

- The test tubes were all clean
- Test tubes were labelled

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June 2018 group 2.

Specimens:

A: Obelia. **E: bread mould**

B: spider. **F: morning glory**

C: fern. **G: wood louse**

D: liverwort. **H: Lycopodium**

1. (a) For each of the specimens (A to H) provided, state:

(i) the major group (phylum) to which it belongs;
(ii) the sub-group (class) to which it belongs;
(iii) the scientific or common name.

Your answer should be written in your book as shown below (8mks)

Specimen	Phylum	Class	Scientific or common name
A	Cnideria	Hydrozoa	Obelia or Obelia geniculata
B	Arthropoda	Arachnida	Spider or Miranda aurantia
C	Filicinophyta	Filicinae	Fern or Dryopteris felix-mas
D	Bryophyta	Hepaticae/Hepaticopsida	Liverwort or Marchantia polymorpha
E	Zygomycota	Zygomycetes	Bread mould or Rhizopus nigricans
F	Angiospermophyta	Dicotyledoneae	Morning glory or Ipomea purpurea
G	Arthropoda	Malacostraca	Wood louse or Oniscus ascellus
H	Lycopodophyta	Lycopodinae	Lycopodium or Lycopodium clavatum

(b) (i) Make a dichotomous key to separate specimens A — H. (7 mks)

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Tentacles present A

ABCDEFGH.

Spinneret present B

Tentacles absent BCDEFGH.

Sori present C

Spinneret absent CDEFGH

Sori absent DEFGH

Gemma cup present D

DEFGH.

Sporangia present E

Gemma cup absent EFGH.

Flower present F

Sporangia absent FGH.

Antennae present G

Flower absent GH

Antennae absent H

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(c) How is specimen E adapted to its mode of life? (5 mks)

- Has rhizoids to penetrate the substratum to obtain nourishment.
- The rhizoids release extracellular enzymes unto the substratum for digestion.
- Sporangiophores carry and support the Sporangia above the ground.
- Sporangial sacs in which Sporangia are produced and discharge in the air.
- The collumala in the Sporangial sac produces a mechanism for spore dispersal.
- The presence of stolon to extend the hyphae.
- The spires are light and can be easily dispersed by wind.

2. (a) How are specimens A (Obelia)and B (spider) adapted for food capture? (4 mks)

A:

- Nematoblasts carrying stinging threads to sting and capture preys.

- Tentacles to capture and carry prey into the mouth.

B:

- It spins webs to capture preys.
- Has fanged chelicelariae with poison to paralyse preys.
- The Pedipalps help to hold preys to be paralysed.

(b) (i) Draw and label the under surface of the frond of specimen C (fern)

(ii) Annotate a structure concerned with reproduction. (8 mks)

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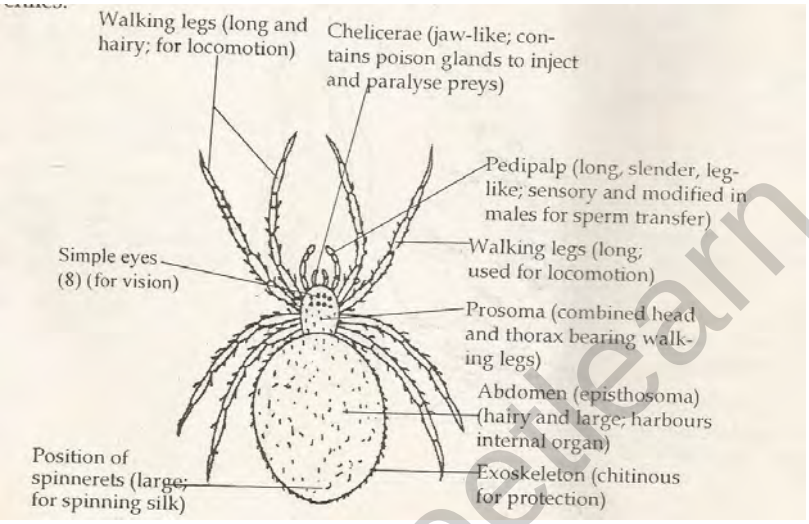
Structure of the under surface of the frond of a fern plant

(c) What is the role of water in the life cycle of specimen C (fern). (2 mks)

- Biflagellate antherozoids from antheridium swims in film of water and through the archegonial neck to fertilize the egg at the base of archegonium.
- The annulus cells with unequal thickened walls easily dries up when there is no water/moisture in the air, thus bending out and breaking the weak stomial cells , releasing spores.
- Water is needed for germination of spores.

(d) Make a labelled diagram of the dorsal view of specimen B (spider). (6 mks)



3. (a) Pin the fish on its back and open the perivisceral cavity to display the alimentary canal and respiratory system. Make a labelled drawing of your dissection. (13 mks)

(b) Annotate the drawing to show the role of three parts in digestion and three in respiration. (12 mks)

Leave your dissection properly displayed for assessment at the end of the examination.

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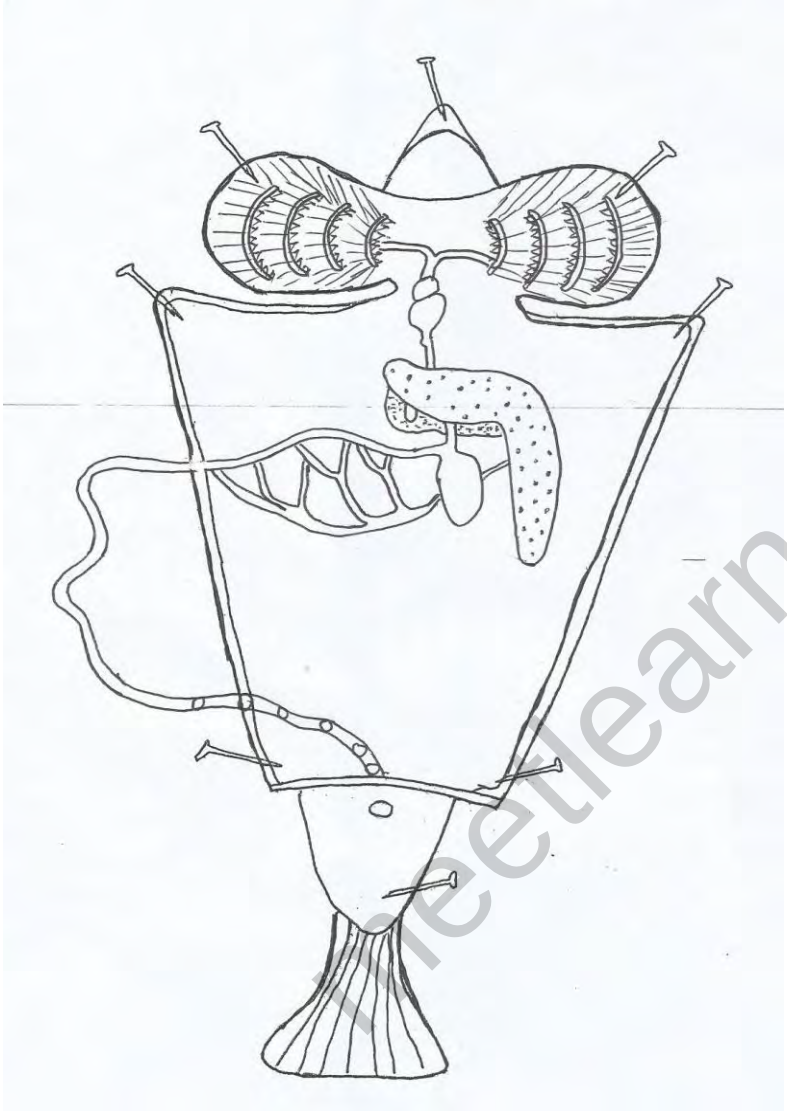

Structure of the alimentary canal and respiratory system of a fish

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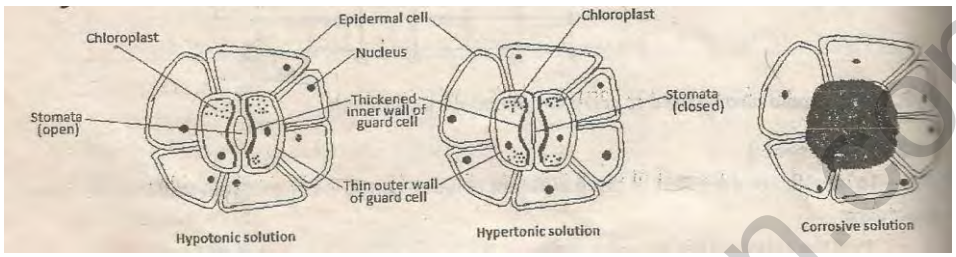
4. Solution X = distilled water, solution Y = saturated salt solution and solution Z = 0.1M NaOH

(a) Strip off three pieces of the lower epidermis from the leaves provided. Mount each piece on a plain slide in a drop of solutions X, Y and Z respectively. Cover each with a cover slid and allow for 15 minutes.

Observe under the microscope and in each case draw at least 6 adjacent cell, including a stoma. (10 mks)



(b) Explain the behavior of the cells in each of the solutions. (15 mks)

*** Cells in solution X (distilled water)**

Procedure	Observation	Inferences
• A small strip was peeled off from the lower epidermis of the tradescantia leave. • The strip wad placed on a clean virgin slide • 5 drops of solution X was put on the strip and covered with a cover slid. • It wad allowed to stand for 15 minutes.	• The cells bulged or larger (turgid) • The guard cells were more curved • The stomata were widely open	• Water entered the cells by osmosis. • The cells became turgid. • Uneven expansion or the outer and inner walls of guard cells caused the guard cells to curve and the stomata to open widely. • Solution X is a hypotonic solution e.g. distilled water,

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• The prepared slide was then put on the microscope stage and viewed under low power (X 40)		while the cell sap is hypertonic.

*** Cells in solution Y (saturated salt solution)**

Procedure	Observation	Inferences
• A small strip was peeled off from the lower epidermis of the tradescantia leave. • The strip wad placed on a clean virgin slide • 5 drops of solution Y was put on the strip and covered with a cover slid. • It wad allowed to stand for 15 minutes. • The prepared slide was then put on the microscope stage and viewed under low power (X 40)	• The cells became smaller (flaccid) • The cells looked weak. • The guard cells became less curved. • The stomata were seen closed.	• Water leaves the cells by osmosis. • The cells became flaccid. • The uneven expansion of the outer and inner walls of the guard cells led to closing of stomatal pores. • Solution Y is a hypertonic solution e.g. saturated salt solution, while the cell sap is hypotonic.

*** Cells in solution Z (0.1M NaOH)**

Procedure	**Observation**	**Inferences**
• A small strip was peeled off from the lower epidermis of the tradescantia leave. • The strip wad placed on a clean virgin slide • 5 drops of solution Z was put on the strip and covered with a cover slid. • It wad allowed to stand for 15 minutes. • The prepared slide was then put on the microscope stage and viewed under low power (X 40)	• The cells became smaller (flaccid) • The cells looked disrupted. • The chloroplasts in the guard cells were destroyed and chlorophyll leaked out, staining all the preparation. • The guard cells and other neighbouring cells were seen wavy in nature.	• The solution Z is a very corrosive solution. E.g. concentrated NaOH. • The cell walls were disrupted and destroyed. • Chloroplasts membranes and cell surface membranes were destroyed. • Osmosis did not take place because there was no selectively permeable membrane, since the cell membranes were destroyed.

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North West Regional mock 2018 group 1

Specimens

A: maize. E: grasshopper

B: paramecium. F: toad

C: earthworm. G: mushroom

D: crayfish. H: fern

1 a). For each of the specimens (A to H) provided, state;

(i) The major group or phylum to which it belongs,
(ii) The sub-group or class to which it belongs,
(iii) The habitat where it is commonly found

Your answer should be written in your answer book in column as shown below (12 mks)

Specimen	**(i)**	**(ii)**	**(iii)**
A	Angiospermophyta	Monocotyledoneae	Farm land / damp humus soil
B	Ciliophora	Ciliata	Fresh stagnant eater, some in marine water
C	Annelida	Oligochaeta	Moist humus soil
D	Arthropoda	Malacostraca	Fresh water / marine water
E	Arthropoda	Insecta	Field/ on green vegetation/ farm
F	Chordata	Amphibia	Damp areas /under stones / swampy areas

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G	Basidiomycota	Basidiomycetes	Moist humus soil/ decaying tree trunks
H	Filicinophyta	Filicinae	Damp shady soil/ on the bark of palm trees

b) Group specimens A-H under the following trophic levels (4 mks)

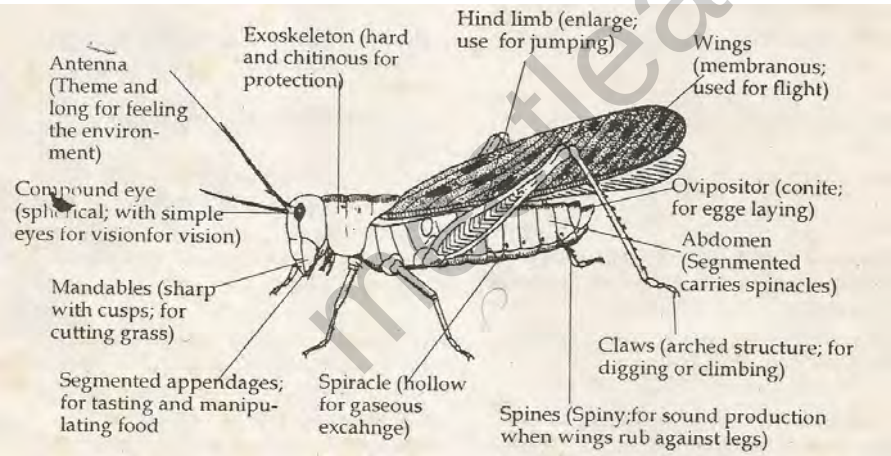
Primary producers: maize, fern plant

Primary consumers: grasshopper, crayfish

Secondary consumers: toad, paramecium, crayfish

Decomposers : earthworm, mushroom

c) Make a fully labelled drawing of specimen E (grasshopper. Annotate two structures (10 mks) concerned with movement



Structure of a grasshopper

d) Explain how specimens A, E, F and G are ecologically related.(4 mks)

- Grasshopper feeds on the leaves of maize

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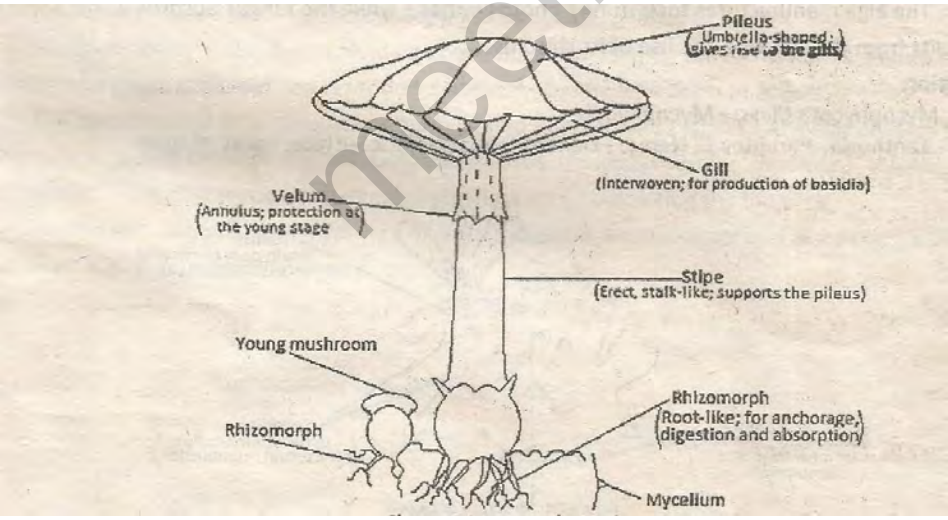
- Maize releases oxygen which is used by grasshopper, toad and mushroom for respiration
- Grasshopper, toad and mushroom release carbon dioxide which is used by maize for photosynthesis
- Mushroom decompose the dead bodies of maize, toad and grasshopper, and improve soil fertility for maize to grow healthy
- Maize serves as shelter and hiding place for grasshopper and toad
- Grasshopper can accidentally pollinate the glowers of maize

2. a)

(i) State three features of biological importance of specimen G (mushroom) (4 mks)

- Bears fruiting body called pileus, which is the reproductive part
- Mycelium grow saprophytically on organic matter in the soil
- Some species are edible
- Posses sheet-like gills that bear basidia
- Has septate hyphae
- Asexual reproduction is by spore formation

(ii) Make a fully labelled drawing of this specimen (mushroom) (6 mks)



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b) Examine specimen F(toad) carefully

(i) State the functions of five external features which adapt it to life in its habitat (5 mks)

- Strong hind limbs for hopping
- Mucous gland on the skin to keep it moist for gaseous exchange and to prevent drying
- Protective colouration for camouflage
- Large bulging eyes to sport danger, food and sex mates
- Eardrums to perceive croaking sounds made by sex mates and various insects

(ii) Tabulate five differences between the hind and fore limbs of specimen F(toad)

Fore limb	**Hind limb**
Short	Long
Digits lack webs	Webbed digits
Less muscular	Very muscular
Four toes/ digits	Five toes/ digits

3. Dissect the bird provided to display the Urinogenital system and its associated organs. Make a drawing of your dissection and label as fully as you can. Annotate any two labels concerned with

(i) Reproduction

(ii) Excretion

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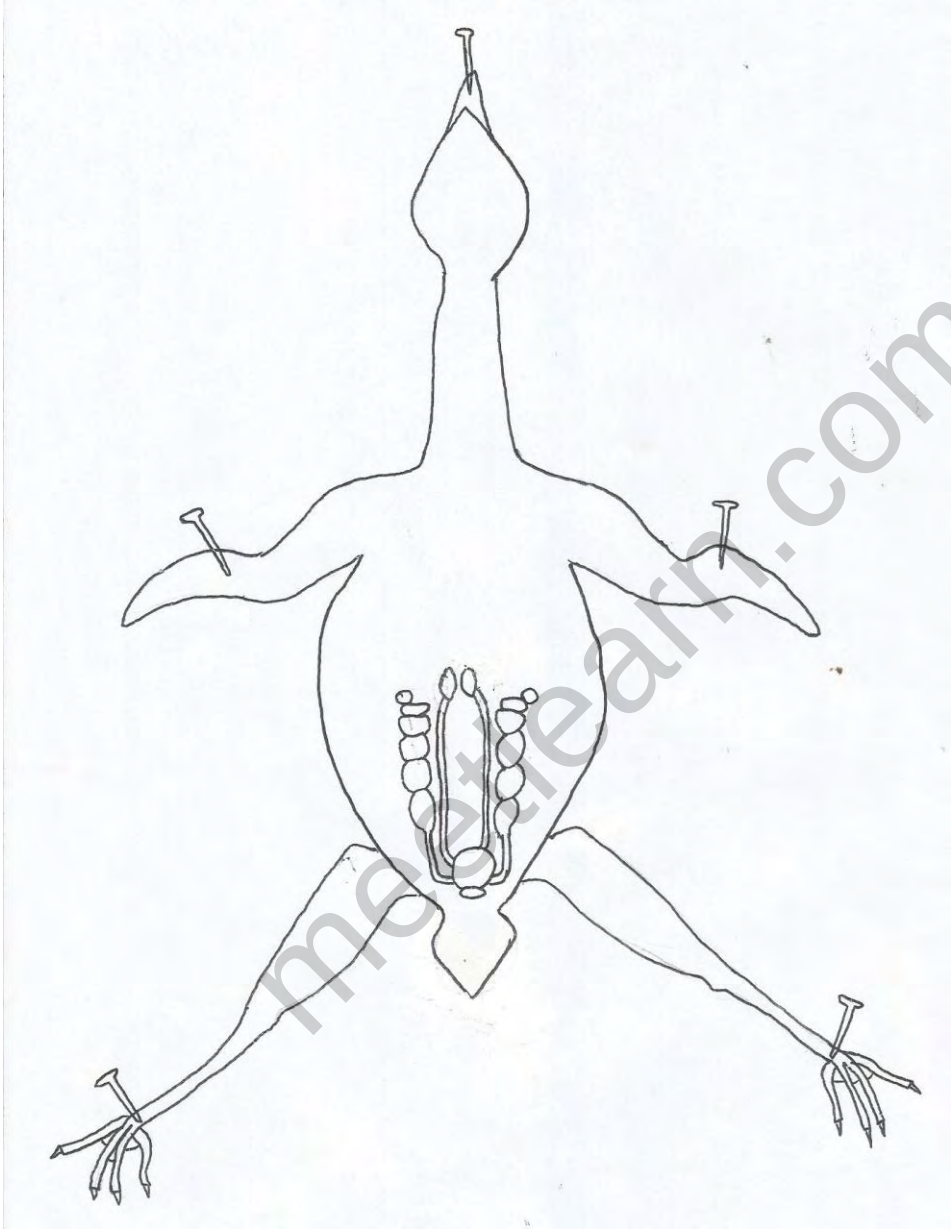

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4. a) Macerate and form a solution of each of the following: liver, ileum and muscle of the bird in question 3 above. Place in three separate test tubes. Investigate the presence of the enzyme Catalase in each of the tissues. Make a report of your experiment. (20 mks)

Answer: see June 2019 group 2

b) Use any of the tissues above, devise a simple experiment to show the effect of heat on the enzyme Catalase. (5mks)

see June 2019 group 2

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North West regional mock 2018 group 2.

Specimens:

A: toad **E: wood louse**

B: mushroom **F: moss plant**

C: pine **G: hibiscus twig**

D: Guinea pig **H: honey bee**

1. (a) For each of the specimens (A to H) provided, state:

(i) the major group (phylum) to which it belongs;
(ii) the sub-group (class) to which it belongs;
(iii) the scientific or common name.

Your answer should be written in your book as shown below (8mks)

Specimen	Phylum	Class	Scientific or common name
A	Chordata	Amphibia	Toad
B	Basidiomycota	Basidiomycetes	Mushroom
C	Coniferophyta	Coniferinae	Pine
D	Chordata	Mammalia	Guinea pig

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E	**Arthropoda**	**Malacostraca**	**Wood louse**
F	**Bryophyta**	**Musci or Bryopsida**	**Moss plant**
G	**Angiospermophyta**	**Dicotyledoneae**	**Hibiscus twig**
H	**Arthropoda**	**Insecta**	**Honey bee**

(b) His are specimens B and E. (6 mks)

(i) similar

- Both carry out heterotrophic nutrition
- Both feed on dead organic matter

(ii) different nutritionally?

- In mushroom, digestion takes place outside the body (extracellular digestion) while in wood louse, digestion is internal along the digestive tract.
- Mushroom is a saprotroph, while wood louse is a detritivore

(c) Using only external visible features, excluding colour and size, make a dichotomous key to separate the specimens A — H. (8 mks)

Wing present H

ABCDEFGH. Flower present G

Wing absent ABCDEF. Pileus present B

Flower absent ABCDEF. Capsule present F

Pileus absent ACDEF

Capsule absent ACDE

Pinnae present D

ACDE. Cone present C

Pinnae absent ACE. Antennae present E

Cone absent AE

Antennae absent A

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(d) How are specimens (i) A and (ii) C structurally adapted to their respective habitats? (8 mks)

(i) A (toad)

- **Long sticky tongue to capture preys**
- **Muscular hind limbs for jumping**
- **Moist glandular skin for seat diffusion of gases**
- **Webbed toes for swimming while in water**
- **Poison glands to scare predators**

(ii) C (cypress)

- Scaly leaves tiny reduces surface area for transpiration
- Leaves are covered with thick waxy cuticle to prevent water loss
- Have cone for reproduction
- Light pollen grains for easy dispersal by wind
- Trunk and branches are covered with scales to prevent water loss

2. (a) Observe specimen E (wood louse) carefully and make a large labelled drawing of the ventral view. Annotate the parts used for

(i) sensitivity. (ii) locomotion. (10 mks)

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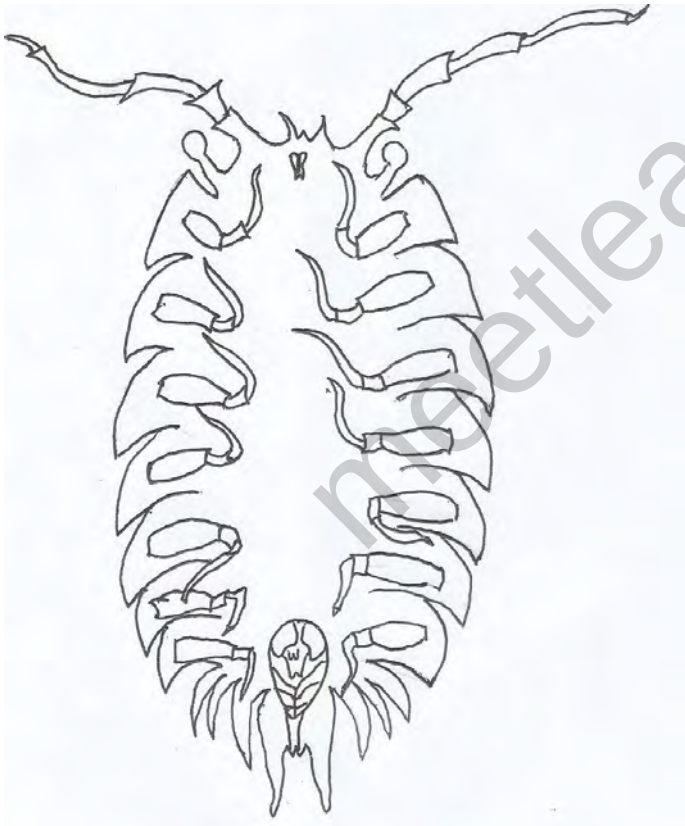

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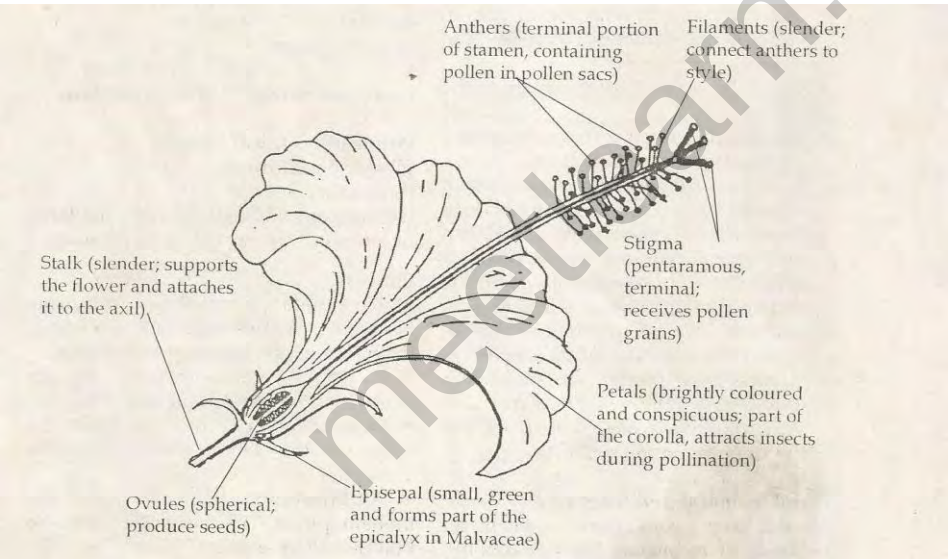
(b) How is specimen H (Honey bee) structurally adapted for food collection? (4 mks)

- Compound eyes to locate flowers
- Membranous wings for flight
- Jointed legs to craw into the flower
- Limbs armed with pollen combs and prongs for collection of pollen
- Hind limbs with pollen basket for storage of pollen

(c) (i) Using a razor blade, carefully cut the flower of specimen G longitudinally. Make a labelled diagram of your dissection.

(ii) on your diagram annotate any structure concerned with pollination. (6 mks)



3. (a) Draw the lateral view of the fish provided to show its external features. Annotate two feathers which are typical of the class to which it belongs. (5 mks)

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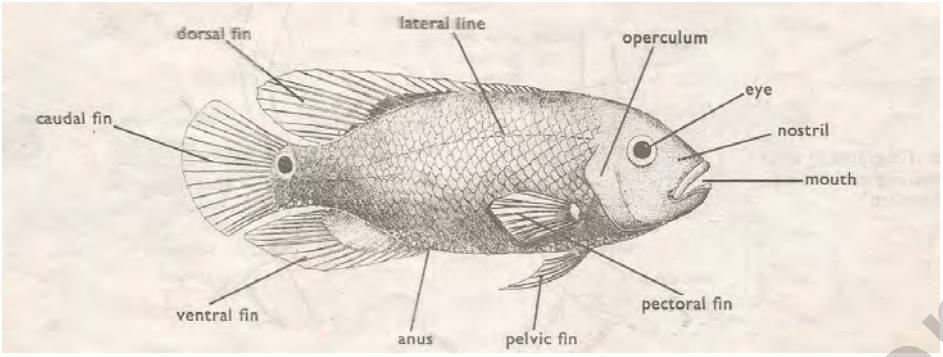

(b) Pin the fish on its back on a dissecting board. Cut into the muscle wall and into the perivisceral cavity to display the respiratory, alimentary and Urinogenital systems. Make a fully labelled drawing of your dissection. (12 mks)

(c) Annotate two structures concerned with each of the following.

(i) digestion

(ii) excretion. (8 mks)

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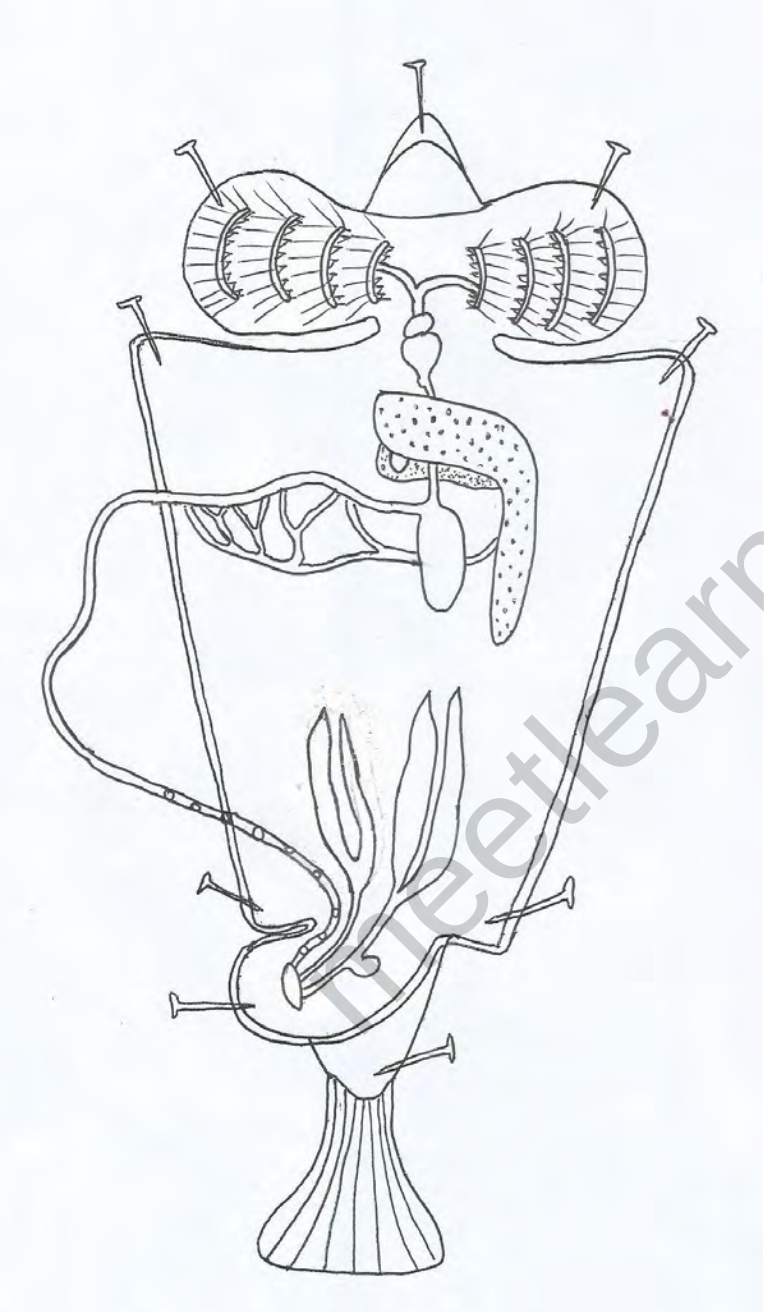

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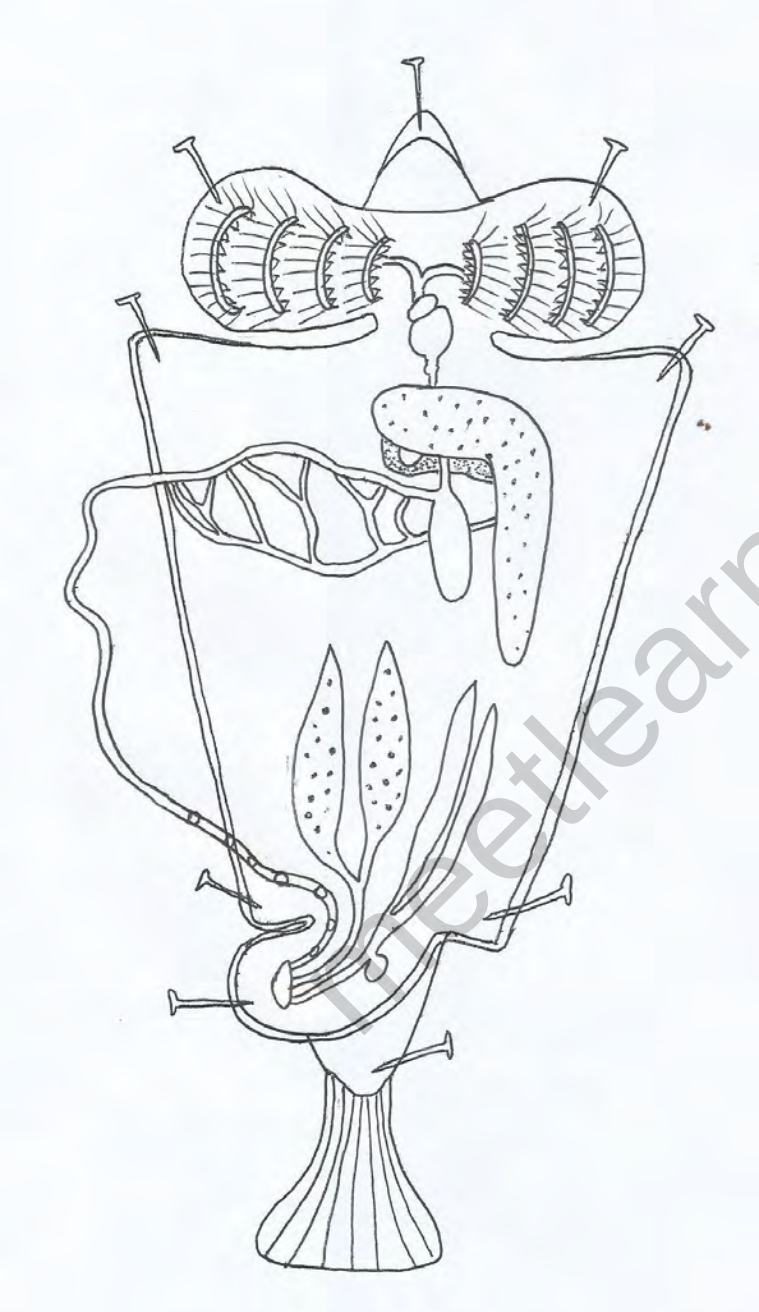

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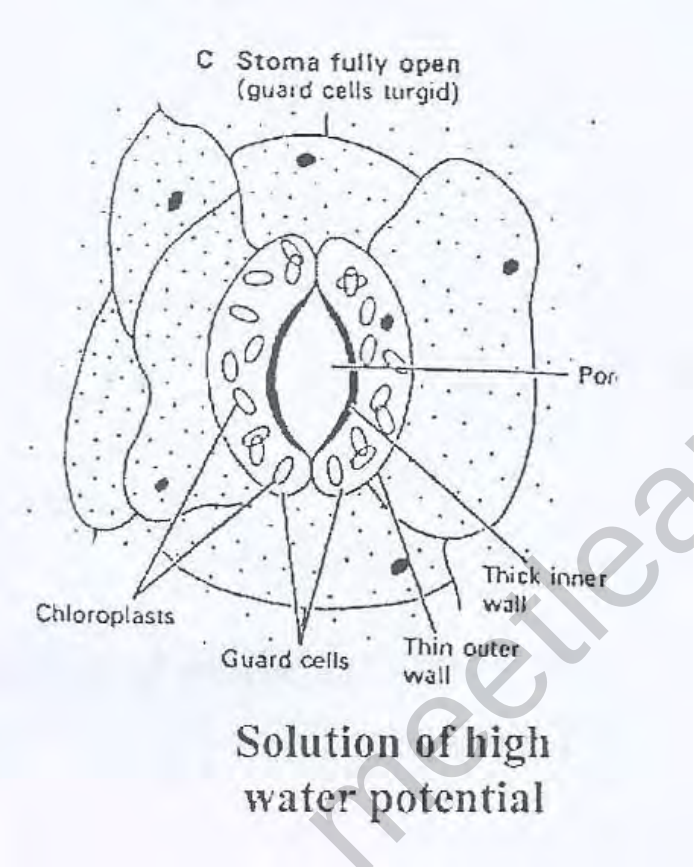
4. (a) Remove a strip of the lower (coloured) epidermis from the leaf P. Mount a small piece of it in distilled water. Observe under low power of a microscope. Draw and label six adjacent cells including a stoma. (4 mks)



(b) You are provided with a potato tuber and solutions S and T. Investigate the effects of immersing strips of potato tuber in solutions S and T. Describe your methods, record your results and plot them on the graph paper provided. (12 mks)

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Solution S= distilled water. **Solution T= NaCl solution**

Method

- The potato tubers were peeled using a knife.
- Two stops of measurement 50mm X 5mm X 5 mm were prepared.
- The texture of the strips was noted.
- The steps were blotted using a tissue Paper.
- Equal volumes of solutions S and T were put into separate labelled beakers.
- The strips were put into the separate labelled beakers at the same time.
- The strips were allowed in the solutions for 30 minutes.
- After 30 minutes, the strips were removed and blotted again.
- The strips were re-measured and their new length noted.
- Their new textures were also noted.

Result

Solution	**Initial texture**	**Final texture**	**Initial length**	**Final length**	**Change in length**
S	Turgid	Very turgid	50mm	53mm	3mm
T	Turgid	Flaccid/soft	50mm	48mm	-2mm

graph

(c) What process is responsible for the changes in the potato strips. (1 mk)

• The process is osmosis

(d) Explain the changes that have taken place in the strips immersed in the different solutions. (8 mks)

Commented [2]:

- Sap vacuole contains solution with dissolved salts and organic solutes
- The living cell membranes act as the selectively permeable membrane.
- Osmosis occurred in both strips.
- In S, water moved from the solution into the potato cells by osmosis, and caused the cells to become turgid.
- The entry of water elongate the cells, leading to an overall increase in the length of strip
- Solution S is therefore a solution of higher water potential (hypotonic solution) than the potato cell sap. E.g. distilled water.
- In T, water moved out of the cells into the solution by osmosis. This made the cells to become flaccid.
- The sizes of cells decrease, leading to an overall decrease in the length of strip.

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- Solution T is therefore a solution of lower water potential (hypertonic solution) than the potato cell sap. E.g. concentrated NaCl solution.

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June 2019 group 1.

Specimens:

A: paramecium on slide. **E: earthworm**

B: maize plant. **F: toad**

C:grasshopper. **G: fern plant with leaves and sori**

D:crayfish. **H: mushroom**

1. (a) For each of the specimens (A to H) provided, state:

(i) the major group (phylum) to which it belongs;
(ii) the sub-group (class) to which it belongs;
(iii) the scientific or common name.

Your answer should be written in your book as shown below (8mks)

Specimen	Phylum	Class	Scientific or common name
A	Ciliophora	Ciliata	Paramecium
B	Angiospermophyta	Monocotyledoneae	Maize plant
C	Arthropoda	Insecta	Grasshopper
D	Arthropoda	Malacostraca	Crayfish
E	Annelida	Oligochaeta	Earthworm
F	Chordata	Amphibia	Toad
G	Filicinophyta	Filicinae	Fern plant
H	Basidiomycota	Basidiomycetes	Mushroom

(b) Group specimen A-H under the following trophic levels (6 mks)

Primary producer: maize plant, fern.

Primary consumer: grasshopper, paramecium, crayfish

Secondary consumer: toad, crayfish, paramecium

Decomposer: mushroom, earthworm

(c) Explain how B, C, F and H are ecologically related.(6 mks)

• The maize plant produces oxygen during photosynthesis, used by the grasshopper, toad and mushroom for respiration.

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• The grasshopper, toad and mushroom produce carbon dioxide during respiration, used by maize plant for photosynthesis.

• The grasshopper feeds on the leaves of maize plant.

• The toad feeds on grasshopper.

• The mushroom feeds and grows on the dead remains of grasshopper, toad and maize.

• The grasshopper lays eggs on the leaves of maize plant.

• The maize plant serve as habitat for grasshopper.

• The maize plant serves as the hiding site for grasshopper and toad.

• The dead remains of toad, grasshopper and mushroom, together with the decomposer action of the mushroom improve soil fertility for maize plant to grow.

• The nitrogenous wastes of toad and grasshopper help to improve soil nitrogen used by maize plant for growth

2. (a) (i) state three features of biological importance of specimen H (mushroom) (4 mks)

• It bears fruiting body called sporophore.

• The fruiting body has a cap called pileus, which gives rise to the gills on the undersurface.

• The sheet-like gills produce basidia on which the basidiospores are formed.

• It has a septate hyphae.

• Hyphae called rhizomorph grow into the substrate for anchorage and absorption of nutrients.

• They have an erect stalk called stipe, which helps to suspend the pileus and gills

(ii) Make a large fully labelled drawing of this specimen (mushroom) (7 mks)

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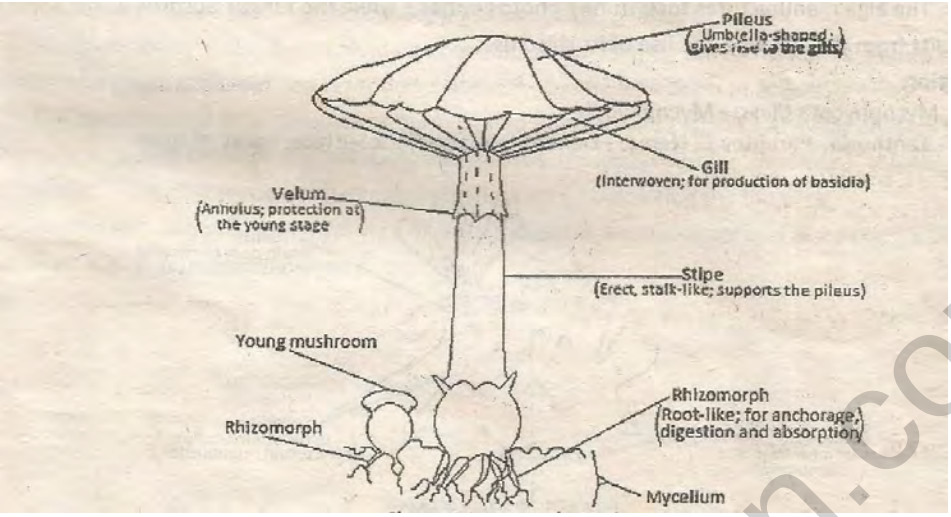

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(b) Examine specimen F (toad) carefully

(i) Describe five external features which adapt it to life in its habitat. (5 mks)

• Has strong and long muscular hind legs for hopping.

• Short muscular fore legs acts as shock absorber during landing.

• Has Webbed toes for swimming.

• Mucus glands in the skin to keep it moist and prevent drying

• Warty body for camouflage.

• Bulging eyes to locate food, enemies and mates.

• Wide mouth containing long and sticky tongue for feeding.

• Eardrum with tympanum to perceive sound.

• Nictitating membrane capable of moving up and down, moistening and protecting the eyes from mechanical injury.

• Poison glands secrete poisonous substances that prevents them being eaten by predators. It makes them to taste baí • Moist body for gaseous exchange.

(ii) Tabulate four differences between the hind and fore limbs of specimen F. (4mks)

Hand limbs	**Fore limbs**
Long	Short
Very muscular	Less muscular
Webbed digits	Digits lack webs
Five digits	Four digits
Used for swimming and jumping	Used for landing, crawling and also absorb shock during landing

3. (a) Dissect the bird provided to display the main organs concerned with reproduction and excretion.

(i) In the space provided below, make a fully labelled drawing of your dissection. (13 mks)

(b) Annotate three features for each of the processes in (a) (i) above (12 mks)

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Leave your dissection properly displayed for assessment at the end of the examination.

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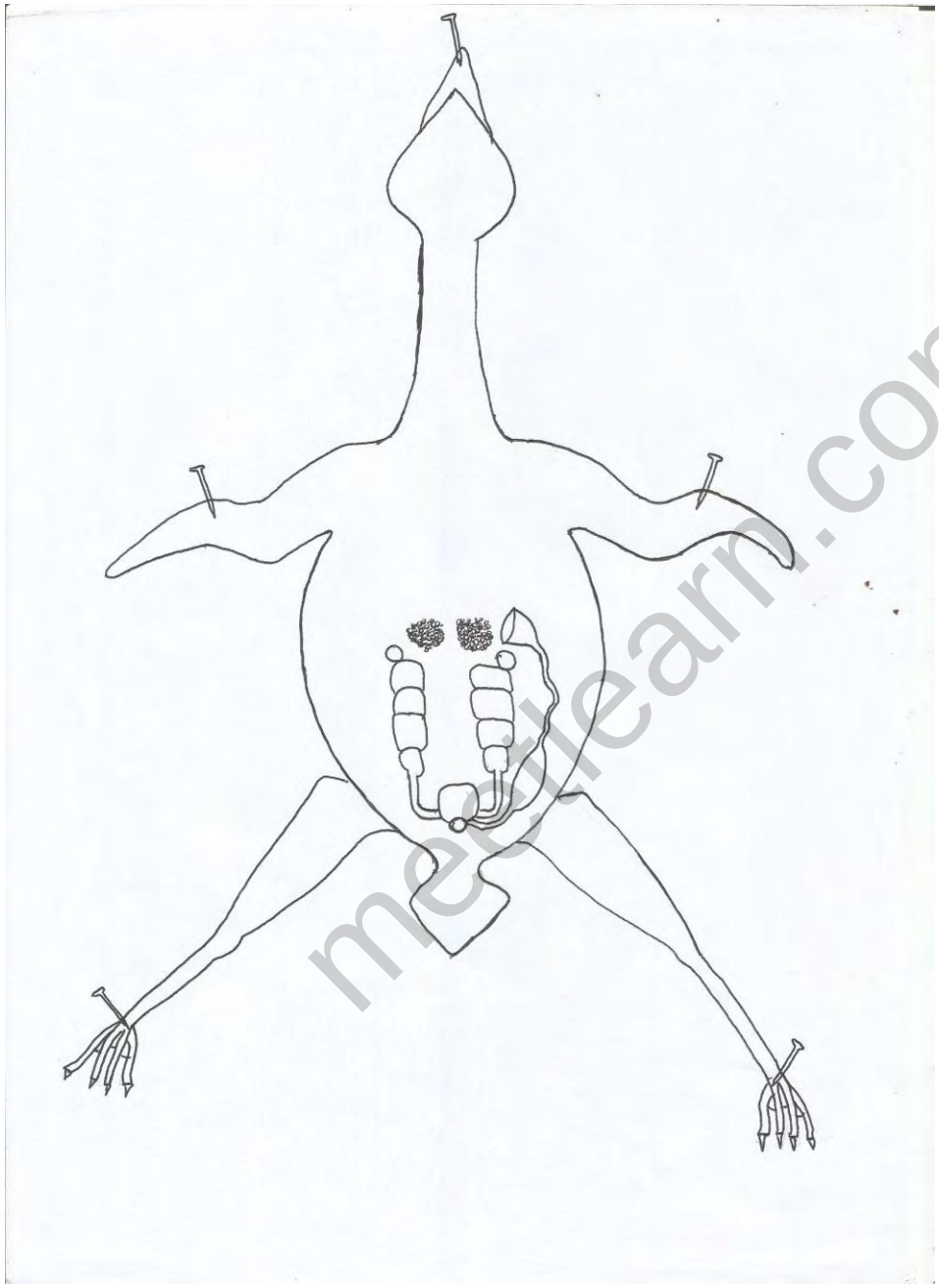

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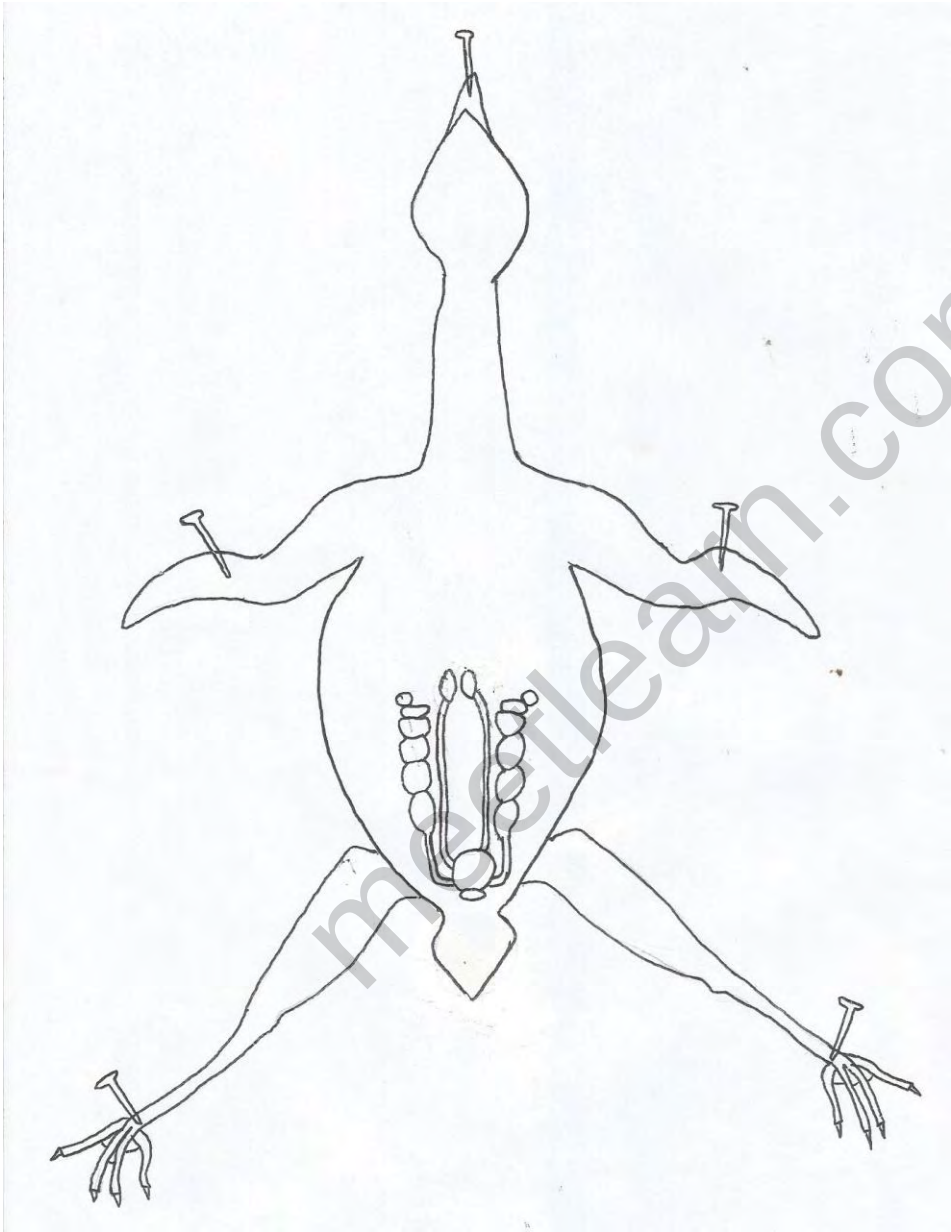

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4. (a) Macerate and form a solution of the muscle, liver and gizzard of the bird used in question three above. Put in three separate test tubes. Investigate the distribution of the enzyme Catalase in each tissue. Make a report of your experiment in the table below. (16 mks)

Procedure	Observation	Inference
• Equal quantities of pieces of muscle, liver and gizzard were removed. • They were macerated separately using a mortar and a pestle. • 6ml of distilled water was added into each macerated tissue sample for form a solution • The tissue solutions were filtered into separate beakers using a filter paper and a funnel. • 2ml of each tissue solution was transferred into separate clean labelled test tubes using a syringe. • 5 drops of hydrogen peroxide was added into each test tube. • The test tubes were allow to stand for 3 minutes on a test tube rack at room temperature.	• Fizzing occurred in all the test tubes. • The highest fizzing occurred in the test tube containing liver (++++++++) • Higher rate in gizzard (+++++) • High rate in muscle (++++)	• Active tissues produce hydrogen peroxide as a by-product of tissue respiration • Hydrogen peroxide is very toxic, and must be converted to non- toxic products. • Catalase is an enzyme produced by all living cells that helps to convert hydrogen peroxide to water and oxygen • $2H_2O_2 \xrightarrow{Catalase} 2H_2O + O_2$ Fizzing indicates the release of oxygen gas • Liver, gizzard and muscle all produce hydrogen peroxide as a by-product of metabolism. • The highest fizzing occurred in liver because it is metabolically very active. It is the site for most metabolic activities. • The gizzard is also active in grinding food, while the muscle helps in movement.

Commented [3]:

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(b) What conclusions can you draw about the amount of Catalase present in each tissue? (4 mks)

• The amount of fizzing or oxygen production in each tissue gives an idea of the amount of Catalase present in the tissue.

• The liver has the highest rate of metabolic activities, and thus the highest amount of Catalase, thus highest amount of fizzing

• The muscle has the higher rate of metabolic activities, and thus the higher amount of Catalase.

• The gizzard has a high rate of metabolic activities, and thus the high amount of Catalase.

• All the tissues contain Catalase, and all also produce hydrogen peroxide as a by-product of metabolism.

(c) what precautionary measures can you undertake to ensure accurate results? (5 mks)

- The same quantities of the different tissues were used.
- The test tubes were of the same size.
- The test tubes and syringes were cleaned and rinsed out with the different tissues.
- Equal volumes of water (6ml) was added to the macerated tissues to form the different solutions.
- Equal volumes of the tissue solutions (enzyme pool) were used.
- Equal volumes of hydrogen peroxide (substrate) were added into the test tubes.
- The test tubes were labelled to avoid confusion.
- The test tubes were allowed to stand for the same length of time.

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June 2019 group 2

Specimens:

A: Obelia **E: maize plant**

B: sea anemone. **F: earthworm**

C: honey bee. **G: toad**

D: bean plant. **H: garden snail**

1. (a) For each of the specimens (A to H) provided, state:

(i) the major group (phylum) to which it belongs;
(ii) the sub-group (class) to which it belongs;
(iii) the habitat in which it is commonly found.

Your answer should be written in your book as shown below (8mks)

Specimen	**Phylum**	**Class**	**Habitat**
A	Cnideria	Hydrozoa	On rock, wooden piles in seas and oceans
B	Cnideria	Anthozoa	On hard surfaces like shells in oceans and seas
C	Arthropoda	Insecta	Bee hive, on flowers, trunks of trees

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D	Angiospermophyta	Dicotyledoneae	Gardens, farms, moist humus soil
E	Angiospermophyta	Monocotyledoneae	Gardens, farms, moist humus soil
F	Annelida	Oligochaeta	Moist humus soil
G	Chordata	Amphibia	Under stones, under old block, in swampy and marsh places
H	Mollusca	Gastropoda	Forest floors, under dead leaves, gardens, farms

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(b) Show the ecological relationships that can exist between specimens C, D, E, G and H. (6 mks)

• Toad feed on honey bee.

• Honey bee feed on the nectar of bean flowers.

• Snail feeds on the fallen leaves of beans and maize

• Honey bee, toad and snail release carbon dioxide during respiration that is used by maize plant for photosynthesis.

• Bean plant sends out carbon oxygen during photosynthesis that is used by toad, snail and honey bee for respiration.

• Honey bee pollutes the flowers of bean plant.

• Beans has root nodules which house nitrogen fixing bacteria that help to soil fertility for the growth of maize plant.

• The nitrogenous wastes and faeces of toads, honey bee and snails help to improve soil fertility for the growth of maize and beans plant.

• The leaves of beans and maize serve as shelter for honey bee, toad and snail.

• When honey bee, snail and toad die, their remains add humus to the soil used by beans and maize plants for growth.

(c) State six differences between specimens D and E. (6 mks)

Specimen D (bean plant)	**Specimen E (maize plant)**
Leaves have net venation	Leaves have parallel venation
Broad leaves	Long narrow and elongated leaves
Flowers are bright in colour	Flowers are dull in colour
Produce pods	Produce cobs
Tap root system	Fibrous root system
Leaves have distinct petioles or stalks	Leaves have sheath bases
Single flowers	Flowers occur as inflorescence
The seeds have two cotyledons	Seeds have one cotyledon
Floral parts in 4 or 5 or multiple of 4 or 5	Floral parts in 3 or multiple of 3
Root nodules present	Root nodules absent

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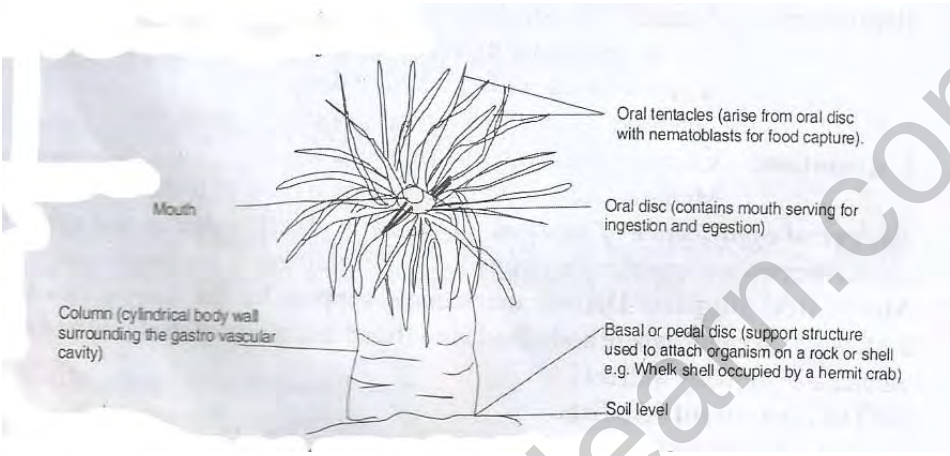
2. (a)

(i) In the space provided, make a large labelled diagram of specimen B (sea anemone) (5 mks)

(ii) Annotate two structures to show their role in nutrition. (4 mks)



(b) What differences exist between specimen A and B. (4 mks)

Specimen A (Obelia)	**Specimen B (sea anemone)**
Colonial	Solitary
both polyp and medusoid forms	Only polyp form exist
Long polyp	Short and stout polyp
Fewer tentacles on the hydranth	Many tentacles
Has both reproductive and feeding polyps	Has only one polyp
Has hydrorhiza for attachment on the substratum	Has basal disc for attachment on the substratum
Body covered with a perisarc	No perisarc

(c) How is specimen A (Obelia)structurally adapted to its life cycle (7 mks)

• It has the perisarc is chitinous and hard to house and protect the colony.

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• It has the hydranth or gastrozoid having tentacles with stinging cells to capture and paralyse preys.

• It has the Gonangium or gonozoid which is the reproductive polyp containing reproductive cells.

• It has whip-like tentacles with stinging cells, which trap food and send into the mouth.

• The Nematoblasts or stinging cells in the tentacles help to paralyse preys and also for defense.

• It has a hollow internal tube called coenosarc, which serves as the alimentary canal that circulate digested food.

• It has the blastostyle, which is an extension of the coenosarc inside the gonangium , which buds off medusa.

• The presence of gonopore, which is a small opening at the tip of gonangium through which medusa bud off from the blastostyle is sent off.

• The gonotheca protects the gonangium or medusae.

• The presence of bell-shaped medusa which swims and produce gametes.

• The presence of hydrorhiza, which is the branching portion of the colony which attaches it to the substratum.

3. (a) Dissect the bird provided to display the Urinogenital system and its associated organs. In the space provided below, make a fully labelled drawing of your dissection. (13 mks)

(b) Annotate any three labels to show how they are adapted to their role in excretion and three to show their in reproduction (12 mks)

Leave your dissection properly displayed for assessment after the examination

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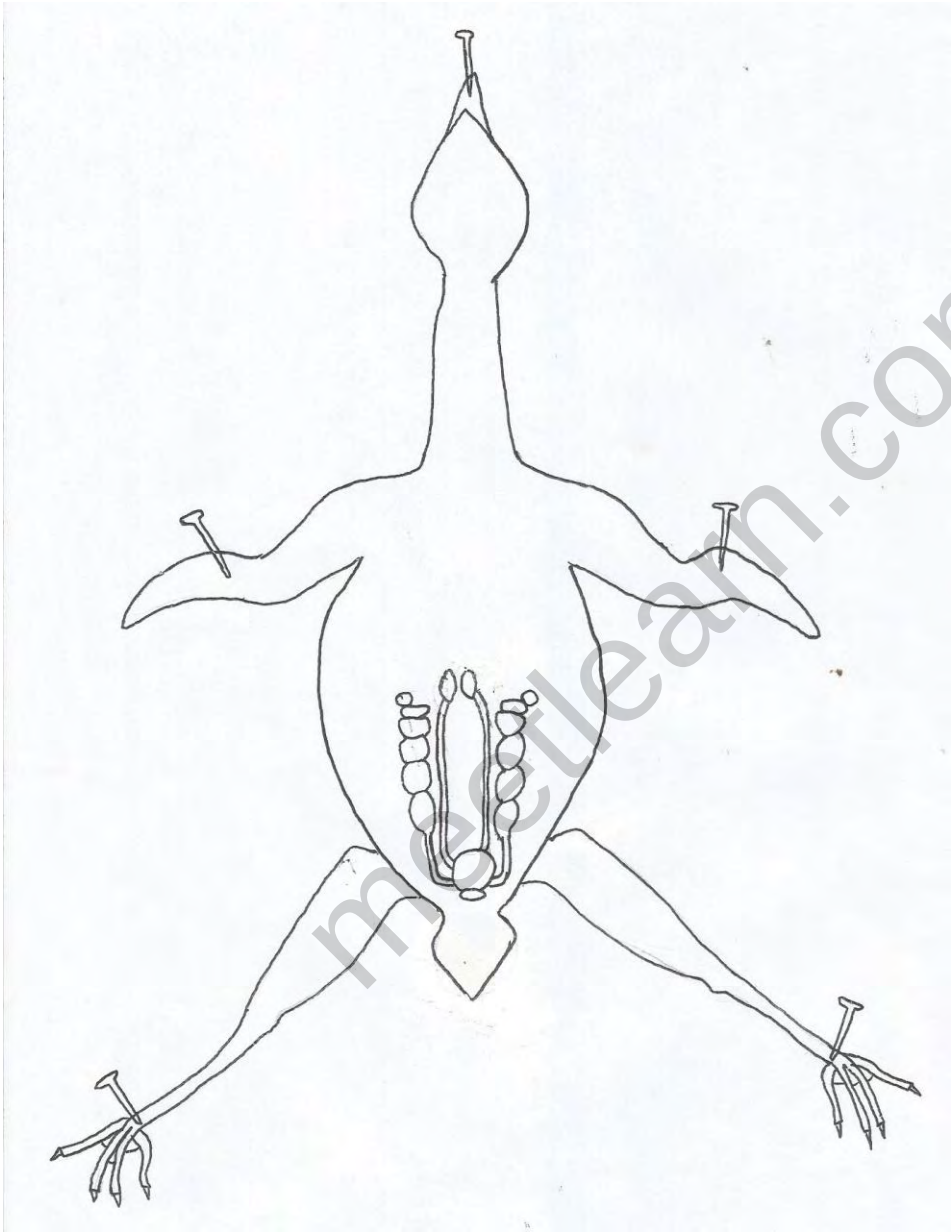

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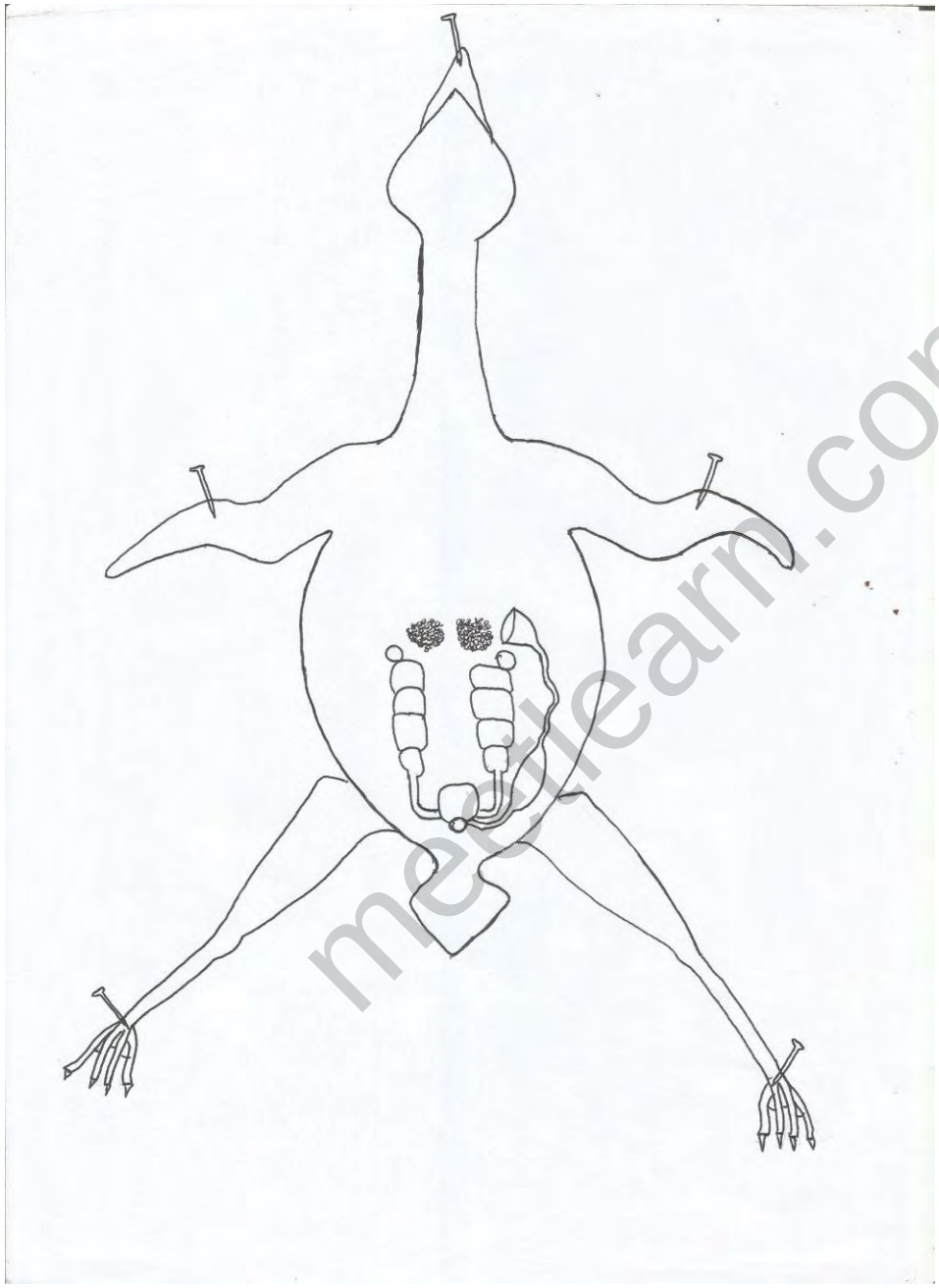

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4. (a). Macerate and collect the juice obtained from liver, ileum and muscle of the bird provided in question 3 above. Put them in three separate test tubes. Investigate the distribution of the enzyme Catalase in each tissue. Make a report of your experiment in the table below. (20 mks)

Procedure	Observation	Inference
• Equal quantities of pieces of liver, ileum and muscle were removed. • They were macerated separately using a mortar and a pestle. • 6ml of distilled water was added into each macerated tissue sample for form a solution • The tissue solutions were filtered into separate beakers using a filter paper and a funnel. • 2ml of each tissue solution was transferred into separate clean labelled test tubes using a syringe. • 5 drops of hydrogen peroxide was added into each test tube. • The test tubes were allow to stand for 3 minutes on a test tube rack at room temperature.	• Fizzing occurred in all the test tubes. • The highest fizzing occurred in the test tube containing liver (++++++++). • Higher rate in ileum (+++++) • High rate in muscle (++++)	• Active tissues produce hydrogen peroxide as a by-product of tissue respiration • Hydrogen peroxide is very toxic, and must be converted to non- toxic products. • Catalase is an enzyme produced by all living cells that helps to convert hydrogen peroxide to water and oxygen • $2H_2O_2 \xrightarrow{Catalase} 2H_2O + O_2$ Fizzing indicates the release of oxygen gas • Liver, gizzard and muscle all produce hydrogen peroxide as a by-product of metabolism. • The highest fizzing occurred in liver because it is metabolically very active. It is the site for most metabolic activities. It has the highest amount of Catalase • The ileum is also active in digestion and absorption of food. It has the higher amount of catalase . The muscle helps in movement and is less active than liver and ileum. Muscle has high amount of Catalase • The amount of fizzing or oxygen produced in each tissue gives an idea of the amount of Catalase present in each tissue.

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(b)Using any of the tissues above and only the materials provided devise a simple experiment to show the effect of heat on the enzyme Catalase. Report your experiment in the table below. (5 mks)

Method	Observation	Explanation
• 2ml of each liver solution was transferred into clean labelled test tubes using a syringe. • The test tube was heated for 5 minutes in a hot water bath. • Test tube was cooled under running tap water. • 5 drops of hydrogen peroxide was added into each test tube. • The test tubes were allow to stand for 3 minutes on a test tube rack at room temperature	• No fizzing was observed.	• The enzyme Catalase was denatured by the heat. • Enzymes lost their conformation and could no longer bind to the substrate (hydrogen peroxide). • No enzyme-substrate complex was formed. • No reactions took place and products were not formed, hence no fizzing was observed.

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<u>June 2019 group 3</u>

<u>Specimens:</u>

A: hibiscus twig with leaves and flowers. **E: star fish**

B: toads. **F: sandworm**

C: fern plant with leaves **G: leeks**

D: cypress **H: dogfish**

1. (a) For each of the specimens (A to H) provided, state:

(i) the major group (phylum) to which it belongs;
(ii) the sub-group (class) to which it belongs;
(iii) the scientific or common name.

Your answer should be written in your book as shown below (8mks)

Specimen	Phylum	Class	Scientific or common name
A	Angiospermophyta	Dicotyledoneae	Hibiscus
B	Chordata	Amphibia	Toad
C	Filicinophyta	Filicinae	Fern
D	Coniferophyta	Coniferae / Coniferinae	Cypress
E	Echinodermata	Stelleroidea	Starfish
F	Annelida	Polychaeta	Nereis / Sandworm
G	Angiospermophyta	Monocotyledoneae	Leeks
H	Chordata	Chondrichthyes	Dogfish

(b) State one feature each possessed by specimen B and C that can enable you place it in major group and one feature that can enable you place it in its sub-group. (4 mks)

B (toad)

Major group:

- Presence of tubular dorsal nerve chord
- Endoskeleton made up of bone or cartilage
- Presence of pharyngeal gill slits
- Bilateral symmetry and are metamerically segmented.
  They have a closed circulatory system.

Sub-group:

- Rough warty skin with no scales.
- Have two pairs of limbs with four to five digits.
- The hind limbs end with webbed toes for swimming.

C (fern)

Main group:

- The spores are homosporous
- The gametophyte is reduced to the prothallus.
- They have true root, true stems and true leaves.

Sub-group:

- presence of fond with sori which bear Sporangia
- Presence of underground stem called rhizome from which fronds develop and grow.
- Sori covered with indusium.

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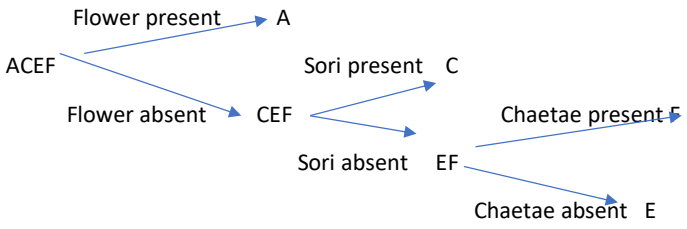
- The gametophyte is heart-shaped and its called prothallus, which bears both antheridia and archegonia

(c) (i) Construct a dichotomous key to separate specimens A, C, E and F. (3 mks)



(ii) Describe five adaptations shown by specimen C fern) to its habitat. (5 mks)

- Presence of true vascular tissues for movement of materials.
- Posses roots for anchorage.
- Sporophyte with vascular bundle is the dominant generation
- Sporophyte does not depend on gametophyte for support and nutrition.
- Presence of annulus cells for catapulting spores out of the Sporangium.
- Indusium protects sori from dehydration.
- Presence of rhizome for vegetative propagation.

2. Examine specimen D (cypress) carefully.

(a) State four features which you consider to be of biological interest as far as the specimen is concerned.

- It produces reproductive male and female cones.
- Has scaly leaves to reduce water loss.
- The leaves are always green all year round for photosynthesis.
- Have very tap root system for anchorage and absorption of water from deeper soil layers.

(b) In the space provided below, make an annotated diagram of the reproductive part of specimen A (hibiscus) to show any four features concerned with reproduction. (11 mks)

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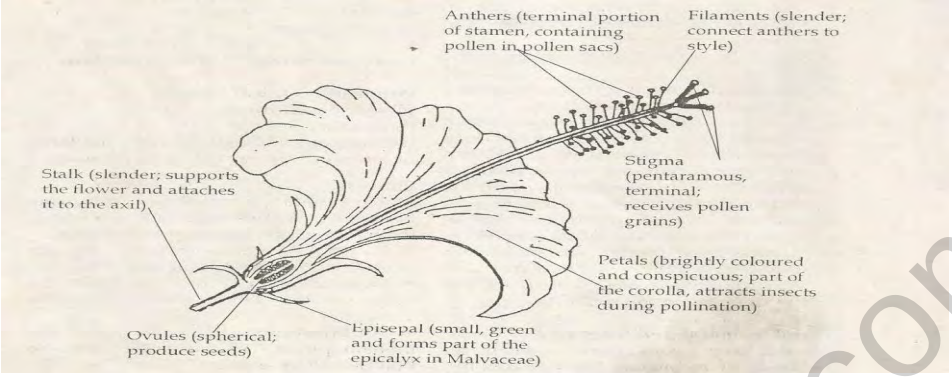

(c) Make a table to give any five differences that are visible between specimens A and G. (5 mks)

Specimen A (hibiscus)	**Specimen G (leaks)**
Net venation	Parallel venation
Tap root system	Fibrous root system
Hard and branched stems	Soft and unbranched stem
Floral parts occur in fives	Floral parts arranged in threes
Broad leaves	Narrow elongated leaves

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3. (a) Dissect the fish provided and show clearly the following features: (i) Stomach (ii) Liver (iii) Gallbladder (iv) Spleen (v) Tail fin (vi) Testes/Ovary (vii) Kidney (viii) Heart (ix) Left gill (x) Ureter (xi) Ileum (xii) Duodenum.

In the space provided below, make a fully labelled drawing of your dissection. (13 mks)

(b) Annotate 6 of the labels to show the function of two in circulation and two in excretion and two in digestion. (12 mks)

Leave your dissection properly displayed for assessment at the end of the examination.

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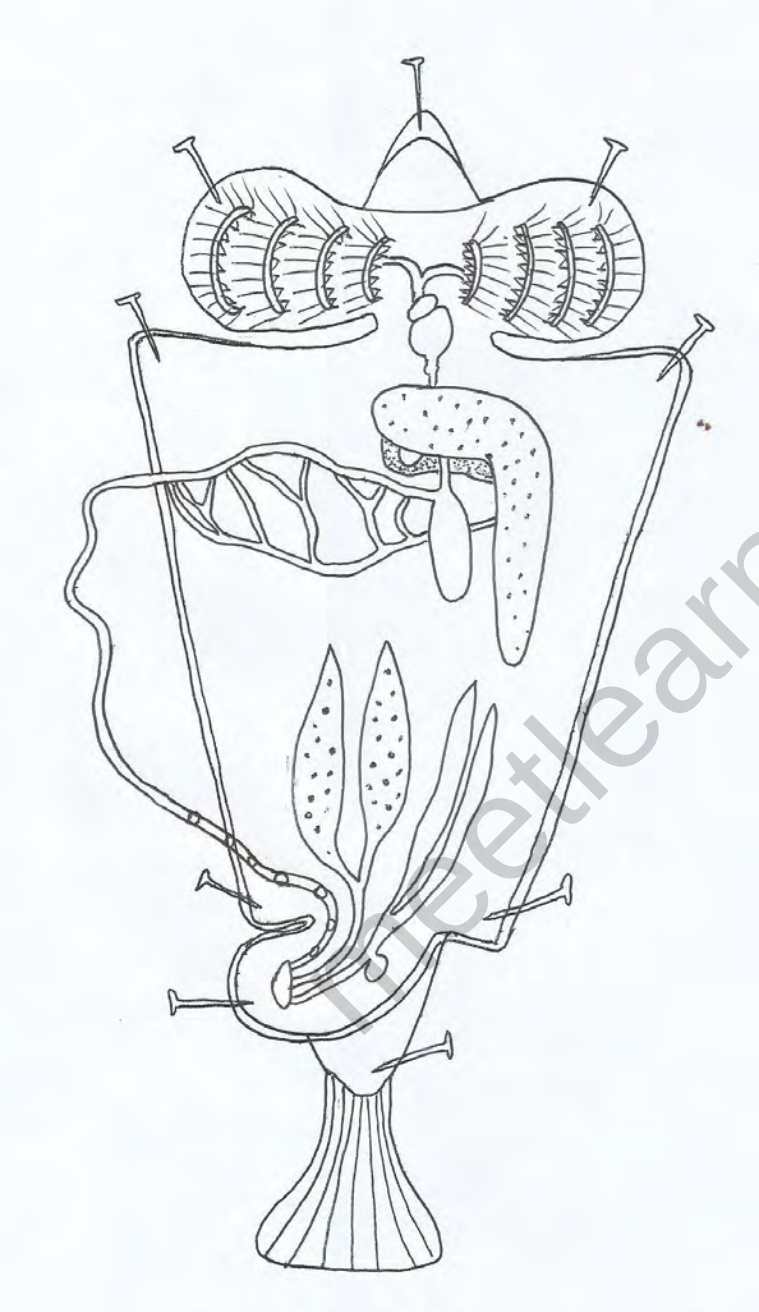

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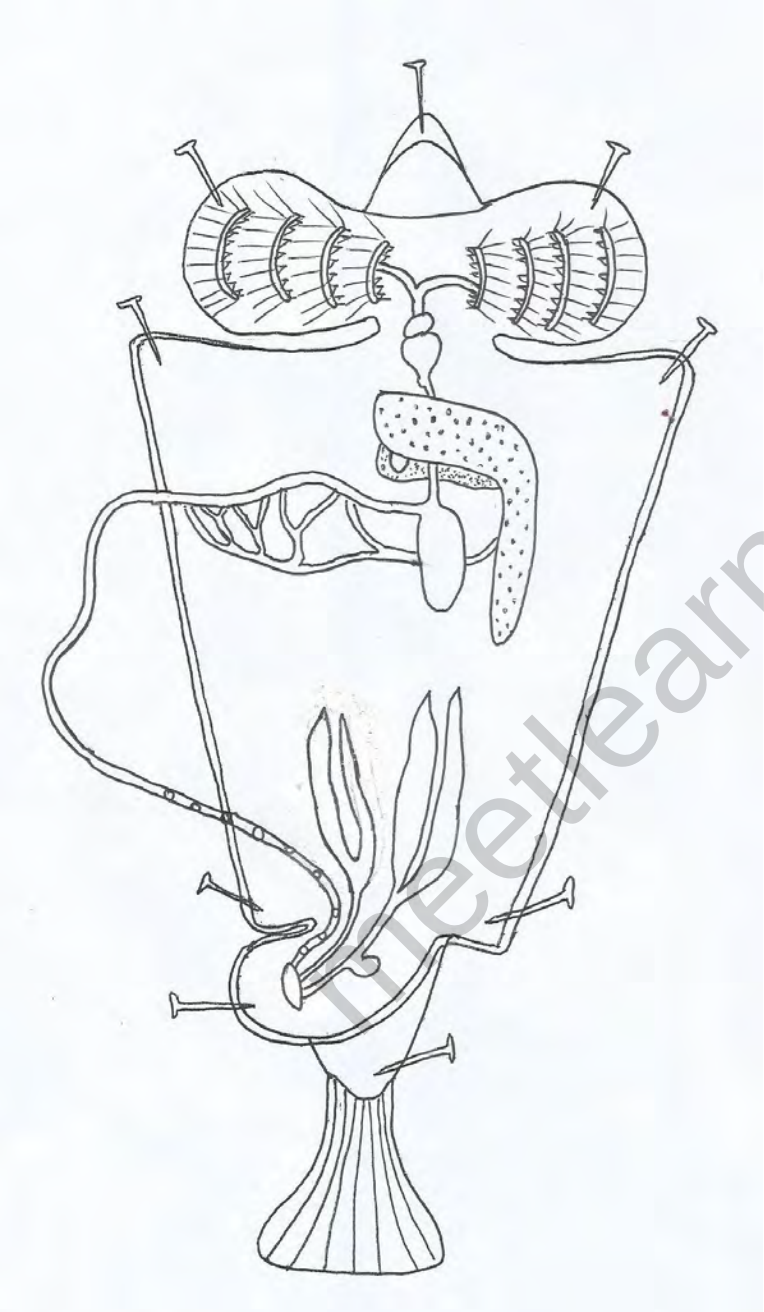

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4. (a) Macerate and come out with a solution from juice obtained from the cotyledon, testa and embryo of ten soaked germinating bean seeds provided. Put in three separate test tubes. Investigate the distribution of the enzyme Catalase in each tissue. Make a report of your experiment in the table below. (16 mks)

Procedure	Observation	Inferences
• The cotyledons, testa and embryos of ten germinating bean seeds were separated. • The tissues were macerated separately using a mortar and a pestle. • 5ml of distilled water was added into each crushed tissue to form solutions. • They were filtered separately into separate labelled beakers using filter paper and funnel. • 1ml of each tissue solution was transferred into separate clean labelled test tubes using a clean syringe. • 1ml of hydrogen peroxide was added into each test tube. • The test tubes were allowed to stand for 3 minutes. • The lengths of the foams produced from each test tube were measured.	**• All the three solutions produce foams.** **• The highest foaming was seen in the test tube containing cotyledon ++++++++ or 10cm.** **• This was followed by the embryo ++++ or 4cm** **• The least foaming was produced by testa + or 1cm**	• Active tissues produce hydrogen peroxide as a by-product of tissue respiration • Hydrogen peroxide is very toxic, and must be converted to non- toxic products. • Catalase is an enzyme produced by all living cells that helps to convert toxic hydrogen peroxide to water and oxygen • $2H_2O_2 \xrightarrow{Catalase} 2H_2O + O_2$ Fizzing indicates the release of oxygen gas • The highest foaming was produced by cotyledon because of high metabolic activities. E.g. digestion of stored food for the embryo. • This was followed by embryo because the cells are actively dividing and building up new organelles. • The least foaming was seen in testa because it is made up mainly of dead cells with no metabolic activities.

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Conclusion: (4 mks)

(b) Make a bar chat using your results. (5mks)

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North West Mock 2019 group 2

Specimens:

A: mushroom. **E: liverwort**

B: leeks. **F: caterpillar**

C: clam. **G: moth.**

D: hibiscus **H: honeybee**

1. (a) For each of the specimens (A to H) provided, state:

(i) the major group (phylum) to which it belongs;
(ii) the sub-group (class) to which it belongs;
(iii) the scientific or common name.

Your answer should be written in your book as shown below (8mks)

Specimen	**Phylum**	**Class**	**Scientific or common name**
A	Basidiomycota	Basidiomycetes	Mushroom
B	Angiospermophyta	Monocotyledoneae	Leeks
C	Mollusca	Bivalvia	Clam
D	Angiospermophyta	Dicotyledoneae	Hibiscus
E	Bryophyta	Hepaticae	Liverwort
F	Arthropoda	Insecta	Caterpillar
G	Arthropoda	Insecta	Moth
H	Arthropoda	Insecta	Honeybee

(b) Examine specimens A to E and state precisely where each can be found. (5 mks)

Specimen	**Habitat**
A	Rotting logs of wood / moist humus soil

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B	Gardens / farm lands
C	Fresh water , remains in burrows under sand
D	Gardens / moist humus soil
E	On moist soil / tree trunk / old damp blocks

(c) State the economic importance of each of specimens F and H. (7 mks)

Specimen F:

- It serves as food to man and other animals
- Agent of decay.
- Its nitrogenous wastes and dead body add soil humus, increasing soil fertility.
- It feeds on plant leaves and destroy crops.
- Produces silk used to make cloths

Specimen H: (7 mks)

- Produces honey used as food and medicine
- Bee wax is used to produce candles, shoe polish and varnish.
- Helps to pollinate flowers.
- Serve as food to man and other animals.
- Its royal jelly is used in the field of medicine to treat asthma, hay fever and pancreatitis

2. (a)

(i) Make a fully labelled diagram of specimen G

(ii) Annotate two structures which adapt it to its mode of feeding. (9 mks)

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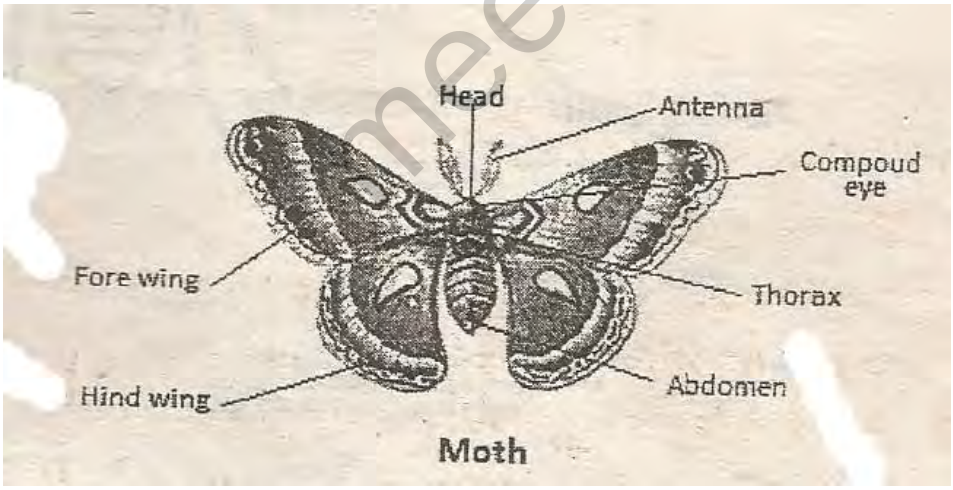
Structure of a moth



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(b) (i) Give the similarities between specimens F and G. (3 mks)

- Both have spiracles for breathing.
- Both have a head that bear a mouth.
- Both bodies are divided into head, thorax and abdomen
- Both have 3 pairs of jointed legs attached on the thoracic segments.

(ii) Bring out 3 differences between specimen F and G (3 mks)

Caterpillar (F)	**Moth (G)**
No wings	Wings present
Prolegs present	Prolegs absent
Presence of silk glands	Silk glands absent
Presence of osmeterium which exudes a reddish substance for defense	Absence of osmeterium

(c) Give the differences between specimen G and butterfly. (5 mks)

Specimen G (moth)	**Butterfly**
Fly or are more active at night	Fly or are more active during the day
At rest, the wings are outspread and horizontal	At rest, the wings are held folded upwards together (vertical)
Most are dull in colour	Wings are brightly coloured
Overlapping scales on the surface of wings (scaly wings)	No scales of wings
Antennae are long and slender, thickened towards the middle or may be branched and feathery.	Antennae are club-shaped and thickened towards the tip
Fore and wings are connected	Fore and hind limbs are separated
Thicker and heavier body	Light and thin delicate body

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3. (a) Dissect fish provided and display the alimentary canal. Make a fully labelled drawing of your dissection. (12 mks)

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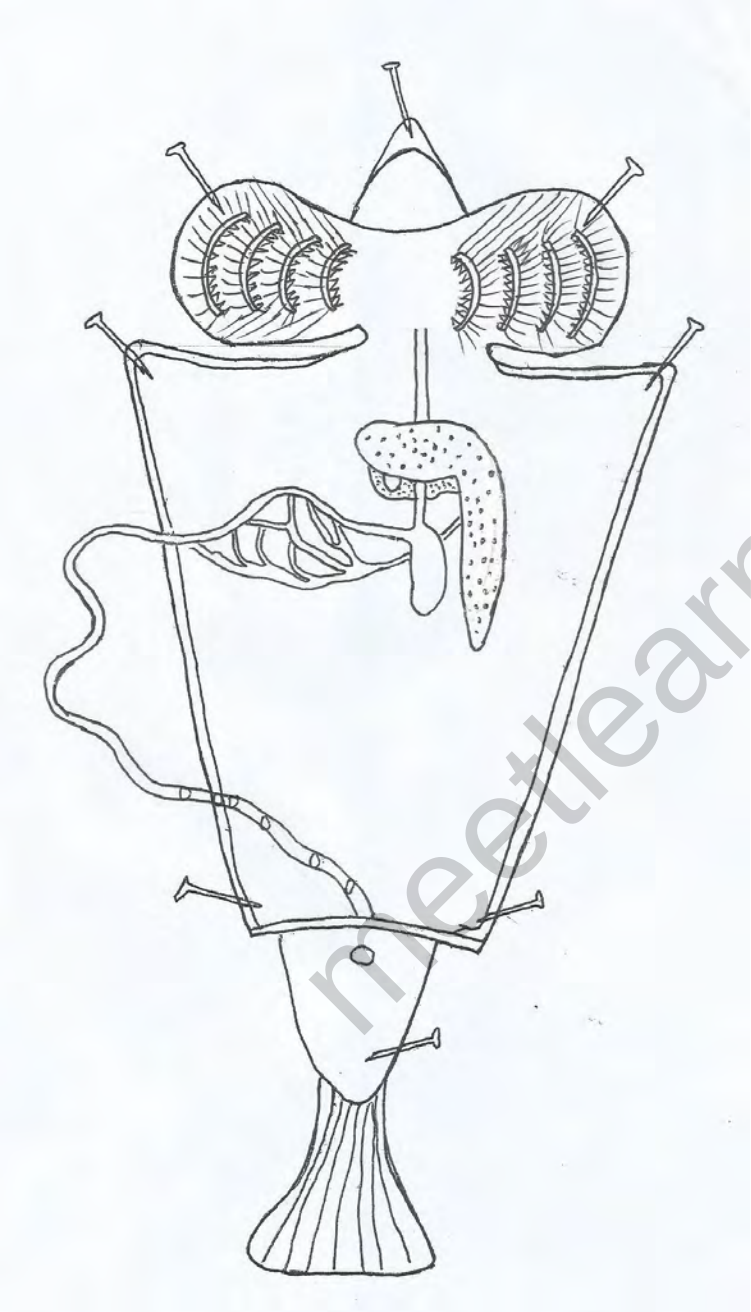

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(b) How is the alimentary canal adapted to the diet of the animal. 4 mks)

- Large muscular U-shaped stomach that secretes gastric juice and acid for digestion own storage of food
- Thin fatty membranous pancreas which secretes pancreatic juice with enzymes for digestion.
- Long and coil ileum that secretes intestinal juice with enzymes for digestion
- Long ileum for large surface area for absorption
- Large lobed yellowish red liver for bile production
- Sac-like gall bladder that releases bile into the duodenum for fat digestion and neutralization of acidic chyme.

(c) What the differences between the alimentary canal of mammal and that of a fish? (4 mks)

Mammalian alimentary canal	**Fish alimentary canal**
Appendix present	Appendix absent
U-shaped duodenum	Flat appendix
Large well developed Caecum	Finger-like Caecum
Flat liver lobes	J-shaped liver
Pancreas soft and fatty	Leaf-like pancreas
Very long ileum	Relatively short ileum
Long oesophagus	Very short oesophagus

Leave your dissection properly displayed for assessment at the end of the examination.

4. (a) Dissect the germinating maize seed provided and draw a large labelled drawing of the longitudinal section. Using iodine solution, annotate the parts to show the concentration of starch in the different tissues. (9 mks)

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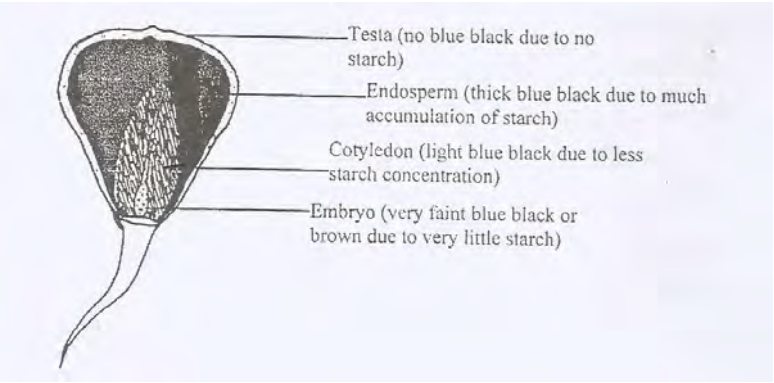

(b) Macerate the germinating maize seeds provided and form a solution. Use the solution to test for the presence of

(i) starch.

(ii) reducing sugars

(iii) proteins.

(iv) lipids. (19 mks)

Preparation of the maize solution

- Ten **germinating maize seeds were macerated using a mortar and a pestle**
- **10ml of distilled water was added to form a solution**
- **The solution was filtered using a filter paper and a funnel.**
- **The filtrate was used for the following test**

Test for starch

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Procedure	Observation and conclusion
•2ml of the maize solution was transfer into clean test tube using a clean syringe • 2 drops of iodine solution was added, while shaking after each drop.	• A blue black was seen, indicating the presence of starch

Test for reducing sugars

Procedure	**Observation and conclusion**
• 2ml of the filtrate was transferred into a clean test tube using a clean syringe. **• 2ml of Benedict's solution was added.** **• The mixture was heated for 5 minutes in a warm water bath, while shaking at interval.**	**• The colour changed from blue to green, yellow, orange and finally to brick red, indicating the presence of reducing sugars**

Test for proteins

Procedure	**Observation and conclusion**
• 2ml of the filtrate was transferred into a clean test tube using a clean syringe. **• 1ml of NaOH solution was added.** **3 drops of $CuSO_4$, while shaking after each drop.**	**• A purple colour was not seen, indicating the absence of starch.**

Test for lipids

Procedure	Observation and conclusion
• 2ml of the filtrate was transferred into a clean test tube using a clean syringe. • 3 drops of Sudan III solution was added. • 2ml of water was added and shakened well	• The oil separates from the water, indicating the presence of lipids

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